Illiquid Heterogeneous Banks and Endogenous Market Structure*

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Abstract

This paper studies banks' market structure and the redistribution of liquidity regulations in an incomplete market economy with an assignment matching game. I develop a theory in which liquidity frictions and illiquidity costs generate negative assortative matching in the banking sector and a redistribution of assets between heterogeneous banks and across assets that occurs through an imperfect substitution of resources. I demonstrate that through the price of illiquid assets, the model gives rise to an endogenous marginal propensity to lend different between large and small banks and between liquid and illiquid banks. The theory does not take the federal funds market as given, but rather it arises endogenously from the nature of the liquidity market. I estimate the model using microdata on US commercial banks and I find a heterogeneous cross-sectional effect on the banks' marginal propensity to lend. Empirically, I find that post-financial crisis the market structure of banks becomes more selective towards small banks with positive marginal propensity to lend, matching with large banks, generating a higher value surplus. I find a remarkable size effect of the marginal propensity to lend in the lower quantiles of the banks' distribution post-financial crisis, three times higher than the pre-financial crisis value. In quantifying the model, I derive a number of important results for the elasticity of illiquid assets. Finally, I use the theory to examine the impact of the Bank Term Funding Program. The program led to a 1.95 percent increase in illiquid assets and a 77.85 percent rise in the marginal propensity to lend.

Keywords: Assignment game, inequality, assortative matching, marginal propensity to lend, assets redistribution, regulations, Tobit model, quantile model, computational methods
JEL Classification: D41, D47, D49, E44, E58, C34, C31, C63, G21, L51

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1 Introduction

The market structure is the channel through which banks' regulations determine the supply of lending. When convex adjustment costs and submodular regulatory constraints lead large banks to merge with small ones, the size distribution of banks compresses to a medium size distribution. Because the marginal propensity to lend is heterogeneous across banks and varies systematically with size, the stationary market structure generated by the regulation becomes the key determinant of aggregate lending. Regulation redefines the industry, and the shape of the industry determines how much banks lend. This paper is motivated by the evidence of heterogeneous marginal propensity to lend across banks documented in this paper and the mechanism through which regulation and adjustment costs in banks' illiquid assets change the market structure of banks. Its importance relies on the observation that over time the steady state of illiquid assets has moved to an increasing value and therefore, the endogenous change of regulations has modified the market structure through banks' fundamentals with outcomes for the stability of the banking system.

Recent empirical studies have overwhelmingly proven the existence of large deposits differences amongst banks (Corbae and D'Erasmo, 2021, 2022; Oberfield et al., 2024; Gigante, 2026). The majority of these deposits is concentrated in small banks within a cohort of banks posing disparities across banks in their banking choices, liquid and illiquid assets. This literature has emphasised deposits as the underpinning source of idiosyncratic uncertainty driving bank size expansion. Yet, little is known in the theory regarding banks' responses of illiquid assets following changes in liquidity regulations and, how regulation and matching technology jointly determine the market structure even when the number of banks is fixed.

The experience of Silicon Valley Bank (hereafter, SVB) has put at center stage the importance of bank regulation in the financial system. Since banks are continuously exposed to short-term liquidity problems, the collapse of SVB has fostered the need of tighter liquidity regulations.[1] This paper studies the costs and benefits of banks' programs and regulations following the financial crisis. It addresses three important questions. First, how does regulation change the market structure, and therefore, the distribution of banks through matching? Second, how does the matching technology determine the composition between small, medium and large banks and what does that imply for the marginal propensity to lend? Third, how does the interaction of regulatory thresholds and matching shift the distribution of banks and how do they translate into changes in lending behaviors and marginal propensity to lend? I address these questions theoretically and quantitatively.

[1] In a recent article published on Reuters on September 26, 2024 US Treasury Secretary Janet Yellen advocated the need of thoughtful financial regulations and emphasized the importance of non reduction of bank capital requirements. In calling thoughtful banks' regulations on all grounds, she states: "A resilient financial system is critical to a strong economy. And strengthening it requires insisting on thoughtful regulation, including in the face of challenges from those who advocate to roll back policies and regulations". The article is available at https://www.reuters.com/markets/us/yellen-call-more-financial-stability-work-thoughtful-regulation-2024-09-26/.

In this paper, I develop a novel theory of banks' regulations and market structure where banks are subject to uninsurable deposit withdrawal shocks which expose them to a liquidity risk. I analyse the ripple effects due to fear of turmoil in one depository institution that can lead depositors to withdraw their deposits from small banks altogether. I study this type of ripple effects from the perspective of a prudent banking sector in holding liquidity. The model disentangles the benefits of holding liquidity and underlines the adjustment costs properties of redistributing liquidity to big banks in period of crises. The mechanism relies on the price of illiquid assets which has to be counterbalanced with the marginal propensity to lend. In particular, the matching stage is an assignment problem in which the surplus from pairing any two banks is determined by convex adjustment costs and submodular regulatory constraints.

Empirically, I dispense direct evidence of the redistribution mechanism in the banking sector, the marginal propensity to lend at the individual bank level and size heterogeneity of banks driven by regulatory constraints. Next, I develop a novel theoretical model that generates an endogenous market structure and I quantify the effects of the mechanism by estimating it to the US commercial banking sector. I estimate the marginal propensity to lend and I validate the model in a series of counterfactual experiments on the matching technology and the regulatory requirements and, by evaluating the effectiveness of the Bank Term Funding Program, a liquidity program introduced by the Federal Reserve System (hereafter, Fed) in March 2023. With this program, the Fed made available additional funding to eligible depository institutions.[2] Therefore, in this part of the model the paper examines the quantitative importance of this program on the marginal propensity to lend and illiquid assets from the bank's perspective.

I begin the analysis by providing descriptive evidence using administrative microdata on banks for the US economy. I collect the data from the Consolidated Report of Condition and Income from the Federal Reserve Bank of Chicago and provided by Wharton Research Data Services (2024). I find that the descriptive evidence are peculiar characteristics of an economic environment that I show to be negative assortative matching in the banking sector, whereby large banks match with small banks and the matching mechanism operates in the size space with a different fixed point of the matching process. The market structure is then characterised by banks whose mass is in a region where they hold more liquid assets, less illiquid assets and therefore, lower marginal propensity to lend.

The main contributions of the paper are theoretical and quantitative. Grounded within the heterogeneity in the banking sector, I develop a novel dynamic model with incomplete markets, an idiosyncratic deposits withdrawal risk, regulatory constraints and the price of illiquid

[2]This program was created as a means to restore confidence in the financial market and assure banks that they have enough liquid funds in the event of withdrawals by depositors. The program offered loans to commercial banks, savings associations, credit unions, and other eligible depository institutions who could pledge some collateral for the purchase of loans. The primary aim of the program was to eliminate an institution's need to quickly sell high quality liquid assets such as securities in times of turmoils. A detailed explanation of the Bank Term Funding Program is available at https://www.federalreserve.gov/financial-stability/bank-term-funding-program.htm and Appendix K provides further detail on the program.

assets, where I microfound an assignment matching game that explains the negative assortative matching heavily documented in the data. Motivated by the fact that loans are largely an illiquid asset for banks, my theory features heterogeneous banks who can trade two types of assets: liquid assets and illiquid assets. Central to the price of loans are adjustment costs in illiquid assets that lead to fewer investment opportunities to banks and guide the redistribution of assets between small and large banks. In the model banks pledge liquidity as a collateral in obtaining additional funding. At the end of each period a fraction of banks exits. The matching technology is characterised by adjustment costs in illiquid assets and submodular regulatory constraints that leads to the endogenous market structure where negative assortative matching is the optimal outcome.

The decentralised economy inherits an inefficiency due to the deposit outflows that leaves banks susceptible to runs. The market structure makes matchings between banks of the same type inefficient since they have the same incentives towards liquid and illiquid assets decisions. However, I show that the redistribution mechanism that occurs from small banks to large banks has important efficiency properties. Regulatory environments improve when matchings occur between large and small banks. However, this inefficiency calls for further policy intervention. I embed adjustment costs in illiquid assets to highlight the importance of frictions in determining the marginal propensity to lend at the individual bank level. My theory engenders an endogenous marginal propensity to lend and I show that it changes in the space of deposits according to banks' size and varies across banks.

The model provides a mechanism for banks' matching that occurs through adjustment costs in illiquid assets interacting with regulatory constraints. They jointly change the distribution of banks upward where the banking sector becomes dominated by medium banks with consequences on the amount of lending, moving upward the stationary value of illiquid assets. In particular, the same primitives that change the market structure also determine the marginal propensity to lend. There is an intermediate stage called the matching stage where banks match according to the joint surplus function leading to a banking sector with more liquidity intensive banks. In light of the guidance of this theoretical environment I show that illiquid assets of large banks increase less than the supply of illiquid assets of small banks more reliant on deposits.

The last theoretical result evaluates the size expansion properties determined by the outflow of deposits between commercial banks. I show that the largest banks match with the smallest banks expanding in size. This is the negative assortative matching that arises in this model and that endogenously changes the market structure giving rise to the federal funds market. The matching stage is conceived along the lines of the seminal articles of Becker (1973) and Shapley and Shubik (1971). The adjustment costs in illiquid assets and submodular regulatory constraints make the matching between large and small banks strictly surplus dominant and the optimal matching. The move upward of the fixed point of the assortative matching between banks implies that the space sector is now made by larger banks with more liquid assets, less

illiquid assets, lower loan-to-deposit ratio and therefore, lower marginal propensity to lend. These findings highlight the importance of tackling the type of ripple effect in the banking sector, since a shock to one type of banks can have profound different consequences to the market structure.

In testing the theory, I estimate the marginal propensity to lend by a Tobit regression specification and the inequality in the marginal propensity to lend across banks via a quantile regression. In doing this analysis, I test the prediction of the model that banks reduce their loans when facing additional funding. I calibrate a set of parameters from US banks' microdata to fit important moments. The estimations through a Tobit model reveal a size-dependent pattern of illiquid assets with a substitution and submodularity effect. While large banks increase illiquid assets following upward changes in liquid assets, small banks acting at the constraint substitute away from illiquid assets. The adjustment costs highlight a redistribution of illiquid assets between banks, whereby illiquid assets decreases by 20.02% and 26.95% post-financial crisis relative to the pre-crisis year, suggesting the imposition of tighter liquidity regulations.

The marginal propensity to lend is estimated to decrease for large banks when they are faced with more idiosyncratic illiquidity, whereas it increases by approximately 869.058% for small banks in 2010 relative to the pre-crisis period. By raising the marginal propensity to lend, small banks amplify the shock of illiquidity risk in the market. When small banks match with large banks, this shock is stabilised since large banks do not expand lending, and negative assortative matching between banks arises. The matching between large and small banks drives selection into the marginal propensity to lend. The unconditional marginal effects corroborates a change in the market structure that is negative assortative post-financial crisis. With more liquid assets, small banks reduce the marginal propensity to lend with an estimated effect that is 10 times larger than the response of large banks.

The market structure switches from a complementarities sector between banks where the surplus of the match was spread between the full size distribution of banks in 1998, to a more selected market structure represented by small banks with positive marginal propensity to lend matching with large banks and whose matches deliver a higher value surplus. In validating the model, I discuss the importance of the identifying assumptions.

In terms of inequality in banks' marginal propensity to lend, I implement a quantile model and I find that liquidity has a larger effect at the upper quantiles of the banks' distribution post-financial crisis, while the opposite holds pre-financial crisis. Since in 2010 the upper tail of the distribution is dominated by constrained small banks, liquid assets strongly increase the marginal propensity to lend of medium large and medium small banks. Negative assortative matching is highlighted by the result that increases in illiquid assets generate a strong heterogeneous response in the tails of the distribution particularly the 25th and 90th quantiles with a sign reversal between 1998 and 2010.

This result suggests that while pre-financial crisis large banks in the 25th quantile strongly reduce the marginal propensity to lend, post-financial crisis they expand loans and their marginal

propensity to lend. This expansion is three times higher than the pre-financial crisis counterpart and is economically meaningful and statistically significant, highlighting a remarkable size effect in the marginal propensity to lend. Small banks in the 90th quantile experience a decline in the marginal propensity to lend post-financial crisis that is almost sixfold higher than the rise in the marginal propensity to lend pre-financial crisis. These findings mark a highly heterogeneous marginal propensity to lend between banks and across years. The inequality in the marginal propensity to lend is mainly due to small banks' fundamentals since large banks show a more uniform behavior across quantiles.

Quantitatively, I use the developed model to study the importance of the assortative matching mechanism for the market structure of banks, of the heterogeneous marginal propensity to lend across banks, and the effectiveness of the Bank Term Funding Program implemented to depository institutions in March 2023.

The quantitative results highlight a very unequal distribution of the marginal propensity to lend that accounts for 56.3% of differences between large and small banks. This distribution is largely centered towards large banks. The results indicate a 0.05% elasticity in the marginal propensity to lend out of upward changes in the interest rate on loans that is mainly driven by matching between large and small banks, since the elasticity at the tails of the distribution is highly unequal. In particular, the elasticity is remarkably high for small banks while it is moderate for large banks. I find that medium large banks increase illiquid assets by 8.5% following a 1% increase in the interest rate on loans, while small banks are highly sensitive to upward changes in the return on loans. The model reconciles the low (high) marginal propensity to lend at high (low) deposits banks in average terms, driven by the negative assortative matching.

My second main finding shows negative assortative matching in the banking sector. More precisely, banks with low deposits have on average more liquid assets, smaller expansion in illiquid assets, low and negative marginal propensity to lend and dividends. The submodularity in the liquidity constraint penalises additional holdings of liquid assets, especially for large banks. The interaction with small banks makes the surplus of the match between large and small banks dominant, since large banks can expand liquid assets by relaxing the constraint and use the additional deposits to expand illiquid assets. Small banks rise illiquid assets when faced with low deposits leading to a positive marginal propensity to lend which turns to decrease as matches between large and small banks occur. I uncovered a V-shaped path of the marginal propensity to lend in the deposits distribution that is dominant for large and medium banks, and a decreasing marginal propensity to lend across banks. Moreover, the marginal propensity to lend is very sensitive to the strength of submodularity, particularly for medium and small banks.

Finally, my third main finding shows a counterfactual experiment that studies the effectiveness of the Bank Term Funding Program. I offer an analysis where the Program is simulated as an unanticipated shock to the liquidity capacity of banks which relaxes their liquidity constraint. The shock is intended to be an aggregate shock along the lines of the approach developed by

Boppart et al. (2018), and I compute stationary equilibrium and transitional dynamics. I find a 1.95% expansion in illiquid assets and a simultaneous 4.51% decay in liquid assets, following a 15% relaxation of the liquidity constraint. More importantly, the results suggest the program would have expanded lending by 77.85% by increasing the banks' marginal propensity to lend on impact. This increase comes mainly from medium and small banks since they have the highest shadow value of relaxing the constraint. The shock creates a redistribution of assets between banks, and the supply of illiquid assets of large banks that are more leverage constrained, increases less than the supply of illiquid assets of small banks.

By studying the heterogeneity in banks' liquid and illiquid assets, the marginal propensity to lend and short term debt for the market structure of banks, the paper highlights the importance of cross-sectional distributional differences in banks' liquid assets and illiquid assets to identify large illiquid banks and the redistributive role of banks' liquidity between large and small banks. Overall, the results show that banks' regulation does not only constrain individual banks, but redefines the equilibrium of the assignment model. They underscore the stability of the mechanism driving banks' matching, while the equilibrium market structure has shifted, making banks' regulations the driver for the feasible region of the assignment matching problem.

Outline. The rest of the paper is organised as follows. Section 2 discusses the related literature. Section 3 documents micro level descriptive evidence on banks. Section 4 presents the model, theoretical results and the characterisation of the efficiency properties and the assignment matching properties. The Section shows how the market structure arises endogenously in the model. Section 5 estimates empirical evidence on the marginal propensity to lend. Section 6 analyses distributional findings, comparative statics and results on the redistribution mechanism of assets, banks' matching and evaluates policy outcomes. Section 7 concludes. The Appendix provides proofs of propositions, omitted analyses, numerical algorithms, additional empirical results, robustness analyses and data sources.

2 Related Literature

The paper relates to the recent heterogeneous bank literature and financial stability. By underlying the feature of incomplete markets in the deposit market, the paper connects with Corbae and D'Erasmo (2020, 2021) who study bank regulation of the 2010 Dodd-Frank Act in an imperfect competition model of big banks and small banks with banks' market power. They find that the banking industry becomes more concentrated given an increase in banks' capital requirement since it impedes entry of small banks. While maintaining a perfectly competitive banking sector, I contribute with a model of two assets as endogeneous states, liquid and illiquid assets, and adjustment costs in illiquid assets to study the redistribution mechanism of assets. In the model, loans are illiquid assets which expose banks to costs if they need funds before the maturity of loans (Diamond and Rajan 2001). My model generates an endogenous market structure due to

adjustment costs in illiquid assets and submodular regulatory constraints.

Relative to the previous literature, in my model adjustment costs in illiquid assets lead to an assignment problem whereby large banks match with small banks. I call this matching pattern negative assortative matching in the banking sector which changes the market structure endogenously raising its stability. Consistently with early theories on banks' illiquid assets (Diamond and Dybvig, 1983; Diamond and Rajan, 2001, 2005), banks finance illiquid assets with deposits. Yet, in my model deposits are idiosyncratic.

Holmstrom and Tirole (1998) study the demand of liquidity from the consumer viewpoint and the welfare and redistribution effects provided by the insurance role of intermediation. They give a role for the government in managing liquidity. Likewise, Farhi et al. (2009) focus on the role of financial intermediaries in allocating liquidity to consumers. They study unobservable trades and the efficient allocation in the presence of a social planner within a model that features limited risk sharing. In my model, banks cannot share the risk of idiosyncratic deposit withdrawal shocks and the fundamental redistribution mechanism between banks and across assets occurs because of the imperfect substitution of resources.

Similarly to Bianchi and Bigio (2022), illiquid assets bring a larger return than liquid assets and liquidity frictions lead banks to build a buffer of liquidity. However, differently from them in my model the distributions of liquid and illiquid assets remain endogeneous creating imperfect substitution amongst them. Therefore, I show the relevance of submodular liquidity constraints, while estimating the marginal propensity to lend. Research guided by the heterogeneity principle from an empirical viewpoint has consistently shown that the marginal propensity to lend is largely heterogeneous across banks and tends to be limited due to banks' high leverage constraints that induce a fall in loans, thereby impairing banks' profits (Agarwal et al., 2018).

Dempsey (2025) develops a heterogeneous bank model with two-sided heterogeneity to study the rises in bank capital requirements. He finds that, although tighter capital requirements have small effects on macroeconomic aggregates they are welfare improving and reduce costly bank failures. While focussing on the loan supply, I share with Dempsey and Faria-e-Castro (2025) the heterogeneous adjustment costs across banks, yet they conceive these costs through deep habits between borrowers and lenders thereby departing from my model. Likewise, Jamilov (2020) and Jamilov and Monacelli (2025) study banks market power in loans and deposits within a heterogeneous banks model. However, their idiosyncratic source of heterogeneity is in the return on capital to study business cycle fluctuations.

The role of financial stability is studied in Rios-Rull et al. (2020) who consider heterogeneity in banks' return rates and indicate that countercyclical capital requirements reduce bank failures and the dynamics of aggregate loans. Their model however is studied within a one asset model. Recent contributions that either consider only ex-ante bank heterogeneity or banks that aggregate to a representative bank with a degenerate distribution include Coimbra and Rey

(2023) and Begenau and Landvoigt (2021).[3]

The paper contributes to the rich literature of the bank lending channel of monetary policy (Bernanke and Blinder, 1988; Bernanke and Gertler, 1995; Kashyap and Stein, 1994, 1995; Jimenez et al., 2012; Curdia and Woodford, 2010, 2016; Farhi and Werning, 2016; He and Krishnamurthy, 2019). However, I focus on this channel by examining the redistribution mechanism between liquid and illiquid assets within the full distribution of banks.

Theoretically, I build on the seminal articles by Becker (1973) and Shapley and Shubik (1971) who formalise properties of the assortative matching and the assignment game with transferable utility where submodularity is necessary for the optimal matching pattern, whereas more recently Chiappori et al. (2017) provide a modern treatment of matching. I microfound an assignment matching game into the model and provide a novel theoretical demonstration where convex adjustment costs and submodular regulatory constraints in the banking sector lead to an endogenous market structure. I interpret adjustment costs in illiquid assets in my model as the costs of matching with submodularity in regulatory constraints, following Gowrisankaran (1999) who shows the importance of convexity in the merger costs.

Table A.1 in Appendix A summarises the contribution to the literature of this paper relative to the closest articles, points out the key differences, highlights the research gaps and the importance of addressing the research gaps in my paper.

3 Motivating Micro Banks Evidence

In this Section I highlight key micro features of the commercial banking sector for the US economy. Moreover, I document facts on the heterogeneity and concentration of banks' liquid and illiquid assets, the banks' size dependency of illiquid assets adjustment costs and the marginal propensity to lend. To this end, I also report statistics that I use to evaluate the quantitative performance of the model.

I collect quarterly data from the Consolidated Reports of Condition and Income submitted to the Federal Reserve by commercial banks from 1998 to 2016 provided by Wharton Research Data Services (2024). This enables me to explore a large panel of banks to identify *i)* the relationship of liquid and illiquid assets with banks' deposits, *ii)* the concentration in illiquid assets and banks' adjustment costs, *iii)* the redistribution mechanism of banks' assets with the marginal propensity to lend and *iv)* the cross-sectional variation in banks' differentials to additional liquidity.

To identify the portfolio redistribution of banks' assets I use an extended measure for banks' liquidity and illiquid assets reported in Appendix F. I define large banks as banks in the 10th

[3]The paper is related to the early literature on banking. Examples are Adrian and Shin (2010), Gertler and Kiyotaki (2010), Gertler and Karadi (2011), Gertler, Kiyotaki and Queralto (2012), Adrian and Shin (2014), Brunnermeier and Sannikov (2014), Fernández-Villaverde, Hurtado and Nuño (2023), Gertler, Kiyotaki and Prestipino (2020). These models however rationalise banks as a representative bank, thereby not featuring ex-post heterogeneous banks.

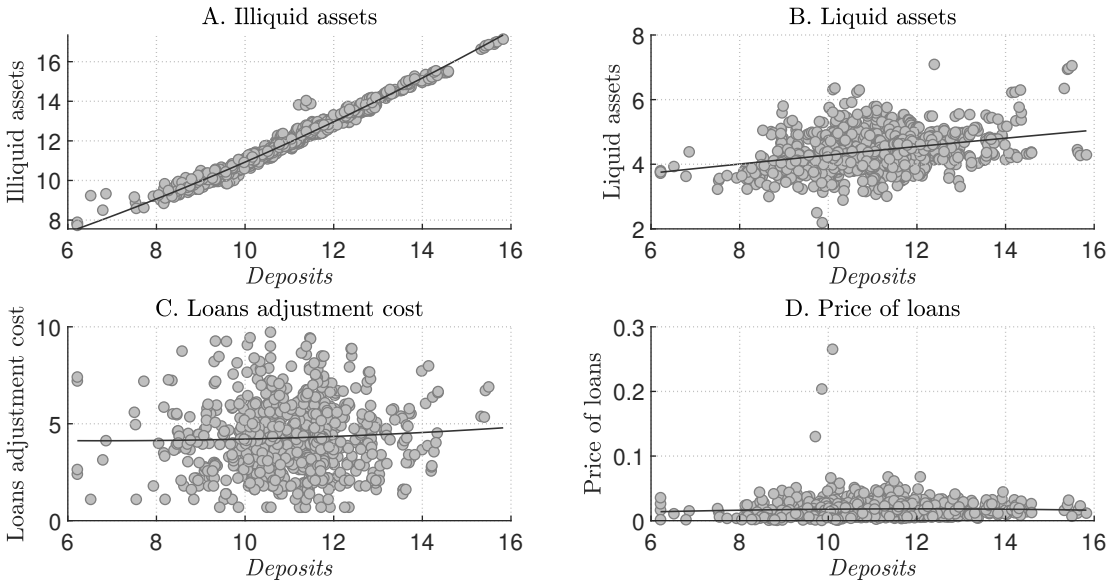


FIG. 1.—Banks' cross-sectional patterns in 1998.

NOTE. Scatter plots of cross-sectional patterns of banks' illiquid assets, liquid assets, loans adjustment costs and price of loans in 1998. Data source and details are provided in Appendix L.

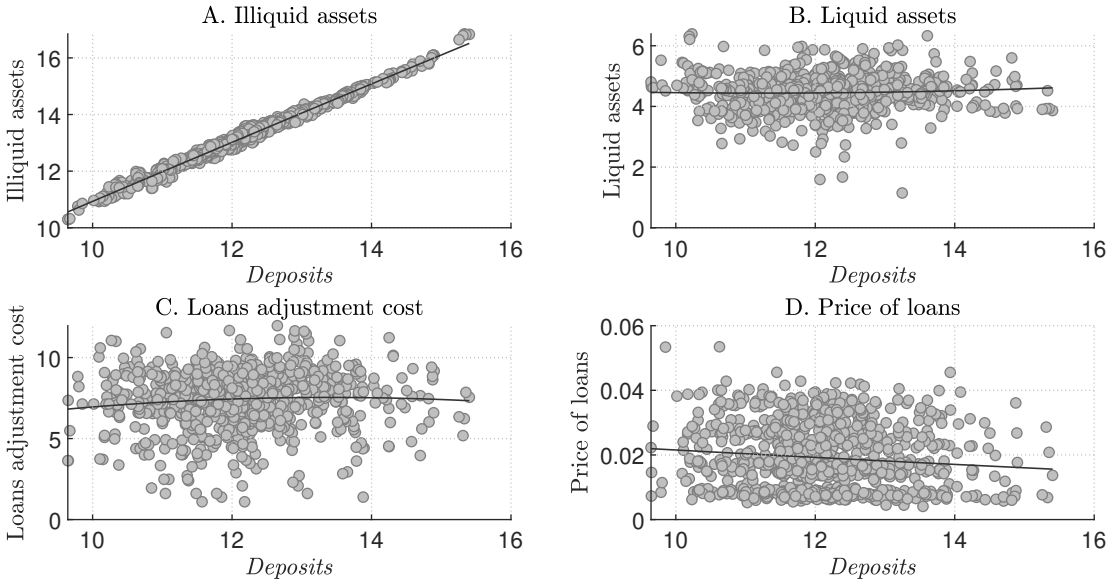


FIG. 2.—Banks' cross-sectional patterns in 2010.

NOTE. Scatter plots of cross-sectional patterns of banks' illiquid assets, liquid assets, loans adjustment costs and price of loans in 2010. Data source and details are provided in Appendix L.

percentile of the distribution of banks and small and medium banks in the 90th percentile of the distribution of banks. These evidence complement previous studies on the heterogeneity surrounding the banking sector and contribute with new facts about the size heterogeneity of banks in terms of deposits and the marginal propensity to lend in a context of abundant liquidity. In Section 5 of the paper I provide empirical estimates associated with them.

I begin by presenting stylised facts on the relation between deposits with banks' balance

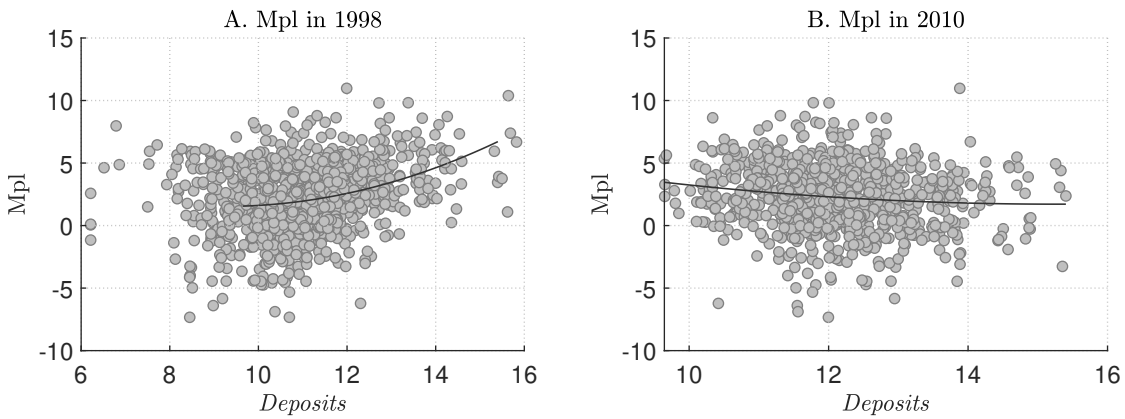


FIG. 3.—Banks' cross-sectional marginal propensity to lend in 1998 and 2010.

NOTE. Scatter plots of cross-sectional patterns of banks' marginal propensity to lend in 1998 and 2010 in panel A and B, respectively. Data source and details are provided in Appendix L.

sheet and income statement variables. Figures 1 and 2 present cross-sectional patterns between deposits and liquid assets, illiquid assets, loans adjustment costs and the price of loans in 1998 and 2010, respectively. Loans adjustment costs are defined as the difference between charge off loans and recovery loans, whereas the price of loans is the sum of interest income from loans to total illiquid assets similarly to Corbae and D' Erasmo (2021). In this regard, the price of loans is considered as a measure of profitability for banks. Both figures show the scatterplots of the log of these variables. The size of banks is measured with banks' deposits.

Figure 1 panel A illustrates that illiquid assets concentrate in the upper tail of the distribution and increase with deposits. Large banks hold more illiquid assets than small banks. By 2010 there was a steadily shift of this distribution upward as shown in panel A of Figure 2. Hence, the largest banks increased in size. Since large banks now hold more illiquid assets, the fixed point of the market structure moves upward with a new equilibrium composition of banks.

Loans adjustment costs have been increasing less over time, reflecting the fact that large banks by expanding in size can sustain lower costs through economies of scale. In facing slower balance sheets adjustments, the credit elasticity shrinks leading to a fall in the marginal propensity to lend. While liquid assets became more widely spread across the banking sector by 2010, the price of loans has been decreasing over time and the fall has been driven by small banks. With more liquidity towards small and medium banks, balance sheets become less risky and the market structure stable. The fall in the price of loans drives down the return to illiquid assets and simultaneously reduces the marginal propensity to lend. Overall these evidence point toward an increasingly costly loan production since only for large banks illiquid assets are profitable at the margin, and the equilibrium market structure is made by banks with high adjustment costs and lower return on loans.

Figure 3 illustrates the relation of the marginal propensity to lend with deposits in 1998 and 2010 in panel A and B, respectively. The marginal propensity to lend is computed as the

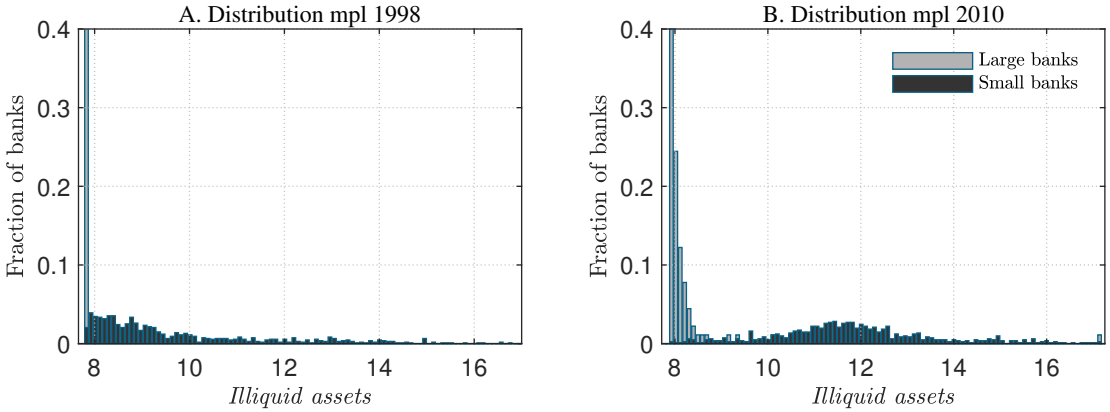


FIG. 4.—Distribution banks' marginal propensity to lend.

NOTE. Distribution of the banks' marginal propensity to lend in 1998 in panel A and in 2010 in panel B by bank size. The marginal propensity to lend is computed as the change in loans given a change in the return on loans. Data source is provided in Appendix L.

change in loans given a unit change in the return on loans. The return on loans is represented by the difference between the interest income from loans and loans adjustment costs. Formally, I define the marginal propensity to lend as follows:

$$\text{mpl}_{i,t} = \frac{(s_{i,t} - s_{i,t-1})}{(r_{s,i,t} - r_{s,i,t-1})}, \qquad \forall i, \forall t, \tag{1}$$

where $s_{i,t}$ identifies loans for bank i in period t, and $r_{s,i,t}$ is the return on loans in period t for each bank i. Equation (1) then specifies how many units of loans any one bank supplies given a unit increase in the return on loans.

Panel A documents an increasing relationship between deposits and the marginal propensity to lend, with large banks having a large propensity to issue a line of loan. The pattern reverts in 2010 where the marginal propensity to lend in panel B decreases due to the consequences of the financial crisis. By 2010 large banks tend to have less incentives to issue loans owning to the reliability of the borrower. Moreover, while more large banks appear by 2010 resulting from merging of these banks, the distribution of small banks shifts towards the middle of the illiquid assets distribution resembling a flat Normal distribution where most of the small banks are concentrated towards the middle value of the illiquid assets distribution. With a downward movement of the marginal propensity to lend by 2010, banks become less responsive to change in the return on loans. This is also the expected result from the cross-sectional patterns in Figures 1 and 2, where the less increasing adjustment costs make banks adjust loans less optimally. Moreover, the fall in the price of loans makes the optimal response to any increase in the return on loans to be small, the more equally distributed distribution of liquidity in the banking sector leads to a falling return on loans at the margin relative to holding liquidity, pushing the marginal propensity to lend further downward for all banks.

Figure 4 illustrates the histogram distribution of the marginal propensity to lend in 1998

and 2010, respectively. In 1998 more than 40 percent of large banks had a low marginal propensity to lend that was concentrated in one bin consisting of the 100 largest banks. By 2010 the distribution of the marginal propensity to lend of large banks became less concentrated, although banks with more illiquid assets maintained a low marginal propensity to lend. As more banks entered the upper tail of the distribution making the distribution thicker than the 1998 counterpart, the marginal propensity to lend for large banks became more heterogeneous leading to an endogenous shift in the market structure. Likewise, the marginal propensity to lend of small banks shifted towards banks with more illiquid assets as a consequence of the fact that a consistent amount of small banks exited or merged with other banks following the financial crisis. While illiquid assets are more limited, the lower tail of the distribution of the marginal propensity to lend becomes more equally spread. These distributional movements highlight that the marginal propensity to lend is decreasing in illiquid assets regardless of the type of banks.

In Appendix F Table F.1 provides summary statistics of the main variables of interest for banks that shows considerable heterogeneity between banks. Figures F.1 and F.2 show the distribution of illiquid assets and the price of loans. Figure F.3 shows the evolution of the average distribution of the loan-to-deposits ratio and the marginal propensity to lend from 1998 to 2011 by bank size. The figure indicates a market structure characterised by an upper tail dominated equilibrium skewed to the right and, a redistribution of assets between banks from illiquid assets to liquid assets, since post financial crisis over time both variables decreased.

To summarise this evidence underscores a considerable size heterogeneity in banks' deposits that guides concentration in banks' illiquid assets, and can be summarised as follows.

i) Large banks hold more illiquid assets than small banks.

ii) Over time large banks increased in size with a new equilibrium composition of banks.

iii) Liquid assets become more widely spread between small and medium banks, with less risky balance sheets and stable market structure.

iv) Balance sheet adjustments with slower loan adjustment costs, led to a more costly loan production where holding illiquid assets is profitable at the margin only for large banks.

v) The relationship between deposits and the marginal propensity to lend reverted to a downward path where banks are less responsive to changes in the return on loans.

vi) The distribution of large banks illiquid assets shifted rightward, while a fatter left tail distribution emerged for small banks.

vii) The distribution of the price of loans shifted to the right where the fraction of large banks halved, thereby charging a high loan price.

viii) The marginal propensity to lend is decreasing in illiquid assets. The distribution of the marginal propensity to lend for large banks became less concentrated and more heterogeneous over time, whereas the distribution of small banks shifted towards banks with more illiquid assets.

ix) Illiquid assets and liquidity in the top percentiles of banks are higher than their counter-

parts at the bottom percentiles of banks.

x) The return on loans in the top percentile exhibited a significant fall by 2010, becoming eight times larger than the return of banks in the bottom percentile.

xi) There is a positive relationship between the loan-to-deposits ratio and the marginal propensity to lend prior to the financial crisis, at which point this relationship becomes negative especially for large banks.

xii) The market structure becomes characterised by an equilibrium with a right skewed illiquid assets distribution dominated by large banks.

xiii) The fall in the loan-to-deposit ratio and the marginal propensity to lend for all banks support the hypothesis that banks redistributed their assets between them and from illiquid assets to liquid assets. The magnitude of the redistribution effect was driven by large banks.

The theoretical model developed below embeds these distinctive features of the banking sector and studies an environment where these facts provide the grounds for the endogenous market structure. The econometric analysis in Section 5 explores more cleanly the relationship between illiquid assets, liquid assets, deposits and the marginal propensity to lend.

4 Mathematical Model

This section presents an economy in which markets are incomplete and banks invest in illiquid assets at an endogenous price, save in liquid assets that pay a risk free interest rate, and are faced with idiosyncratic withdrawal risk on deposits.

The model builds on Gigante (2026) and contains four features of the banking sector that are necessary to generate heterogeneous lending choices, the required share of medium but liquidity constrained banks and a portfolio redistribution mechanism for assets. These features are: ex-post heterogeneity with idiosyncratic deposit withdrawals, a liquidity constraint that limits self-insurance, a leverage constraint on illiquid assets that constrains the banks' ability to lend, two types of assets: liquid and illiquid assets, and adjustment costs in illiquid assets that make banks track their past illiquid assets choices.

The aim of the theoretical model is to answer three questions. First, how does regulation change the structure of the market, and therefore, the distribution of banks through matching? Second, how does matching technology determine the composition between medium and large banks and what does that imply for the marginal propensity to lend? Third, how does deposit withdrawal risk affect the redistribution of banks' assets and the marginal propensity to lend?

4.1 Mathematical Environment of Heterogeneous Banks

Time is discrete, of infinite horizon and indexed by t. I assume a perfectly competitive banking sector.[4] At each date, banks invest in liquid assets and illiquid assets. Banks face individual

[4]This is in line with the observation that market for loans and deposits are the manifestation of a single primitive, the demand for liquid assets (Kashyap et al., 2002).

deposit shocks and cannot issue external equity. A bank facing a negative deposit shock, can use its saving in liquid assets and internal equity. The economy is populated by a continuum of bankers with unit mass indexed by the subscript $i \in [0, 1]$ and they are in charge of a bank.

Deposits. Banks are heterogeneous in their deposits holding. Large banks hold lower deposits than small banks. Therefore, banks of different sizes hold different value of deposits.[5] I assume that deposits circulate in an unpredictable way by letting them to follow a first order Markov process with transition probabilities defined over the probability matrix Ψ_t. Therefore, banks cannot exchange deposits contracts within each period. Consequently, deposits, $d_{i,t}$, follow an exogenous history dependent process with specific history given by d^t, where $d^t = \{d_0, ..., d_t\}$ is defined as a series of deposit shocks from time 0 to time t.

The process is the same for all banks and drawn from the finite set $D \subset \mathbb{R}_+$. The initial value of deposits, d_0, is nonnegative such that $d_0 \in D$. The Markov chain $\psi(d_{i,t+1}, d_{i,t})$ consists of realisations that are iid across banks. Each deposit value is drawn with probability $\Pr(d_{i,t+1} = d_j | d_{i,t} = d_i) \equiv \lambda_{ij} \geq 0$ and $\sum_{i=1}^{n_d} \lambda_{ij} = 1$ for each $i = 1, ..., n_d$. The probability distribution $\lambda_{i,j}$ represents the mass of banks in each deposit state at a particular date j. Therefore, I define the probability measure of deposits as $\mu^t(v_0, d^t)$.

Deposits pay the gross interest rate, r_d. The history dependent component represents the transitory component of deposits. The level of deposits drives the expansion of bank size. Consequently, large banks are subject to different deposit withdrawals because they enter the period with more favorable initial conditions of deposits.

Banks' preferences. I define bank i total net worth the sum of liquid and illiquid assets minus deposits, $n_{i,t}(d^{t-1}; v_0) = \ell_{i,t}(d^{t-1}; v_0) + s_{i,t}(d^{t-1}; v_0) - d_{i,t}$. The initial value of liquid and illiquid assets are ℓ_0 and s_0 respectively. They are drawn from their finite set $L \subset \mathbb{R}_+$ and $S \subset \mathbb{R}_+$, respectively. The space of all admissible banker individual states is defined with V such that $V = (D \times L \times S)$ with initial state given by v_0.

Bankers have preferences over a stochastic sequence of dividends income:

$$\sum_{t=0}^{\infty} \beta^t \sum_{d^t \in \mathcal{D}^t} \mu^t(v_0, d^t) u(c_{i,t}(d^t; v_0)), \tag{2}$$

where, $\beta \in (0, 1)$ is the time discount factor, $c_{i,t}$ denotes dividends income and $u(c_{i,t}(\cdot))$ represents the utility function over dividends. The utility function is such that $u(c_{i,t}) : \mathbb{R}_+ \longrightarrow \mathbb{R}$ and satisfies conditions for an interior solution by being strictly increasing and strictly concave. Because banks' dividends consumption needs to be nonnegative, this implies limited liability in the banking sector.

Banks' assets. Banks invest in an illiquid project with a loan contract, $s_{i,t}$, that earns the gross interest rate $r_{s,t}$. The interest rate on loans is endogenous and function of deposits,

[5] The assumption is motivated by different volatility of deposits between large and small banks found in the data (Corbae and D'Erasmo, 2021).

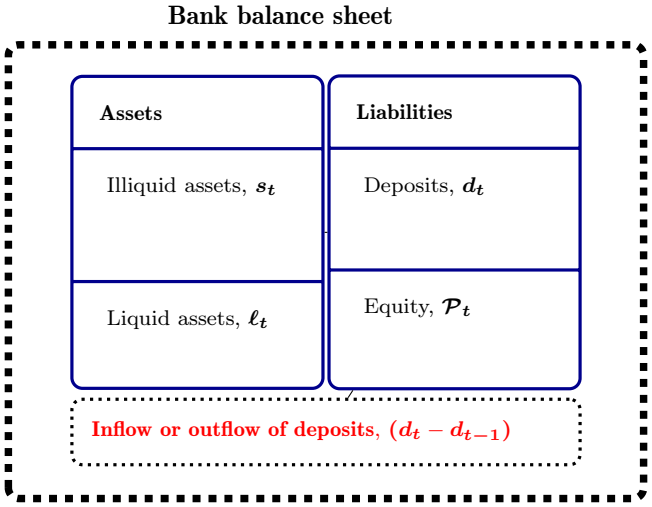


FIG. 5.—Illustration of the ex-ante bank balance sheet

$r_{s,t}(d_{i,t})$. They save in liquid assets, $\ell_{i,t}$, consisting of reserves and government securities that earn the gross interest rate on liquidity, r_ℓ. I assume that $r_\ell < r_d < r_{s,t}$ for each $t \geq 0$, because−in reality−loans are risky and they are long term maturity.

Illiquid assets are subject to a variable quadratic adjustment costs function $\eta(s_{i,t+1}, s_{i,t})$ that differs across banks of different size. The adjustment cost function is strictly increasing and convex such that $\eta'(s_{i,t+1}, s_{i,t}) > 0$ and $\eta''(s_{i,t+1}, s_{i,t}) \geq 0$. Therefore, the marginal value of one additional unit of illiquid assets is decreasing. The adjustment cost parameter is $\alpha > 0$, and it denotes the illiquidity cost. I assume an illiquidity cost since illiquid assets are long term maturity, therefore banks have to bear a cost in case they need to liquidate the asset before maturity. This is consistent with the transaction costs theory in illiquid assets of Diamond and Dybvig (1983) and Diamond and Rajan (2001). Paying the illiquidity cost means that banks terminate the project and it gives rise to a financial friction into the model.

The illustration of the bank balance sheet at the beginning of the period is given in Figure 5. Given the presence of limited liability, investment in liquid and illiquid assets must cover any loss incurred by banks. Moreover, I assume that banks do not issue equity but they accumulate it inasmuch as equity equals the beginning of period t profits, $\mathcal{P}_{i,t}(d^{t-1}; v_0)$. The balance sheet identity at the beginning of period t is given by: $\ell_{i,t} + s_{i,t} = d_{i,t} + \mathcal{P}_{i,t}$.

Liquidity frictions. The deposit withdrawal shock realises before any interest on liquid and illiquid assets is earned and interest on deposits is paid. Moreover, the deposit withdrawal shock realises after banks' decisions are made. Since the deposits withdrawal gives rise to an inflow/outflow of deposits, deposits update to $d_{i,t+1}$ and, banks can experience an inflow or outflow of deposits. Therefore, banks face an intraperiod liquidity requirement as follows:

$$\tilde{\ell}_{i,t+1} = \ell_{i,t}(d^{t-1}; v_0) + (d_{i,t+1} - d_{i,t}), \tag{3}$$

where, liquidity $\ell_{i,t}$ helps to smooth any idiosyncratic variation in the deposit withdrawal, $(d_{i,t+1} - d_{i,t})$. As a consequence, the intraperiod state contingent liquidity constraint is:

$$\ell_{i,t}(d^{t-1}; v_0) + (d_{i,t+1} - d_{i,t}) \geq \phi, \quad \forall d_{i,t+1} \in D, \quad t = 0, 1, ..., \tag{4}$$

where, ϕ is the exogenous liquidity requirement. The presence of the state and time dependent liquidity constraint implies that banks must have enough liquidity by the end of the period to cover any deposit outflow. Therefore, it strengthens the incentive for banks to accumulate liquidity to hedge against the deposit withdrawal shock. As a consequence, the inflow/outflow of deposits can be met with an increase or decrease of liquidity.

Leverage constraint. The project requires a loan contract to be enforced. Since deposits are idiosyncratic, deposit contracts are not enforceable until the shock has been realised. This feature of the model generates a leverage constraint that tights the bank ability to extend loans in the future. The leverage constraint is:

$$\frac{s_{i,t+1}(d^t; v_0)}{d_{i,t+1}} \leq \kappa, \tag{5}$$

where, κ limits the capacity of banks to invest in illiquid assets. Likewise, the higher the deposits the higher the illiquid assets banks can invest. Financing a project will become difficult for highly levered banks if there is the possibility that they will be liquidity constrained in the future. In equilibrium this outcome can occur, therefore saving in liquid assets will be the dominant strategy.

The leverage constraint introduces a moral hazard problem between borrowing banks and lending banks since it exposes banks' balance sheets to maturity mismatch.[6] As the deposit withdrawal shock occurs before the project matures, moral hazard may imply excessive risk taking since deposits is purely idiosyncratic. This is consistent with the theories of Diamond and Rajan (2012), Farhi and Tirole (2012) and Gertler et al. (2012). Moreover, the leverage constraint is imposed to be occasionally binding.[7]

The assumption on the leverage constraint is motivated by lending practices that serve to assess the liquidity of a bank. In particular, it reconciles the loan-to-deposits ratio. While a high value of the illiquid assets to deposits ratio ensures that banks make enough dividends, it also reflects a measure of low liquidity in their balance sheet to cover unexpected withdrawal of deposits.

[6] In this regard, the leverage constraint acts as a collateral constraint (Kiyotaki and Moore, 1997), with the difference that the leverage constraint in this paper does not depend on the market value of illiquid assets.

[7] Models that have studied occasionally binding constraints in a different framework are Mendoza (2010), Brunnermeier and Sannikov (2014), Bianchi (2016), Bocola (2016), He and Krishnamurthy (2019) and Justiniano et al. (2019).

Budget constraint. The beginning of period budget constraint is:

$$\begin{aligned} c_{i,t}(d^t;v_0)+\ell_{i,t+1}(d^t;v_0)+s_{i,t+1}(d^t;v_0)+\eta(s_{i,t+1},s_{i,t})-d_{i,t+1}=r_\ell\tilde{\ell}_{i,t+1} \\ +r_{s,t}(d_{i,t})s_{i,t}(d^{t-1};v_0)-r_d d_{i,t+1}, \quad \forall d_{i,t+1}\in D, \quad t=0,1,..., \end{aligned} \tag{6}$$

in which $\ell_{i,t+1}$ and $s_{i,t+1}$ are the values of liquid and illiquid assets chosen in period t with a sequence of realisations of idiosyncratic deposit uncertainty, $d_{i,t+1}$.

Note that the budget constraint in equation (6) contains the liquidity requirement $\tilde{\ell}_{i,t+1}$ with the beginning of period liquidity. Once banks make the liquidity choice to satisfy the liquidity requirement, then the end of period liquidity constraint banks need to satisfy becomes:

$$\tilde{\ell}^*_{t+1}=\ell_{i,t+1}(d^t;v_0)+(d_{i,t+1}-d_{i,t}), \tag{7}$$

which is the liquidity constraint, $\tilde{\ell}^*_{t+1}\geq\phi$, banks internalise in their individual programme.

Submodular liquidity constraint. The fundamental assumption that characterises the model is the presence of a submodular liquidity constraint. More precisely, this assumption makes liquid assets and deposits substitute each other in the balance sheet. At the margin, in a low deposit state the value of liquid assets is high, which makes banks to value liquidity more in a drying up of deposits and viceversa. Therefore, equation (7) is substituted with a submodular liquidity constraint. It becomes a state dependent liquidity constraint that depends on the current level of liquid assets and contains a penalty term for holding additional liquidity relative to the available short term debt:

$$\tilde{\ell}^*_{t+1}=\frac{(\ell_{i,t+1}(d^t;v_0)+(d_{i,t+1}-d_{i,t})-\delta\ell_{i,t}d_{i,t})}{d_{i,t}}, \tag{8}$$

where the term $(\delta\ell_{i,t}d_{i,t})$, is the cost of managing liquidity buffers against idiosyncratic withdrawal of deposits. Therefore, banks with high liquid assets find that low deposit states are more attractive because they are less valuable and optimally pair with them. When deposits rise, banks value liquid assets less in their balance sheet. Accordingly, the decreasing marginal value of liquidity leads small banks with high deposits to match with large banks with low deposits. This is the negative assortative matching that arises in the model. The parameter δ then measures the strength of submodularity in the negative assortative matching, making the constraint tighter for banks with high liquid assets that are large banks.

Moral hazard. The model features a moral hazard problem induced by the leverage constraint. Therefore, the contract with additional liquid assets may lead banks to take more risk and advance loans to less reliable borrowers, since deposits are idiosyncratic. In particular, when the leverage constraint $s_{i,t+1}-\kappa d_{i,t+1}\leq 0$ binds the shadow value of relaxing the constraint is positive. When the constraint holds with an inequality, the incentive for banks to take on more risk depends on their level of deposits. Banks with high deposits have incentives to increase

investment into illiquid assets, disinvest in liquid assets and hide the risk since they have a lower leverage limit. Banks with low deposits are more likely to be constrained and have incentives to invest in high return assets to relax the constraint. Essentially, leverage is capped and restricts the type of assets holding by shifting moral hazard risk but not the composition.

Interpretation of adjustment costs. There are two reasons for the presence of adjustment costs in illiquid assets, $\eta(s_{i,t+1}, s_{i,t})$. First, they are meant to capture the fatter tail in the distribution of banks' illiquid assets. By displacing the bank cost of illiquid assets when illiquid assets choice changes between banks and any two periods, the adjustment cost effectively influences the price of a loan, the marginal propensity to lend and therefore the quantity of illiquid assets to invest.

In addition, this feature of the model generates the endogeneous redistribution mechanism of assets according to banks' differentials. The mechanism arises because of changes in the price of illiquid assets and the lack of a market for liquid assets that instead originates with a precautionary liquidity motive. Indeed, increasing the amount of deposits the bank needs to keep in the form of liquidity, necessarily decreases the lending it is able to invest. Second, by affecting the liquidity motive adjustment costs are meant to give rise to a redistribution between banks and across assets. Moreover, the adjustment costs in illiquid assets in this model are necessary to generate the endogenous market structure where banks match in an assignment matching game between them that influences the technology under which they operate.

Exit and entry of banks. At the end of the period, banks exit. Exit is exogenous and banks face an exit shock. In particular, each period, a constant fraction of banks exits the economy and exiting banks are followed by a similar fraction of entrant banks. Each period a new mass of banks enters the economy. Exiting banks receive an outside option of $\mathcal{V}(\tilde{\ell}_{i,t}, s_{i,t}, d_{i,t}, d_{i,t-1}) = 0$ and must pay their networth with dividends. Banks exit with probability δ_e. In particular, this assumption implies that the effective discount factor banks use to discount their future value is $\beta(1-\delta_e)$. Effectively, banks cannot stay in operation indefinitely and cannot accumulate infinite liquid and illiquid assets.

The mass of banks entering is denoted with μ_e, and is distributed across banks through the probability distribution of deposits. Essentially, $\delta_e = \mu_e$. A distributional mass of $(1 - \mu_e)$ of banks survive to the next period. Entrant banks start the period with the steady state values of liquid and illiquid assets which indicates the entry point in the distribution. The steady state values of liquid and illiquid assets, and dividends, are assumed to be under the scenario in which there were no entry and exit, the liquidity constraint binds and the leverage constraint is slack.

Timing. The timeline of events is presented in Figure 6. Timing in a given period is specified as follows.

1. Bank i enters the period with its previous period of liquid assets, $\ell_{i,t}$, illiquid assets $s_{i,t}$, deposits $d_{i,t}$, and profits, $\mathcal{P}_{i,t}$.
2. Banks decide how many illiquid assets to invest, $s_{i,t+1}$, how much liquidity, $\ell_{i,t+1}$, to save,

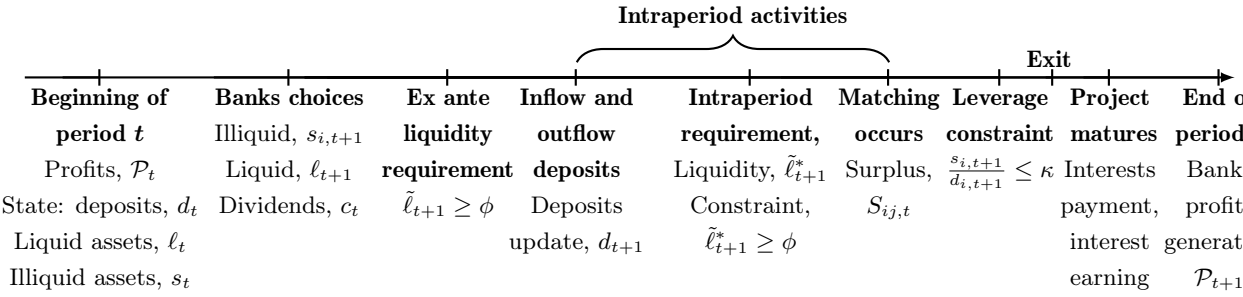


FIG. 6.— Timeline of events

and how many dividends to issue, $c_{i,t}$.

3. *Intraperiod activities*
   i. The deposit shock is realised, banks experience an inflow/outflow of deposits. Therefore, deposits update to $d_{i,t+1}$.
   ii. Banks must satisfy the liquidity requirement, $\tilde{\ell}^*_{i,t+1} \geq \phi$.
   iii. Banks matching occurs, large banks pair with small banks through an assignment matching game.
4. Banks must satisfy the leverage constraint, $\frac{s_{i,t+1}}{d_{i,t+1}} \leq \kappa$.
5. Banks face an exogenous exit shock. Upon exiting banks must pay their networth with dividends and are replaced by a fraction of entering banks.
6. The illiquid assets project matures. Interest rate on deposits are paid, interest rates on liquid and illiquid assets are earned and banks' profits, $\mathcal{P}_{i,t+1}$, are generated.

4.2 Equilibrium Analysis

The end of period budget constraint is derived by substituting the beginning of period liquidity constraint in equation (4) as follows:

$$\begin{aligned} c_{i,t}(d^t; v_0) + \ell_{i,t+1}(d^t; v_0) + s_{i,t+1}(d^t; v_0) + \eta(s_{i,t+1}, s_{i,t}) - d_{i,t+1} = \\ r_\ell(\ell_{i,t}(d^{t-1}; v_0) + (d_{i,t+1} - d_{i,t})) + r_{s,t}(d_{i,t}) s_{i,t}(d^{t-1}; v_0) - r_d d_{i,t+1}, \end{aligned} \tag{9}$$

that yields:

$$\begin{aligned} c_{i,t}(d^t; v_0) + \ell_{i,t+1}(d^t; v_0) + s_{i,t+1}(d^t; v_0) + \eta(s_{i,t+1}, s_{i,t}) - d_{i,t+1} = \\ r_\ell \ell_{i,t}(d^{t-1}; v_0) + r_{s,t}(d_{i,t}) s_{i,t}(d^{t-1}; v_0) - r_d d_{i,t} + (r_\ell - r_d)(d_{i,t+1} - d_{i,t}). \end{aligned} \tag{10}$$

The effect of the intraperiod liquidity requirement materialises in the inflow/outflow effect, $(r_\ell - r_d)(d_{i,t+1} - d_{i,t})$ that is internalised in the bank budget constraint. Because of the timing assumption, banks receiving the outflow of deposits pay the interest rate on deposits.

Let $\mathcal{V}(\tilde{\ell}_{i,t}, s_{i,t}, d_{i,t}, d_{i,t-1})$ be the value of the bank that begins with $\tilde{\ell}_{i,t}$ units of liquid assets, $s_{i,t}$ units of illiquid assets, with deposits states $d_{i,t-1}$ and $d_{i,t}$. The programme of individual banks is to maximise the Bellman equation subject to the budget constraint in equation (6),

the liquidity constraint (7) or equation (8) and the leverage constraint (5), taking as given the interest rate on liquidity, the interest rate on deposits and the initial state for liquid and illiquid assets:

$$\mathcal{V}(\tilde{\ell}_{i,t}, s_{i,t}, d_{i,t}, d_{i,t-1}) = \max_{c_{i,t}\geq 0, \underline{\ell}\leq \ell_{i,t+1}\leq \mathbb{R}_+, s_{i,t+1}\in\mathbb{R}_+} \left\{ u(c_{i,t}) + \beta(1-\delta_e) \sum_{d^t\in\mathcal{D}^t} \mu^t(v_0, d^t)\mathcal{V}(\tilde{\ell}_{i,t+1}, s_{i,t+1}, d_{i,t+1}, d_{i,t}) \right\}, \tag{11}$$

$$\begin{aligned} c_{i,t}(d^t; v_0) + \ell_{i,t+1}(d^t; v_0) + s_{i,t+1}(d^t; v_0) + \eta(s_{i,t+1}, s_{i,t}) - d_{i,t+1} = \\ r_\ell \ell_{i,t}(d^{t-1}; v_0) + r_{s,t}(d_{i,t}) s_{i,t}(d^{t-1}; v_0) - r_d d_{i,t} + (r_\ell - r_d)(d_{i,t+1} - d_{i,t}), \end{aligned} \tag{12}$$

$$\ell_{i,t+1}(d^t; v_0) + (d_{i,t+1} - d_{i,t}) \geq \phi, \tag{13}$$

$$\frac{s_{i,t+1}(d^t; v_0)}{d_{i,t+1}} \leq \kappa; \tag{14}$$

where, $\underline{\ell}$, sets a lower bound in the support of liquid assets decision rule.

The optimal bank plan yields optimalities for dividends, liquid and illiquid assets as follows:

$$-u'(c_{i,t}(d^{t-1}; v_0)) + \beta(1-\delta_e) \sum_{d^t\in\mathcal{D}^t} \mu^t(v_0, d^t)[r_\ell u'(c_{i,t+1}(d^t; v_0))] - \mu_{i,t}(d^{t-1}; v_0) = 0, \tag{15}$$

$$-u'(c_{i,t}(d^{t-1}; v_0)) + \beta(1-\delta_e) \sum_{d^t\in\mathcal{D}^t} \mu^t(v_0, d^t) \left\{ u'(c_{i,t+1}(d^t; v_0))[r_{s,t+1}(d_{i,t+1}) - \eta'(s_{i,t+1}, s_{i,t})] - \frac{\omega_{i,t}(d^{t-1}; v_0)}{d_{i,t+1}} \right\} = 0, \tag{16}$$

$$\mu_{i,t}(\ell_{i,t+1}(d^t; v_0) + (d_{i,t+1} - d_{i,t}) - \phi) = 0, \quad \mu_{i,t} \geq 0, \tag{17}$$

$$\omega_{i,t} \left(\frac{s_{i,t+1}(d^t; v_0)}{d_{i,t+1}} - \kappa \right) = 0, \quad \omega_{i,t} \geq 0; \tag{18}$$

where $\mu_{i,t}$ and $\omega_{i,t}$ identify the multipliers on the liquidity constraint and the leverage constraint. Equations (17) and (18) are the complementary slackness conditions associated with the liquidity and leverage constraint, respectively. Equations (15) and (16) are the Euler equations for liquid and illiquid assets. Both equations equate the bank marginal costs of renouncing to issue dividends in the present with the marginal benefit of one more unit of liquid assets and investment of illiquid assets into the future, respectively. I exclude the possibility of $r_{s,t} < r_\ell$ from the model so as to ensure that banks make sufficient profits and they can keep their position in the market. Equation (15) with the submodular liquidity constraint becomes:

$$-u'(c_{i,t}(d^{t-1}; v_0)) + \beta(1-\delta_e) \sum_{d^t\in\mathcal{D}^t} \mu^t(v_0, d^t)[r_\ell u'(c_{i,t+1}(d^t; v_0))] - \mu_{i,t}(d^{t-1}; v_0)\frac{1}{d_{i,t}} = 0, \tag{19}$$

where equation(19) includes the effect of short term debt on the complementary slackness condition of liquid assets.

The redistribution of assets arises for three reasons. One, the presence of adjustment costs implies that banks can adjust illiquid assets gradually. As it is costly for banks to invest in illiquid

assets if their associated return is not sufficiently high, banks prefer to rebalance their positions towards liquid assets. The adjustment costs give the price of illiquid assets the possibility to punish fast changes in illiquid assets or deviations from a fixed amount of illiquid assets. Both adjustment costs in illiquid assets and the liquidity constraint jointly define the technology of banks. Two, by setting a minimum amount of liquid assets, the liquidity constraint gives further incentives to banks to accumulate liquid assets. Three, a binding leverage constraint based on a low value of illiquid assets acts as a second order effect on the redistribution of assets towards liquidity. Furthermore, large and small banks experience different types of deposits withdrawals.

Incomplete markets for deposit contracts imply that the established result in these economies is $\beta(1-\delta_e)r_\ell < 1$, to ensure a precautionary liquidity motive by banks and a well definite distribution of liquid assets in equilibrium. This assumption guarantees an upper bound on the accumulation of liquid assets. The mass of entrants is then given by, $\delta_e \mu_e \bar{\psi}(d_{t+1}, d_t)$, and exiting banks obtain zero value function.

The distribution of banks evolves according to $\mu_{t+1}(\ell_{t+1}, s_{t+1}, d_{t+1}, d_t)$ such that:

$$\mu_{t+1}(\ell_{t+1}, s_{t+1}, d_{t+1}, d_t) = \int_{L\times S\times D} (1-\delta_e)\psi(d_{t+1}, d_t) d\mu_t(\ell_t, s_t, d_t, d_{t-1}) + \delta_e \mu_e \bar{\psi}(d_{t+1}, d_t), \tag{20}$$

where I omitted subscript i to simplify notation. In equilibrium the market for liquid assets and illiquid assets clear according to the following equations:

$$\int_{\ell:\ell_{i,t+1}\in L} \sum_{d_{t+1}\in D} \ell_{i,t+1} d\mu_{t+1}(\ell_{i,t+1}, s_{i,t+1}, d_{i,t+1}, d_{i,t}) = L, \quad t \geq 1, \tag{21}$$

$$\int_{s:s_{i,t+1}\in S} \sum_{d_{t+1}\in D} s_{i,t+1} d\mu_{t+1}(\ell_{i,t+1}, s_{i,t+1}, d_{i,t+1}, d_{i,t}) = \mathcal{S}, \quad t \geq 1, \tag{22}$$

where L and $\mathcal{S}$ are the aggregate level of liquid assets and illiquid assets, respectively. The market for liquid assets and illiquid assets close in fixed supply.

Definition 1 A *recursive competitive equilibrium* for this economy is given by banks' value function $\mathcal{V}(\ell, s, d, d_{-1}; \mu) : L\times S\times\Gamma \to \mathbb{R}$, banks' decision rules $c(\ell, s, d, d_{-1}; \mu) : L\times S\times\Gamma \to \mathbb{R}_+$, $\ell'(\ell, s, d, d_{-1}; \mu) : L\times S\times\Gamma \to \mathbb{R}_+$, $s'(\ell, s, d, d_{-1}; \mu) : L\times S\times\Gamma \to \mathbb{R}$, price of illiquid assets $q(\ell, s, d, d_{-1}; \mu) : \Gamma \to \mathbb{R}_+$, probability measure Γ of the sets $L\times S\times D$, a cross-sectional distribution $\mu(\ell, s, d, d_{-1})$, strictly positive interest rates $r_{s,t}$, r_ℓ, r_d such that:

1. Given the interest rates and the illiquid assets price function, the banks' value function satisfies the Bellman equation in equation (11), and the associated decision rules solve the optimisation programme specified by equations (11), (12), (13) and (14).
2. The sequence of cross-sectional distributions satisfies its law of motion and transition probability $\psi(d', d)$, the mass of entry μ_e, the probability of exiting δ_e, and is consistent with individual decision rules as specified in equation (20).
3. The liquid assets and illiquid assets markets clear according to equations (21) and (22),

respectively.

4. Exit of banks is satisfied with an injection of entrant banks.

4.3 Theoretical Properties of the Model

This Section presents the main theoretical results of the paper. First, I show that the economy admits a constrained steady state close form representation under some assumptions. Second, I demonstrate that the marginal propensity to lend can be characterised as a decomposition of a redistribution effect, price effect and shadow price effect. Lastly, I demonstrate the assignment matching game that leads to the endogenous federal fund market. The analysis underscores the costs and benefits of liquidity. In Appendix C I characterise the types of heterogeneous banks that arise in this model and I show the redistribution mechanism within assets and across banks. I show that the model can give rise to the share of two types constrained medium banks. These are constrained liquidity medium banks and constrained leverage medium banks.

4.3.1 Equilibria Characterization

In this Section, I analyse two variants of the model that are meant to study the equilibrium properties of the model. First, I consider a case in which the cost of holding illiquid assets is convex. Second, I inspect a case in which adjustment costs are partially convex. More precisely, they are infinitely illiquid because the convex adjustment cost function is set as deviation from the long run level of illiquid assets. I show that under this characterisation the model can nest the economy where banks hold liquid assets only.

Convex adjustment costs. Let assume that a bank can adjust its portfolio some time in a year such that trades take place with delay and that new investment is costly. Banks' optimal condition on illiquid assets is represented by equation (16) under a binding and non binding leverage constraint. In particular, when the leverage constraint holds with strict equality, the shadow value of illiquid assets is strictly positive, $\omega_{i,t}(d^{t-1}; v_0) > 0$. This reduces the marginal benefit of supplying one unit of loans in the future which at the optimum must equal one unit increase in dividends in terms of current marginal utility. Conversely, when the leverage constraint holds with an inequality and it is loose then the shadow value of one unit increment of illiquid assets is null, $\omega_{i,t}(d^{t-1}; v_0) = 0$.

Under this characterisation of adjustment costs and under the assumption that the leverage constraint is loose, the liquidity constraint binds and there is no entry and exit it can be shown that a constrained steady state for this economy exists. The existence of the steady state is guaranteed under constant deposits and by the existence of a stationary distribution that satisfies Blackwell sufficient conditions for a contraction. Thus, the decision rules for assets and dividends are bounded on the feasible set, and the Bellman operator is a contraction mapping that satisfies continuity and compactness, with condition to be enforced $\beta(1 - \delta_e)r_\ell < 1$. The constrained steady state is then along the stationary distribution. Proposition 1 provides the

formal argument.

Proposition 1 *Let define with c^*, ℓ^* and s^*, the steady state values of dividends, liquid assets and illiquid assets. A constrained steady state for this economy with convex adjustment costs exists and all allocations c^*, ℓ^* and s^*, have a closed form representation.*

Proof. In the Appendix B.1. ■

The proof proceeds with the assumption that the functional form for the adjustment costs is $\eta(s_{i,t+1}, s_{i,t}) = \frac{\alpha}{2}(s_{i,t+1} - s_{i,t})^2$, however any functional form of this type suffices.

Partial adjustment costs. I now let illiquid assets to be infinitely illiquid by modifying the adjustment cost rule as, $\eta(s_{i,t+1}, s_{i,t}) = \frac{\alpha}{2}(s_{i,t+1} - \overline{s})^2$. The convexity assumption is maintained, but I let the adjustment cost of illiquid assets to be with respect to a fixed level of illiquid assets, $\overline{s}$, that represents a stock of loans. Therefore, this adjustment cost form allows banks to penalise deviation from a fixed amount.

The interpretation for $\overline{s}$, being the steady state value of illiquid assets is that in a set of non stochastic equilibria and a set of stochastic equilibria, $\overline{s}$ can be conceived as the bifurcation point in the set of stochastic equilibria that intersects with the set of stochastic equilibria.[8] Therefore, banks adjust relative to a quasi-fixed stock of loans. It can be shown that when $\alpha \Longrightarrow \infty$, then $s_{i,t+1}(d^t; v_0) \Longrightarrow \overline{s}$ *a.s.*, hence $\Delta^+ s_t = 0$. The optimal solution is a fixed value of illiquid assets, that captures fixed capital in banks' balance sheet.

Proposition 2 *When adjustment costs in illiquid assets take the functional form $\frac{\alpha}{2}(s_{i,t+1} - \overline{s})^2$, the existence of the optimal illiquid assets solution, $s^*_{i,t+1}(d^t; v_0)$, is sensitive to the adjustment costs parameter, α. There is no equilibrium in which $\alpha = 0$ and illiquid assets choice becomes unbounded.*

Proof. In the Appendix B.2. ■

The intuition is that by keeping the adjustment cost parameter close to infinity it returns the economy with liquid assets as the only single banks' asset choice. As illiquid assets lose their relevance because they become expensive, banks substitute away to the investment in liquid assets as their sole motive to invest. Moreover, given that for any $\alpha > 0$ banks incur larger costs in investing in illiquid assets above the previous level of illiquid assets then α determines the degree of illiquidity of banks' illiquid assets. The optimal choice $\overline{s}$ gives then a lower bound for the illiquid assets set.

By contrast, when $\alpha \Longrightarrow 0$, it removes the cost of holding illiquid assets and being the adjustment cost relative to its long run value, there does not exist a solution for $s_{i,t+1}(d^t; v_0)$ and the bank plan does not admit an equilibrium. The condition holds since the effect of the adjustment cost parameter α is very small that dominates the incentive for banks in holding

[8] Numerical proof of the argument is demonstrated in Judd and Guu (2001).

liquid assets and investment in illiquid assets becomes unbounded. Therefore, these insights lead me to Proposition 3.

Proposition 3 *There is no equilibrium in which* $s_{i,t+1}(d^t; v_0) = 0$ *and the illiquid assets set needs to be bounded from below and from above and it needs not to be empty,* $S \nsubseteq \emptyset$.

Proof. In the Appendix B.3. ■

4.3.2 The Marginal Propensity to Lend

One of the aims of this paper is to provide a model for the marginal propensity to lend at the individual bank level in the space of deposits. While leading empirical articles have provided supporting evidence on the heterogeneous marginal propensity to lend, the theoretical models have been founding the studies on an aggregate quantity rather than a heterogeneous marginal propensity to lend across banks.

To make formal predictions on the marginal propensity to lend, a functional form for the adjustment costs function needs to be specified. I consider the convex adjustment costs function, $\frac{\alpha}{2}(s_{i,t+1} - s_{i,t})^2$, analysed in Section 4.3.1. Therefore, total investment in illiquid assets can be specified as $S_{i,t} = s_{i,t+1}(d^t; v_0) + \frac{\alpha}{2}(s_{i,t+1} - s_{i,t})^2$. This implies that the price of illiquid assets can be characterised as $q_{i,t}(s_{i,t+1}, s_{i,t}) = \frac{\partial S_{i,t}}{\partial s_{i,t+1}}$, hence $q_{i,t}(s_{i,t+1}, s_{i,t}) = 1 + \alpha(s_{i,t+1} - s_{i,t})$. I have conceived the price of illiquid assets heterogeneous across banks as it captures different prices of loans charged by banks.

Proposition 4 *The marginal propensity to lend is endogenous and heterogeneous across banks, varying by banks' size and takes the endogenous form* $\frac{\partial s_{i,t+1}}{\partial r_{s,t+1}}$*, from*

$$\frac{\partial q_{i,t+1}}{\partial r_{s,t+1}} = \beta(1-\delta_e) \sum_{d^t \in \mathcal{D}^t} \mu^t(v_0, d^t) \left[\underbrace{\frac{u''(c_{i,t+1})}{u''(c_{i,t})} \frac{\partial s_{i,t+1}}{\partial r_{s,t+1}}}_{\text{redistribution effect}} \left(1 + \underbrace{\frac{\partial q_{i,t+1}}{\partial q_{i,t}} \frac{\partial q_{i,t}}{\partial s_{i,t+1}}}_{\text{price effect}} \underbrace{\frac{\partial s_{i,t+1}}{\partial r_{s,t+1}}}_{\text{mpl}} \right) - \underbrace{u''(c_{i,t}) \frac{\partial \left(\frac{\omega_{i,t}}{d_{i,t+1}} \right)}{\partial s_{i,t+1}} \frac{\partial s_{i,t+1}}{\partial r_{s,t+1}}}_{\text{shadow price effect}} \right], \tag{23}$$

The marginal propensity to lend is decreasing in deposits for small banks and a V-shaped function of deposits for large banks.

Proof. In Appendix B.4. ■

4.3.3 Endogenous Market Structure and Assortative Matching

In this Section I study the conditions under which the federal funds market arises endogenously into the model, I characterise the assignment matching game and show the assortative matching. In particular, the Section establishes the conditions under which the core of the assignment matching game exists and is a feasible implementation.

I introduce the assignment matching game of Shapley and Shubik (1971) and Becker (1973) into an otherwise dynamic equilibrium model with incomplete markets and in an economy under

uncertainty, where the matching process is endogenised as an equilibrium outcome. The main problem in Shapley and Shubik (1971) is to determine which buyer is matched to which seller to exchange an indivisible object. In my theory I start with the assumption of large banks being the buyers and small banks the sellers of liquidity. I then relax this assumption and show that the optimal assignment matching game still occurs between large and small banks. I call this result duality of the optimal matching.

Matching Environment. The market contains many heterogeneous banks of each type. There are $m \in M$ sellers with i generic seller and $p \in P$ buyers with j generic buyer, where M is the set of all sellers and P the set of all buyers. For tractability, I consider a bilateral market where each bank exchanges liquidity with one other bank. Therefore, this is a two sided market game between banks that generalises to the rest of banks. While liquidity is a perfectly divisible object in banks' budget constraints, it is an indivisible object in value. This applies directly from two properties of the model. First, markets are incomplete, thus liquidity is not perfectly shareable across banks and liquidity risk cannot be insured by itself between banks. Second, the submodularity in the liquidity constraint makes the joint surplus of the match non additive. Moreover, each bank demands one unit of liquidity and supplies a symmetric unit of liquidity. In this sense, markets are inflexible.

Properties of the Game. The game contains three properties on the characteristic function, monogamy and complementarity, in a similar spirit of Shapley and Shubik (1971).

Let assume that at each date t the states of small banks (sellers) and large banks (buyers) can be characterised as follows: $x_{i,t} = (\ell_{i,t}, s_{i,t}, d_{i,t}, d_{i,t-1})$ and $x_{j,t} = (\ell_{j,t}, s_{j,t}, d_{j,t}, d_{j,t-1})$, respectively. The surplus of the match can then be defined as $W(\ell_{i,t}, s_{i,t}, d_{i,t}, d_{i,t-1}; \ell_{j,t}, s_{j,t}, d_{j,t}, d_{j,t-1})$. Hence, within a period banks solve a static matching game, which repeats itself in each period.

Let N be the total number of bank players such that $N = M \cup P$. Let define the dynamic value functions of small banks in the event that there is no match at time t as:

$$v_{\ell^*, s_{min}, \overline{d}}(\tilde{\ell}_{i,t}, s_{i,t}, d_{i,t}, d_{i,t-1}) = \max_{c_{i,t} \geq 0, \ell_{i,t+1}, s_{i,t+1}} \left\{ u(c_{i,t}) + \beta(1-\delta_e) \sum_{d^t \in \mathcal{D}^t} \mu^t(v_0, d^t) v(\tilde{\ell}_{i,t+1}, s_{i,t+1}, d_{i,t+1}, d_{i,t}) \right\}, \tag{24}$$

and of large banks as:

$$v_{\overline{\ell}, s_{max}, \underline{d}}(\tilde{\ell}_{j,t}, s_{j,t}, d_{j,t}, d_{j,t-1}) = \max_{c_{j,t} \geq 0, \ell_{j,t+1}, s_{j,t+1}} \left\{ u(c_{j,t}) + \beta(1-\delta_e) \sum_{d^t \in \mathcal{D}^t} \mu^t(v_0, d^t) v(\tilde{\ell}_{j,t+1}, s_{j,t+1}, d_{j,t+1}, d_{j,t}) \right\}, \tag{25}$$

where equation (24) and (25) are subject to equation (12), (8), (14), and where $L \subseteq [\ell^*, \overline{\ell}]$, $S \subseteq [s_{min}, s_{max}]$, and $D \subseteq [\underline{d}, \overline{d}]$. Because a matching occurs once banks have made their optimal choices of dividends, liquid and illiquid assets and the deposit shock has realised, the types of the matching game are $\sigma_{i,t} = (\ell_{i,t+1}, s_{i,t+1}, d_{i,t+1}, d_{i,t})$ and $\sigma_{j,t} = (\ell_{j,t+1}, s_{j,t+1}, d_{j,t+1}, d_{j,t})$ defined over the set $V = (D \times L \times S)$ and containing all relevant constraints.

Given $\sigma_{i,t}$ and $\sigma_{j,t}$, assume that seller i and buyer j want to form an interbank market contract, then the joint continuation value under the optimal contract can be written as $W_t(\sigma_{i,t}, \sigma_{j,t})$. Then the transferable surplus at time t can be characterised as follows:

$$S_{ij,t} = W_t(\sigma_{i,t}, \sigma_{j,t}) - (v_{\ell^*, s_{min}, \overline{d}}(\ell_{i,t+1}, s_{i,t+1}, d_{i,t+1}, d_{i,t}) + v_{\overline{\ell}, s_{max}, \underline{d}}(\ell_{j,t+1}, s_{j,t+1}, d_{j,t+1}, d_{j,t})). \tag{26}$$

Definition 2 A coalition is any subset $A \subseteq N$ such that any seller $i \in A \cap M$ matches with any buyer $j \in A \cap P$. The coalition feasible matchings are matrices of the type $x_{ij,t} \in \{0, 1\}$.

Having defined a coalition, the property monogamy of the game requires $\sum_{i \in A \cap M} x_{ij,t} \leq 1$ and $\sum_{i \in A \cap P} x_{ij,t} \leq 1$ inside A, where $x_{ij,t}$ are the fractions of states that a bank obtains when a match is formed. Therefore, one bank has to be matched with one other bank. I rule out cases of polygamy matches.

The characteristic function of the game can now be characterised as:

$$v_t(N) = \max_{x_{ij,t}} \sum_{ij,t} S_{ij,t} x_{ij,t}, \tag{27}$$

where equation (27) holds over all possible banks' coalitions $N = M \cup P$ and thus, it determines the maximum payoff value and Pareto set of the game. It holds that $v_t(N) = 0$ if $|N| = 0, 1$, and $v_t(N) = 0$ if $N \subseteq M$ or $N \subseteq P$, effectively banks cannot extract value alone but they must be paired with one other bank. The maximum of $v_t(N)$ is attained when $x_{ij,t} = 0$ or $x_{ij,t} = 1$. The assignment matching game is effectively a pair $(N, v_t(N))$ at time t which will be defined as transferable utility game.

The marginal value of a player to the coalition then depends on the amount of deposits and not the total number of banks of each type. In essence, this means that one would expect a player's valuation to the coalition to be small when deposits are abundant and large when deposits are scarce. Therefore, banks with different level of deposits are complements, whereas banks with the same level of deposits are substitutes. The following definition asserts the definition of the core following Shapley and Shubik (1971) and Becker (1973).

Definition 3 The core is the set of allocations that no coalition $A \subseteq N$ can improve upon. An allocation belongs to the core if it cannot be blocked by any feasible pair of players, and no coalition outside the core can make any bank better off without making another worse off.

Note that matching is then the equilibrium allocations of players that maximises total surplus in the presence of a submodular liquidity constraint. The function of the core is to divide the surplus among a matched banks pair. By the property of the core, I have assumed that banks match if and only if they are both better off and hence, their utility increases.

The Assignment Game. Let the shares of the surplus be defined as b_i and e_j indexed as $b_1, ..., b_m, e_1, ..., e_p$. Let the payoff vector be $(b, e) = (b_1, ..., b_m, e_1, ..., e_p)$. Then, the vector (b, e) is a core allocation if for every $A \subseteq N$, $\sum_{i \in A \cap M} b_i + \sum_{j \in A \cap P} e_j \geq v_t(N)$ that is the no blocking

condition holds for any coalition, and feasibility requires:

$$\sum_{i\in M} b_i + \sum_{j\in P} e_j \geq v_t(N) = v_t(N) = \max_{x_{ij,t}} \sum_{ij,t} S_{ij,t} x_{ij,t}, \tag{28}$$

thus, banks cannot send more liquidity than what they hold.

Proposition 5 *Banks have a marginal value of liquidity that is strictly decreasing in deposits d_t, and is independent of their size.*

Proof. In the Appendix B.5. ■

Proposition 5 states that the marginal value of liquidity is greater the lower the deposits. This property is necessary to the matching game and in practice will enter the surplus of the match and the inequality for submodularity.

Proposition 6 *Any two banks $i \in M$, $j \in P$ match until the marginal values of holding liquidity are equalised through trades $v_i^{\ell}(d_{i,t}) = v_j^{\ell}(d_{j,t})$ and $W_t(\sigma_{i,t}, \sigma_{j,t}) \geq (v_{\ell^*, s_{min}, \overline{d}}(\sigma_{i,t}) + v_{\overline{\ell}, s_{max}, \underline{d}}(\sigma_{i,t}))$.*

Proof. In the Appendix B.6. ■

Proposition 6 shows that matches in the market of liquidity occur by equalising the marginal value of liquidity between banks such that no other coalition improvements can occur. Effectively, banks of different type obtain gains from matching. Therefore, $W_t(\sigma_{i,t}, \sigma_{j,t}) \geq (v_{\ell^*, s_{min}, \overline{d}}(\sigma_{i,t}) + v_{\overline{\ell}, s_{max}, \underline{d}}(\sigma_{i,t}))$.

Assortative Matching. I now consider the optimal assortative matching given the primitives of the model, by considering the banks' surplus function as a mapping from types to value and considering the exogenous trait of banks in deposits. Consider the joint surplus of the match as function of deposits:

$$S_{ij,t}(d_{i,t}, d_{j,t}) = W_t(\sigma_{i,t}, \sigma_{j,t}) - (v_{\ell^*, s_{min}, \overline{d}}(\ell_{i,t+1}, s_{i,t+1}, d_{i,t+1}, d_{i,t}) + v_{\overline{\ell}, s_{max}, \underline{d}}(\ell_{j,t+1}, s_{j,t+1}, d_{j,t+1}, d_{j,t})). \tag{29}$$

Proposition 7 *For any banks' pair with $i \in M$ and $j \in P$ the surplus of the match is submodular in deposits such that $\frac{\partial^2 S_{ij,t}(d_{i,t}, d_{j,t})}{\partial d_{i,t} \partial d_{j,t}} \leq 0$.*

Proof. In the Appendix B.7. ■

Proposition 7 shows that the decreasing marginal value of liquidity between banks lead banks to obtain gains from sharing liquidity when maximising the overall surplus of the match.

Having defined a matching coalition as $x_{ij,t}$, then, the following Proposition holds.

Proposition 8 *The optimal matching between banks, $x_{ij,t}$, that maximises banks' matching payoff, $v_t(N)$, is negative assortative. The gains from banks' matching are such that $(S_{ij'}(d_i, d_{j'}) + S_{i'j}(d_{i'}, d_j)) \geq (S_{ij}(d_i, d_j) + S_{i'j'}(d_{i'}, d_{j'}))$.*

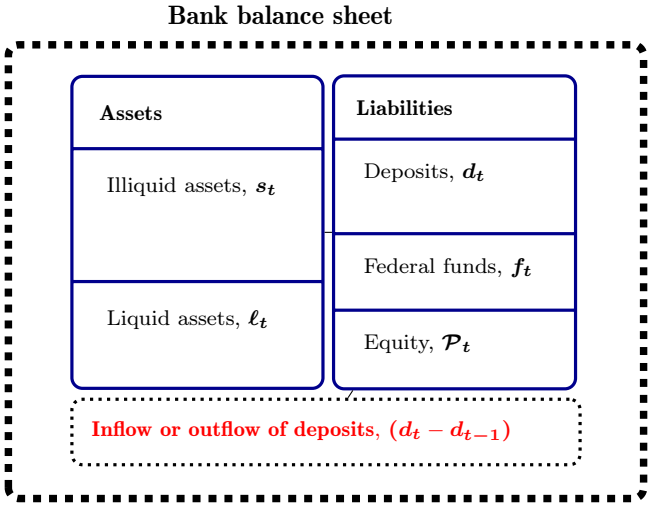


FIG. 7.—Ex-post banks' balance sheet

Proof. In the Appendix B.8. ■

Proposition 8 shows the existence of a banks' matching that maximises total surplus under negative assortative matching. More precisely, by matching banks are exchanging liquidity leading to the endogenous formation of the interbank market. Because negative assortative matching between banks in the federal funds market is the optimal matching, then the more complementary in deposits banks are, the larger are the gains from matching, and the greater is the surplus.

Furthermore, the intergroup allocation of banks' matching is unique because of the submodularity of the liquidity constraint and the cross pairs matching between banks. Effectively, an assortative matching result can be established in the liquidity market. The payoffs from matching can be conceived as transferable utility. Therefore, they are indeterminate similarly to Becker (1973). The ex-post bank balance sheet is given in Figure 7.

Optimal Matching Duality. I now relax the assumptions that large banks are the buyers of liquid assets and small banks the sellers of liquid assets. I show that, while relaxing this assumption, the optimal assignment matching game is always negative assortative.

Proposition 9 *The programme of individuals banks is a matching market where the transfer of liquid assets is* $(d_{i,t+1} - d_{i,t}) > 0$. *The optimal assignment matching game is endogenous and negative assortative such that* $(S_{ij'}(d_i, d_{j'}) + S_{i'j}(d_{i'}, d_j)) \geq (S_{ij}(d_i, d_j) + S_{i'j'}(d_{i'}, d_{j'}))$.

Proof. In the Appendix B.9. ■

Proposition 9 shows that the optimal assignment matching game is endogenous to the model and can be delivered without the imposition of the banks' type as buyers and sellers of liquidity. More precisely, by relaxing this assumption, the market still delivers small banks the sellers of liquidity and large banks the buyers of liquidity with the optimal matching being negative

assortative. A result that I call the optimal matching duality. The proposition then establishes that the optimal matching is such that the largest bank matches with the smallest bank, the second largest bank matches with the second smallest bank and so on, for all banks.

Once matching has occurred and large banks had the possibility to match with a small bank, the marginal propensity to lend is a more convoluted derivative of the total surplus with respect to the interest rate on loans taking into account the price of illiquid assets, subject to the liquidity and leverage constraints. The resulting matching modifies the marginal incentive towards liquidity and the marginal propensity to lend.

These results show that the design of thoughtful liquidity regulations that takes into account features of the liquidity market can lead banks of different size to match one another. This becomes particularly relevant in period of crises when small banks tend to transfer liquidity to large banks or their bank holding company.

Note that matching is an assignment matching game, search and matching frictions would change the properties of the model in two ways. First, in a direct search environment, banks would choose the counterpart they want to match with, and second, a match would be valuable as long as a loan creating condition holds which will pin down banks' assets. I leave this avenue for future research.

5 Empirical Analysis and Testing the Theory

In this Section I estimate the marginal propensity to lend across heterogeneous banks. First, I account for the fact that the marginal propensity to lend is truncated at zero due to negative values. Second, I consider that there is a cluster of observation at zero in the marginal propensity to lend. These observations motivate my empirical analysis to implement a Tobit specification to estimate the marginal propensity to lend at individual bank level. Ultimately, I estimate a quantile regression model to study the determinants of the marginal propensity to lend and trends in loans' inequality across the distribution of banks.

5.1 Tobit Model, the Marginal Propensity to Lend and Assortative Matching

I estimate the marginal propensity to lend at individual bank level from the perspective of the supply of banks. In the estimation I use the last quarter of each year as representative of the year for each bank and I sort bank according to their assets. In particular, to have a sufficient number of observations for the empirical analysis, I define large banks as located in the 10th percentile of the assets distribution, and small banks are all the rest of banks.

5.1.1 Empirical Specification for the Marginal Propensity to Lend

Consider the change in banks' loans from the previous period of operation for each bank. The nature of the accounting principle makes the data characterized by nonnegative values with

some observations clustered at zero, while negative predictions need to be excluded. A linear regression model is not suitable for my analysis because it excludes the fact that the marginal propensity to lend cannot be negative. Furthermore, being the data truncated at zero from below then constant marginal propensity to lend is not plausible making the estimation inaccurate. I interpret the presence of these values truncated at zero as corner solutions since some banks may not experience any change in loans from the previous period.

Therefore, I proceed with a two-step procedure. First, to isolate the effects of size on banks' incentives to hold illiquid assets I residualise illiquid assets by estimating a regression of illiquid assets on total assets and a set of control variables through ordinary least squares estimation (hereafter, OLS), as follows:

$$s_{i,t} = \mu_0 + \mu_1' \mathbf{X}_{i,t} + \mu_2' \mathbf{\Gamma}_{i,t} + \vartheta_{i,t}, \tag{30}$$

where in equation (30) $s_{i,t}$ is illiquid assets per unit of deposits, μ_0 is a constant term, and $\vartheta_{i,t}$ is the error term. The vector $\mathbf{X}_{i,t}$ includes banks' balance sheet variables such as total assets, liquid assets, with associated vector of coefficients, μ_1, while $\mathbf{\Gamma}_{i,t}$ contains variables from banks' income statement such as interest rate on deposits, the cost of federal funds, among others, with μ_2 its vector of coefficients.

The estimation results in the prediction of illiquid assets, $\hat{s}_{i,t}$, and I then use the residual term measured as the difference between the predicted and actual level of illiquid assets per unit of deposits, $\tilde{s}_{i,t} = \hat{s}_{i,t} - s_{i,t}$ in the main estimation of the marginal propensity to lend. This term captures the idiosyncratic component of illiquid assets that is not explained by banks' fundamentals, and can be attributed to unobserved regulatory constraints such as further liquidity frictions or unexpected changes in illiquid assets driving the marginal propensity to lend. By obtaining an orthogonalised measure of illiquid assets, the first step aims at reducing any collinearity between illiquid assets, total assets and deposits and identifies within size heterogeneity between large and small banks. It examines whether additional illiquid assets change the marginal propensity to lend.

In the second step I specify a Tobit model for the purpose of the analysis, since the analysis studies simultaneously the probability of banks having zero marginal propensity to lend and the expected marginal propensity to lend conditional on past positive marginal propensity to lend. In particular, the focus of the analysis is how idiosyncratic variation in deposits influence banks' marginal propensity to lend.[9] The model is fully specified by the joint set of equations:

$$\begin{aligned} \text{mpl}^*_{i,t} = \gamma_0 + \gamma_1 \tilde{s}_{i,t} + \gamma_2' \mathbf{X}_{i,t} + \epsilon_{i,t}, \quad & \forall i, \forall \mathbb{1}(i \in [s_{min}, \overline{s}_{min}]), \\ & \text{mpl}_{i,t} = \max(0, \text{mpl}^*_{i,t}), \end{aligned} \tag{31}$$

[9] I follow the Tobit model specification along the lines of the seminal article by Tobin (1958) and further details can be found in Cameron and Trivedi (2005).

therefore, equation (31) estimates the marginal propensity to lend, $\text{mpl}_{i,t}$ with normally distributed conditional errors $\epsilon_{i,t}|x_{i,t}$ as $\mathcal{N}(0,\sigma^2)$, on the set of control variables $\mathbf{X}_{i,t}$, which includes the variables of the income statement and the balance sheet, with vector of coefficients γ_2'. I include a constant term, γ_0, and the term $\mathbb{1}(i \in [s_{min}, \overline{s}_{min}])$ is an indicator variable that indicates whether a bank is a small bank, hence it equals 1 if a bank is small and 0 otherwise. A bank above the threshold $\overline{s}_{min}$ in the interval $[s_{min}, \overline{s}_{min}] \in S$ of illiquid assets is identified as large bank. The estimation is therefore performed separately for all banks, small and large banks. It follows from equation (31) that:

$$\begin{aligned} \text{mpl}_{i,t} &= \text{mpl}^*_{i,t} \quad \text{if} \quad \text{mpl}^*_{i,t} > 0 \\ \text{mpl}_{i,t} &= 0 \qquad\quad \text{if} \quad \text{mpl}^*_{i,t} \leq 0. \end{aligned} \tag{32}$$

Lastly, the coefficient γ_1 in equation (31) captures the excess part of banks' illiquid assets that can lead to an illiquidity channel in the marginal propensity to lend. In particular, it measures whether those banks that hold more or less illiquid assets than the predicted value change loans' supply and, the marginal propensity to lend within the respective banks' size. If this coefficient is zero, then the idiosyncratic component of illiquid assets has no effect on the amount of banks' illiquid assets and all variation in lending is explained by banks' fundamentals. If, on the other hand, this coefficient is different from zero, then idiosyncratic variation in illiquid assets has a non-negligible effect on illiquid assets and banks adjust lending in response to this variation.

5.1.2 Identification Strategy

The identification of the marginal propensity to lend faces some challenges. The main threats to identification are the endogeneity of banks' balance sheet variables that enter the marginal propensity to lend, and simultaneity since liquid and illiquid assets are in fact banks' choices and can be correlated with the error term in the marginal propensity to lend equation (31). Banks normally adjust their portfolio choices in response to shocks that also affect lending, therefore a standard regression of the marginal propensity to lend may confound balance sheet relationships generating co-movements between variables.

Another challenging threat arises because of the specification of the marginal propensity to lend defined as a change in loans over a change in the return on loans, because it is by construction correlated with illiquid assets. Therefore, including illiquid assets also as a regressor would create spurious dependence of the independent variable on the regressors. This is even more problematic when total assets is included as a regressor.

To address these challenges, I implement a two-step estimation procedure. In the first step, I estimate the effects of banks' balance sheet and income statement variables on the level of illiquid assets per unit of deposits. The residual term is then the idiosyncratic part of illiquid assets that is unobserved and not explained by banks' fundamentals. Because the residual term is orthogonal to the regressors, it automatically removes multicollinearity concerns when the residual term is

introduced in the estimation of the marginal propensity to lend. Moreover, because it estimates a pure measure of illiquid assets, it removes the simultaneity concern between illiquid assets and the regressors included.

In the second step, the Tobit model is estimated using the same control variables of the first step and the residual term of illiquid assets. This will allow the separation of the explained and unexplained component of illiquid assets in order to capture relevant cross-sectional heterogeneity in the marginal propensity to lend. Therefore, by controlling for the unobserved heterogeneity correlated with illiquid assets and the marginal propensity to lend that is absorbed by the residual component, the endogeneity concern is reduced.

I use a two-step estimation procedure instead of an instrumental variable approach because the latter would require a strong exclusion restriction of the instrument, which is difficult to justify when most of banks' controls are determined with loans and normally respond to the same shocks. By implementing a two-step procedure as outlined above, I can be confident that the dependency between illiquid assets and the marginal propensity to lend is pruned out and, concerns of simultaneity and endogeneity are partialled out.

5.1.3 Empirical Results on the Marginal Propensity to Lend and the Assortative Matching

I now report the results of the two-step procedure. Table 1 shows the results of the first step OLS regression of illiquid assets in units of deposits on banks' balance sheet and income statement variables. The variable bank size and liquid assets are included in log level to remove spurious correlation and bias due to assets in the regression, whereas the rest of regressors except the price of loans, the cost of federal funds and the cost of funds are in units of deposits. The regression explains between 75.6%-98.5% of the variation in illiquid assets for large and small banks in both years 1998 and 2010.

The variables included in the analysis demonstrate a size-dependent pattern with a substitution and complementarity effect. In particular, in 1998 for small banks liquid assets substitute illiquid assets per deposits by 0.212 units and the coefficient is strongly significant at 1% critical value. By 2010 this relationship reduces, banks continue to substitute liquid assets for illiquid assets, but only by 0.036 units with a coefficient that is no longer significant. By contrast, large banks expand their balance sheet by increasing illiquid assets given a unit increase in liquid assets between 0.035 and 0.012 units in 1998 and 2010, respectively. While both coefficients are insignificant, the estimates show that large banks are mainly unconstrained and for this type of banks, liquid assets yield a complementarity effect on illiquid assets. The estimates on liquid assets for all banks mask the heterogeneity between and within banks, which are mainly driven by small banks, thereby not accounting for the unconstrained complementarity effect for large banks and the strong substitution effect for small banks.

These results show the important role of liquidity regulations and consolidation for small banks, while they depict a distinguished feature of large banks of using scale economies and a

TABLE 1.—

First step regression estimates of illiquid assets

	1998			2010		
	Large banks	Small banks	All banks	Large banks	Small banks	All banks
Liquid assets	0.035	-0.212***	-0.026	0.012	-0.036	-0.028**
	(0.052)	(0.082)	(0.051)	(0.017)	(0.024)	(0.014)
Bank size	-0.170**	0.128	-0.078	-0.016	0.031	0.025**
	(0.074)	(0.101)	(0.063)	(0.015)	(0.021)	(0.013)
Price loans	-64.593***	-107.052***	-77.832***	-76.027***	-80.169***	-78.429***
	(4.791)	(9.251)	(9.164)	(2.178)	(3.696)	(2.261)
Return loans	28.491***	39.769***	33.743***	28.532***	28.996***	28.786***
	(1.991)	(3.653)	(3.033)	(0.462)	(0.847)	(0.462)
Adjust. cost	36.012***	39.521***	39.335***	28.803***	28.869***	28.796***
	(4.803)	(4.459)	(5.160)	(0.465)	(0.847)	(0.512)
Interest deposits	-34.398***	19.069**	-10.891	10.429***	5.415	6.083
	(7.751)	(9.349)	(15.273)	(4.035)	(8.641)	(3.859)
Constant	4.134***	3.559***	3.755***	2.677***	2.738***	2.698***
	(0.469)	(0.524)	(0.276)	(0.098)	(0.189)	(0.091)
Controls	✓	✓	✓	✓	✓	✓
Observations	138	165	304	94	87	182
Adjusted R^2	0.952	0.756	0.868	0.985	0.973	0.978

NOTE. Results of the first step OLS regression of illiquid assets on banks' controls. 'Adjust. cost' indicates adjustment costs, 'size' identifies the level of total assets, 'controls' indicates the presence of other controls beyond the ones reported in the table. Parentheses show robust standard errors. Significance with ***, **, * indicates p-values at 1%, 5% and 10% critical level, respectively.

diversified funding structure in expanding their balance sheet.

Appendix H Section H.1 describes the results for the remaining coefficients in Table 1. The main takeaway of the first step is that the residual component of this regression disentangles idiosyncratic illiquidity effects and additional size heterogeneity for the study of the marginal propensity to lend which are included in the following analysis.

In Appendix H.1 Table H.1 I report the estimates for the Tobit model. The model is estimated via Maximum Likelihood estimation and conditional marginal effects are computed through the Inverse Mills Ratio. I compute robust standard errors for the Tobit coefficient estimates and for conditional and unconditional marginal effects.

The question I am interested in understanding is how deposits affect the probability of giving one line of loans and how the marginal propensity to lend varies as the returns on loans and deposits vary. The two questions are related. To allow meaningful interpretation of the non linear estimates one needs to recover the Inverse Mills ratio. I report in Appendix G further details on this ratio. Here, it suffices to underline that the Inverse Mills ratio accounts for the bias in the probability of censoring and any distortion caused by the censoring from below. If the Inverse Mills ratio is large then the probability of observing the marginal propensity to lend

TABLE 2.—

Unconditional marginal effects of Tobit estimates of the marginal propensity to lend

	1998			2010		
	Large banks	Small banks	All banks	Large banks	Small banks	All banks
Illiquid res., $\tilde{s}_{i,t}$	-0.000	0.002*	0.002*	-0.0002	0.014	0.007
	(0.000)	(0.001)	(0.001)	(0.0003)	(0.009)	(0.007)
Liquid assets	0.0001**	-0.001	0.001***	-0.0001**	-0.001	-0.0001
	(0.000)	(0.001)	(0.000)	(0.000)	(0.001)	(0.0002)
Bank size	-0.0002**	0.001	-0.002***	0.0001*	-0.0004	-0.0001
	(0.000)	(0.001)	(0.001)	(0.000)	(0.0005)	(0.0003)
Price loans	-0.011**	-0.139	-0.172**	0.0002	0.304	0.133
	(0.005)	(0.104)	(0.078)	(0.002)	(0.197)	(0.117)
Return loans	0.003**	0.009	0.064**	-0.0001	-0.073*	-0.026
	(0.001)	(0.040)	(0.028)	(0.0006)	(0.043)	(0.022)
Adjust. cost	0.004	0.004	0.100**	-0.001	-0.071*	-0.031
	(0.004)	(0.081)	(0.046)	(0.0008)	(0.040)	(0.023)
Interest deposits	0.016**	0.696***	0.353**	-0.003	-0.042	-0.039
	(0.007)	(0.179)	(0.154)	(0.009)	(0.502)	(0.176)
Controls	✓	✓	✓	✓	✓	✓
Observations	138	165	304	94	87	182

NOTE. Results of unconditional marginal effects of the Tobit regression specification of the marginal propensity to lend. 'Adjust. cost' indicates adjustment costs, 'size' identifies the level of total assets, 'illiquid res.' is the residualised illiquid assets, 'controls' indicates the presence of other controls beyond the ones reported in the table. Parentheses show robust standard errors. Significance with ***, **, * indicates p-values at 1%, 5% and 10% critical level, respectively.

greater than zero is small which means that the conditional marginal effects are lower than the Tobit estimates. If, on the other hand, the Inverse Mills ratio is small, then this is indication that the observations are uncensored, thereby having a larger effect on the estimated coefficients.

By computing marginal effects, I allow for two marginal effects on the marginal propensity to lend which are denoted as extensive and intensive margin. The extensive margin is the probability that banks extend a unit of loan as deposits rise interacted with the expected marginal propensity to lend. The intensive margin is the increase in the expected marginal propensity to lend given a rise in deposits interacted with the probability of extending one unit of loan.

To account for the true impact effect on the censored observations, I compute conditional and unconditional marginal effects. Table H.2 in Appendix H.1 shows the conditional marginal effects on the second step Tobit estimates in 1998 and 2010, respectively. Here in the main text I report the unconditional marginal effects in Table 2.

The unconditional marginal effects are lower than the conditional marginal effects for illiquid assets in both considered years since they show the effect for the entire population of banks not only for those with positive marginal propensity to lend. They show the aggregate effect on the marginal propensity to lend. The residual term of illiquid assets underlines a weak heterogeneity in 1998 since in the event banks become more illiquid than what is predicted by

banks' fundamentals, the marginal propensity to lend increases by 0.002 for small banks while it is unchanged for large banks. A clear negative assortative matching between banks is evident in 2010, since in the event of more illiquidity small banks increase more strongly the marginal propensity to lend by 0.014. Although this effect is not statistically significant, the coefficient for all banks highlights the negative assortative matching since it is not the average between large and small banks coefficient, rather the interaction between large banks and small banks. Appendix H.1 explains the rest of unconditional marginal effects estimates in Table 2.

In Appendix I Section I.1 I implement a robustness analysis with an interaction term between illiquid assets and the return on loans as a measure of profitability. Table I.1 shows the results for the Tobit estimates and Tables I.2 and I.3 show the conditional and unconditional marginal effects.

5.1.4 Discussion Tobit Analysis for the Marginal Propensity to Lend and the Market Structure

The findings in Section 5.1 reveal a size dependent technology effect with size heterogeneity driving the assignment matching game envisaged in the theoretical analysis.

The conditional marginal effects highlight the results for the uncensored observations on the intensive margin of the marginal propensity to lend. The rise in the marginal propensity to lend of small banks due to a rise in the idiosyncratic illiquid assets takes them to a region where the marginal return on loans are higher. For large banks, albeit with the insignificant coefficient, the sign of the effect of the idiosyncratic illiquid assets indicates a reduction in the marginal propensity to lend since large banks are already operating near the technology frontier and any increase in illiquid assets also raises the adjustment costs.

The response of the marginal propensity to lend is particularly sensitive to the return on loans. This effect decreases sharply in 2010 making the banking system weakly responsive to the return on loans, while it was return sensitive in 1998. For large banks, this effect is strongly positive in 1998 while it turns modestly negative in 2010 indicating that large banks show tiny responsiveness to the return on loans and therefore, do not drive the path of aggregate lending. By contrast, small banks exhibit a remarkable decrease in the marginal propensity to lend in 2010, that combined with the Inverse Mills ratio indicate not only that some small banks exited lending activities, but also that a subset of these banks responded strongly with the marginal propensity to lend to changes in the return on loans.

The liquidity effect on the marginal propensity to lend changes considerably over time and highlights the regulatory constraints banks are subject by 2010. In 1998 liquidity was almost a buffer for all banks. During this time, it decreased the marginal propensity to lend of small banks by 0.103, while it increased the marginal propensity to lend of large banks by 0.128, with an aggregate positive effect of all banks' marginal propensity to lend driven by large banks. This finding means that in 1998 liquidity was abundant and mainly relaxed balance sheet constraints and facilitated lending responses. In 2010 liquidity has a negative impact on the marginal

propensity to lend, since it becomes a regulatory constraint that forces banks to save in liquidity and reduce lending. The effect of all banks is more than 10 times lower than in 1998, reflecting the regulatory environment for all banks. In addition to the Inverse Mills Ratio, these results highlight that liquidity becomes binding and regulation mainly constrain small banks while large banks are less responsive to liquidity conditions.

The relevant market structure is negative assortative matching, but the composition of the matches changes over time. While in 1998 many matches between large and small banks were productive, in 2010 only a few sets of banks maintain a high surplus with a positive marginal propensity to lend. Only small banks with a large response to the return on loans and illiquid assets create a high surplus with which large banks want to match. Therefore, there is a selection effect by which large banks prefer to match with small banks with high and positive marginal propensity to lend, and the second largest banks want to match with the second smallest banks in the system and similarly for the remaining banks. These findings are consistent with the results envisaged in the theoretical analysis.

The economic conditions in 1998 were such that negative assortative matching was efficient in the sense that matches between large and small banks obtained a large contribution by the high response of the marginal propensity to lend and its elasticity of small banks. In 2010 only a few banks have a positive high marginal propensity to lend, which makes selection of these small banks with a high marginal propensity to lend the selected subset of the assignment matching game with a positive surplus. Therefore, the market structure switches from a complementarities sector where the surplus of a match was spread across the size distribution of banks to a more concentrated and selected market structure represented by a few small banks with high marginal propensity to lend and therefore, high value for matches.

The unconditional marginal effects corroborate the results of the conditional marginal effects. Small banks are the main driver of lending in 1998, where the marginal propensity to lend is very elastic to idiosyncratic illiquid assets with positive return on loans. By 2010 the composition of the banking sector changes, where the technology for all banks, not only for large banks, becomes flat and lending becomes more concentrated in a selected subset of small banks that bring a high marginal propensity to lend. As liquidity requirements become tighter post-financial crisis, then liquidity is binding for small banks.

These results have direct implications for the negative assortative matching. In 1998 these matches create a positive surplus because small banks bring a high marginal propensity to lend, through different factors such as idiosyncratic illiquid assets, banks' size, return on loans and adjustment costs in illiquid assets. Because the size distribution features complementarities between banks, matching between this type of banks is attractive. By 2010, the match between large and small banks generates a lower surplus, since these factors have a reduced effect on the marginal propensity to lend of small banks with the exception of illiquid assets. By contrast, they have a small or also an insignificant effect on large banks. Therefore, the negative assortative

matching is more selective since the number of small banks with positive marginal propensity to lend shrinks, and only those match with a high value surplus from the distribution of the marginal propensity to lend are present in the market.

More precisely, with regulation in place and under the submodular liquidity constraint designed in equation (8), in 1998 the match between large and small banks was surplus dominant because the joint liquidity capacity relaxed the constraint and was translated into a contemporaneous increase in lending. In 2010, liquid assets and banks' size decrease the marginal propensity to lend for the majority of small banks with only a few matches between large and small banks being surplus enhancing. Therefore, the submodularity makes the negative assortative matching to persist, but it becomes more selective with only a few small banks with high marginal propensity to lend being the preferred matching assignment for large banks.

By estimating the marginal effects of the primitives underling the assignment matching game between banks, the results underscore the type of match between banks which is negative assortative matching and showcase a change in the surplus of the matching where the negative assortative matching becomes more selective towards the set of small banks underpinned by a high marginal propensity to lend.

5.2 Quantile Regression for Inequality in the Marginal Propensity to Lend

This Section studies the marginal propensity to lend along the distribution of banks through a quantile regression analysis to investigate inequality in the marginal propensity to lend driven by banks' size.

5.2.1 Empirical Specification for Inequality in the Marginal Propensity to Lend

The analysis estimates conditional mean functions by estimating the distribution of quantile functions of the marginal propensity to lend and studies how each quantile changes given each regressor. The quantile regression is implemented along the lines of Koenker (1978, 2005).[10]

Assume the τ quantile of the distribution of the marginal propensity to lend is $q_{\text{MPL}}(\tau)$, then it can be recovered as the solution to $F_{\text{MPL}}(q) = \tau$ where $F_{\text{MPL}}(\text{mpl}) = P(\text{MPL} \leq \text{mpl})$ is the distribution function of the marginal propensity to lend. I implement the following quantile regression specification:

$$\text{mpl}_{i,t} = \gamma_0 + \gamma_1' \mathbf{X}_{i,t} + \epsilon_{i,t}, \quad \forall i, \forall \mathbb{1}(i \in [s_{min}, \overline{s}_{min}]), \tag{33}$$

where, i, identifies a bank, t is the specific year considered, γ_0 is the intercept, γ_1, is a vector of coefficients on the set of regressors $\mathbf{X}_{i,t}$, the term $\epsilon_{i,t}$ represents normally distributed conditional error term, and the term $\mathbb{1}(i \in [s_{min}, \overline{s}_{min}])$ is an indicator variable that equals one if the quantile

[10] For a literature review on quantile regression analysis see Koenker (2001) and an examination of goodness of fit measure applied to quantile regression is available in Koenker and Machado (1999).

regression is performed for small banks and viceversa. The regression is then performed for all banks. Controls include measure of banks' illiquid assets to deposits, liquid assets to deposits, banks' size in log of deposits, the return of loans. Computing the quantile distribution functions requires to undertake the following minimisation problem:

$$\min_{\gamma_1} [\tau \int_{\text{mpl}_{i,t} > \gamma_1' \mathbf{X}_{i,t}} |\text{mpl}_{i,t} - \gamma_0 - \gamma_1' \mathbf{X}_{i,t}| dF_{\text{MPL}|\mathbf{X}_{i,t}}(\text{mpl}_{i,t}) + (1-\tau) \int_{\text{mpl}_{i,t} < \gamma_1' \mathbf{X}_{i,t}} |\text{mpl}_{i,t} - \gamma_0 - \gamma_1' \mathbf{X}_{i,t}| dF_{\text{MPL}|\mathbf{X}_{i,t}}(\text{mpl}_{i,t}) \Bigg], \tag{34}$$

where the solution to the quantile minimisation problem depends on the quantile function, $\gamma_{1,\tau}$ such that it is possible to obtain the τ-th conditional quantile $q_{\text{MPL}_{i,t}|\mathbf{X}_{i,t}}(\tau) = \gamma_{1,\tau}' \mathbf{X}_{i,t}$. The estimator can be obtained via the sample counterpart that is the average of weighted absolute errors asymmetrically computed with weights τ and $(1-\tau)$ on positive and negative errors, respectively. In particular, it equals:

$$N_n(\gamma_1, \gamma_0; \tau) := \frac{1}{n} \Bigg[\tau \sum_{i:\text{mpl}_{i,t} > \gamma_1' \mathbf{X}_{i,t}} |\text{mpl}_{i,t} - \gamma_0 - \gamma_1' \mathbf{X}_{i,t}| + (1-\tau) \times \sum_{i:\text{mpl}_{i,t} < \gamma_1' \mathbf{X}_{i,t}} |\text{mpl}_{i,t} - \gamma_0 - \gamma_1' \mathbf{X}_{i,t}| \Bigg], \tag{35}$$

where n indicates the sample size. I solve for this unconstrained optimisation problem with Quasi-Newton algorithm to obtain the coefficient estimates for the vector γ_1' and γ_0. To compute robust standard errors I implement bootstrapping methods over $5,000$ resampling repetitions with replacement.

5.2.2 Empirical Results on Heterogeneity in the Marginal Propensity to Lend

Table 3 and 4 show the results on the distribution of the marginal propensity to lend for all banks for the year 1998 and 2010, respectively. I report the results for all banks' marginal propensity to lend since in the quantile analysis they disentagle the presence of negative assortative matching. In Appendix I I relegate the results for large and small banks' marginal propensity to lend.

The results indicate that illiquid assets have a larger effect at larger quantiles of the marginal propensity to lend represented by small banks. In particular, in response to an increase in illiquid assets, the marginal propensity to lend is very heterogeneous at different points of the distribution of the marginal propensity to lend.

In 1998, more illiquid assets generate a strongly negative marginal propensity to lend for banks in the 25th quantile of the distribution with a decline of 15.522 in the marginal propensity to lend. Banks in this location (at the bottom of the distribution) are large banks who hold liquid assets and do not reduce lending when becoming more illiquid and the risk in their balance

TABLE 3.—

Inequality in the marginal propensity to lend of all banks in 1998

	1998			
	25th quantile	50th quantile	75th quantile	90th quantile
Illiquid assets	-15.522***	-5.429	-1.272	3.561
	(4.599)	(4.141)	(4.191)	(22.217)
Liquid assets	12.590**	6.664	5.159	-11.552
	(6.281)	(4.720)	(5.357)	(47.228)
Size deposits	3.325***	0.718	-0.553	-11.905
	(1.201)	(1.108)	(1.697)	(12.388)
Constant	-2.979	15.755	36.016*	211.393
	(15.996)	(13.396)	(20.097)	(139.984)
Controls	✓	✓	✓	✓
Observations	304	304	304	304
Pseudo R^2	0.021	0.009	0.008	0.017

NOTE. Results of quantile regression analysis of the marginal propensity to lend in 1998. 'Size' identifies the log of deposits, 'controls' indicates the presence of other controls beyond the ones reported in the table. Parentheses show robust standard errors computed with bootstrapping method in 5,000 resampling with replacement. Significance with ***, **, * indicates p-values at 1%, 5% and 10% critical level, respectively.

TABLE 4.—

Inequality in the marginal propensity to lend of all banks in 2010

	2010			
	25th quantile	50th quantile	75th quantile	90th quantile
Illiquid assets	5.351	-3.066	-7.121	-20.879
	(8.103)	(5.660)	(7.980)	(37.977)
Liquid assets	-10.406	24.721*	47.237**	96.711
	(34.778)	(12.507)	(20.394)	(124.191)
Size deposits	-1.571	2.193	-1.252	-6.144
	(2.289)	(1.566)	(2.411)	(13.422)
Constant	2.937	-22.206	30.697	124.407
	(38.198)	(16.128)	(39.472)	(234.538)
Controls	✓	✓	✓	✓
Observations	182	182	182	182
Pseudo R^2	0.005	0.011	0.022	0.034

NOTE. Results of quantile regression analysis of the marginal propensity to lend in 2010. 'Size' identifies the log of deposits, 'controls' indicates the presence of other controls beyond the ones reported in the table. Parentheses show robust standard errors computed with bootstrapping method in 5,000 resampling with replacement. Significance with ***, **, * indicates p-values at 1%, 5% and 10% critical level, respectively.

sheet rises. The magnitude of the effect falls sharply from the 25th to the 75th quantile, at the 90th quantile the effect becomes positive indicating that small banks aggressively increase the marginal propensity to lend by 3.561. As the negative effect weakens, the slope of the marginal propensity to lend becomes flatter. While the coefficients on the 50th, 75th and 90th quantile are not significant, they are informative of the heterogeneity in the marginal propensity to lend.

The marginal propensity to lend response of small banks that increases with illiquid assets indicates that small banks face tighter liquidity constraints and increases loans when they become more illiquid. Medium large banks and medium small banks in the 50th and 75th quantile, respectively, tend to be less constrained and respond to increases in illiquid assets by reducing the marginal propensity to lend.

In 2010 the upper tail of the distribution is dominated by constrained small banks and the sign of impact of an increase in illiquid assets reverts along the distribution. Large banks in the lower 25th quantile of the distribution rise lending and the marginal propensity to lend in response to an increase in illiquid assets. This rise is three times higher than the 1998 figure of large banks, which is economically meaningful and statistically significant and highlights a remarkable size effect in the marginal propensity to lend in the lower tail of the distribution of large banks. By contrast, small banks strongly reduce the marginal propensity to lend by 20.879 in response to the same rise, all else equal. The fall is almost sixfold the response in 1998 which is non negligible because it reverts direction.

In Appendix H.2 I explain the remaining coefficient estimates in Table 3 and 4.

These results, and particularly the coefficient on the 90th quantile, are consistent with the Tobit results in Section 5.1 where the responses of small banks drive the shape of the marginal propensity to lend. Moreover, the lower tail of the distribution expands the marginal propensity to lend when banks in this tail are more liquid and expand in size with deposits, by contrast the upper tail of the distribution is characterised by upward changes in the marginal propensity to lend only when they become more illiquid. This is evidence that corroborates negative assortative matching in the banking sector, where large banks match with small banks and not within their type, changing the market structure through the assignment matching game since banks' fundamentals generate opposite effects at the tails of the marginal propensity to lend.

Appendix I Section I.2 performs a robustness analysis and undertakes quantile regressions separately for large banks and small banks. In these experiments, the quantile results are indicative of the heterogeneity within each type of banks.

6 Quantitative Analysis

I now study the quantitative properties of the model. In particular, in this Section I start by calibrating the model to match moments for the US commercial banking sector and I then analyse the results. The model results consist of the examination of the redistribution mechanism resulting from the additional liquidity and idiosyncratic deposits shocks, the endogenous market structure, the redistribution mechanism of assets and the distributional consequences of adjustment costs on banks and liquidity regulations. I conclude by examining the policy outcomes of the Bank Term Funding Program introduced by the Fed.

6.1 Calibration

A period in the model is a quarter. I use the US economy as a reference to calibrate the model. The choice of the US is based on the empirical evidence that the deposits market is highly concentrated and following the financial crisis liquidity of small banks increased relatively more than liquidity of large banks. This provides the ground for the redistribution mechanism of assets. Likewise, the choice of the US economy is motivated by the adjustment costs in illiquid assets that are disproportionately higher for small banks.[11]

I use two data sources for the calibration of the model: the Consolidated Reports of Conditions and Income of commercial banks provided by the Federal Reserve and collected from Wharton Research Data Services (2024) for micro data and the Federal Reserve Bank of St. Louis Economic Research Database for macro aggregate data.

I calibrate the model from 2008q4 to 2019q1. The choice of the time period is based on the fact that the Federal Reserve started paying the interest rate on excess reserves in October 2008. The calibration consists of two subsets of parameters. The first subset is taken directly from the data or the literature, and the second subset is chosen to match key data targets.

To map the model to the data I need to classify aggregate banks' liquid and illiquid assets. I define liquid assets as total excess reserves held by depository institutions, whilst illiquid assets consists of commercial industrial loans, total loans and leases, real estate loans, total assets, other borrowed money, commercial real estate construction, loans to individuals for households, family, personal expenditure, credit cards and revolving credit.[12]

Table 5 provides summary of the structural and calibrated parameters.

My calibration strategy consists of choosing three parameters to match three data targets. The calibrated parameters are the liquidity constraint parameter, ϕ, the standard deviation and autocorrelation of the transitory component of the deposits process, σ_d and ρ_d, respectively. These parameters match *i)* the liquidity to deposit ratio of 0.183, *ii)* the annual standard deviation of deposits 0.00366 and *iii)* the annual autocorrelation of deposits 0.882, respectively.

Deposit process. I approximate the deposit process with a transitory component, $d_{i,t} = \psi_{i,t}(d_{i,t+1}, d_{i,t})$. The transitory component that follows a Markov chain $\psi(d_{i,t+1}, d_{i,t})$, is approximated with an AR(1) process in the log of deposits as estimated in Corbae and D'Erasmo (2021). The specific functional form is:

$$\log(d_{i,t+1}) = (1-\rho_d)\eta_d + \rho_d \log(d_{i,t}) + \xi_{i,t+1}, \tag{36}$$

where the idiosyncratic component $\xi_{i,t+1}$ is iid and normally distributed $\mathcal{N}(0, \sigma_\xi)$. The calibration of the deposit process follows Gigante (2026). The model is at quarterly frequency. I take

[11] This fact is in accordance with the economies of scale by which average adjustment costs tend to increase as size grows and after banks have reached a medium size adjustment costs are decreasing in banks' size (Berger and Mester, 1997; Berger and Hannan, 1998).

[12] Appendix L provides further data details.

TABLE 5.—Structural and Calibrated Parameters

Parameter	Description	Value	Source/Target moment
	Structural parameters		
γ	Risk aversion parameter	1	Bianchi and Bigio (2022)
r_ℓ	Liquidity interest rate	1.001	Excess reserves rate 0.673%
r_d	Deposit interest rate	1.006	Drechsler et al. (2017)
r_s	Loan interest rate	1.109	Assigned
α	Adjustment costs	2.17	Literature
ψ	Debt elastic rule	0.001	Schmitt-Grohé et al. (2003)
κ	Leverage requirement	1.50	Observed
δ	Liquid assets state dependency	0.05	Assigned
δ_e	Exit probability	0.12	Corbae and D'Erasmo (2021)
	Calibrated parameters		
β	Subjective discount factor	0.99896	Constraint, $\beta r_\ell < 1$
ϕ	Liquidity requirement	0.118	Target, $\ell/d, 0.184$
ρ_d	Autocorrelation withdrawal shock	0.944	Annual target, ρ_d
σ_d	Standard deviation deposits	0.002	Annual target, σ_d

NOTE. This table reports parameters and calibration of the model. Calculations and targets are reported in the main text.

as targets the values of mean, standard deviation and persistence of deposits, $\{\eta_d, \sigma_d, \rho_d\}$, as estimated in Corbae and D'Erasmo (2021). These values, however, are at a annual frequency. Therefore, I derive time-aggregated annual parameters for $\{\eta_d, \sigma_d, \rho_d\}$, to match annual data moments. The obtained values are 0.002 and 0.944 for standard deviation and persistence of the deposit process, respectively.

To obtain a finite state representation of deposits I discretise the idiosyncratic process following Tauchen and Hussey (1991) discretisation with three states $d_j \in \{d_h, d_m, d_l\}$, high, medium and low such that $\{d_l < d_m < d_h\}$. With this procedure I therefore obtain a (3×3) transition probability matrix. Given the nature of the intraperiod liquidity requirement and the deposit withdrawal shock, the output of the Tauchen and Hussey' discretisation is modified with a 9×9 transition probability matrix encoding 9 states for the process of deposits.

In the quantitative model with full convex adjustment costs, entry and exit, I calibrate the deposit process in the original (3×3) transition probability matrix to retain tractability due to the presence of sparsity in the modified transition matrix and the computation of the marginal propensity to lend as derivative over a policy function which is an infinite dimensional object. Thus, I solve for the inflow/outflow term in the bank budget constraint as $[\psi_{d_{t+1},d_t}(d_{i,t+1}) - d_{i,t}]$.

Bankers' preferences. I parameterize bankers' utility function with a CRRA functional form:

$$u(c) = \frac{c^{1-\gamma}}{1-\gamma}, \quad \gamma > 0. \tag{37}$$

This functional form is consistent with recent studies on banks' preferences (Bianchi and Bigio, 2022). I set the risk aversion coefficient to $\gamma = 1$, consistent with an IES equals to 1.

Interest rates. I set the gross interest rate on liquidity to $r_\ell = 1.001$ in line with the annualised interest rate on excess reserves 0.673%. The gross interest rate on deposits is set to $r_d = 1.006$, as provided in Drechsler et al. (2017). The gross interest rate on loans is parameterised to $r_s = 1.109$ to obtain a well definite distribution of liquid assets.

Liquidity constraint. The parameter value that governs the liquidity constraint, ϕ, is calibrated to 0.118 to match the data moment target for the liquidity to deposit ratio, $\ell/d = 0.184$. I set bankers' discount factor according to Gigante (2026). Given the interest rate on liquidity, this implies that the parameter value for the subjective discount factor is set to $\beta = 0.99896$.

Submodularity and state dependency of liquid assets. The submodular liquidity constraint encodes a parameter that controls for the state dependency of liquid assets and the submodularity of deposits with liquid assets, δ, and acts as a further tightening of the liquidity constraint. More precisely, it captures the strength of the interactions between liquid assets and deposits in a bank balance sheet. When $\delta > 0$, liquid assets become less costly in low deposit states, and more costly in high deposit realisations, hence it additionally tights the liquidity constraint and liquid assets and deposits are substitute. When $\delta < 0$, liquid assets are less (more) costly in high (low) deposits realisations, thus liquid assets and deposits become complement. I set δ to 0.05 in the baseline economy, and then investigate adjustments to the strength of submodularity.

Entry and Exit. In each period a constant fraction of banks exit the economy. I calibrate the exit probability $\delta_e = 0.12$ following Corbae and D'Erasmo (2021). This probability will also denote the entry mass at the entry state. At the entry state banks locate in the distribution with the steady state values of liquid and illiquid assets computed under the assumption of constant deposits, binding liquidity constraint and slack leverage constraint, with convex adjustment costs; instead of entering with zero liquid and illiquid assets.

Illiquid assets. Illiquid assets are an endogeneous choice. However, a steady state value is computed under the above assumptions to determine the entry point of illiquid assets in the distribution for entrant banks. I parameterise the leverage constraint parameter in accordance with the Riegle-Neal Interstate Banking and Branching Efficiency Act that establishes the basis for the loan-to-deposit ratio. The Act requires banks to have sufficient liquidity to meet deposit withdrawals and that banks have enough deposits to lend. While a high value of the loan-to-deposits ratio is a measure of not enough liquidity, a low value may point towards too little profits. Therefore, I set the leverage constraint parameter to $\kappa = 1.50$, about double the value of illiquid assets, as a good compromise between balanced liquidity in a bank balance sheet and balanced profits.

Adjustment costs in illiquid assets. The key parameter to determine the redistribution of assets between liquid and illiquid assets is the adjustment cost parameter, α. Indeed, it indicates the degree of illiquidity in banks' balance sheet. The literature provides a range for this parameter in the magnitude of $(3.54, 2.17, 2.75)$ including calibration as in Gertler et al.

(2012). Therefore, I set the adjustment cost parameter to $\alpha = 2.17$.

I use two functional forms for the adjustment costs function. They corresponds to the functional forms analysed in the theoretical properties in Section 4.3. The first functional form is the partial adjustment cost function, $\eta(s', s) = \frac{\alpha}{2}(s' - \overline{s})^2$ and the second the full convex adjustment cost function, $\eta(s', s) = \frac{\alpha}{2}(s' - s)^2$.[13] I set the steady state parameter for illiquid assets to $\psi = 0.001$ in line with the literature on the debt elastic interest rate rule (Schmitt-Grohé and Uribe, 2003). In the computation of the model with partial adjustment costs, I modify the adjustment cost parameter to $\alpha = 0.899$ to assure convergence of the model.

Empirically, adjustment costs in illiquid assets are not directly observable and the literature that estimates an adjustment costs function on banks remains limited. The literature have provided estimates either of the adjustment costs friction (Flannery and Rangan, 2008; Berger et al., 2008), or of a structural adjustment costs that focus on capital adjustments of lending dynamics (Gropp et al., 2019; Jokipii and Milne, 2011).

Appendix E describes the solution algorithms implemented to compute the baseline model, its variants, and the MIT algorithm for the unanticipated shock to the liquidity capacity in the banking sector.

6.2 Micro Banks' Assets and the Marginal Propensity to Lend

This Section provides data and distributional moments in comparison with model moments. First, I solve the baseline model which contains full convex adjustment costs, entry and exit, the leverage constraint and the submodular liquidity constraint. Then, I allow for a version of the liquidity constraint which does not embed submodularity in illiquid assets. Likewise, I provide the results without entry and exit.

Table 6 presents some of the main moments generated from the model in comparison with the data. These moments are untargeted. The baseline model does particularly well in mapping the kurtosis of illiquid assets and the price of loans, the mean of liquid assets and the marginal propensity to lend. The model does not do well in generating the average level of illiquid assets which is better captured by the model without exit. The average duration of banks in the models with entry and exit is of approximately 8 quarter periods.

In comparison with the three versions of the model, the baseline model does best in generating the average price of loans close to the data. The mean of the marginal propensity to lend is overestimated in the model without exit and introducing exit helps to reduce this overestimation. The model with the liquidity constraint not in submodular version, does well in generating the mean of liquid assets similarly to the baseline model. Likewise, it matches almost well the average adjustment costs, but it misses their standard deviation and kurtosis.

Since these are non targeted moments, the baseline model does a decent job in matching the approximated pattern observed in the data. Additionally, because the focus is on the marginal

[13] I have dropped time subscript to simplify notation and a prime indicates future variables.

TABLE 6.—

Model moments mapped to data

Moment	Data	Baseline	No submod. constraint	No exit
Illiquid assets				
Mean, s	2.792	0.656	0.655	1.171
St. dev., s	0.355	0.131	0.133	0.129
Kurtosis, s	4.979	3.218	2.934	1.762
Liquid assets				
Mean, ℓ	0.236	0.258	0.258	0.263
St.dev., ℓ	0.109	0.017	0.028	0.012
Kurtosis, ℓ	10.868	1.489	1.423	1.856
Price of loans				
Price of loans mean	0.019	0.776	0.798	0.869
Price of loans st. dev.	0.010	0.250	0.263	0.196
Price of loans kurt.	2.271	1.726	1.814	1.625
Adjustment costs				
Adjust. costs mean	0.043	1.124	0.028	1.109
Adjust. costs st. dev.	0.186	0.000	0.046	0.000
Adjust. costs kurtosis	253.840	3.000	2.957	3.000
Marginal propensity to lend				
Mpl mean	89.267	388.157	316.051	466.675
Mpl st. dev.	2,508	399.302	424.766	515.370
Mpl kurtosis	840.373	1.451	1.586	1.774
Duration banks				
Average periods	–	8.333	8.333	–

NOTE. This table reports main moments in comparison with the data. It also reports the moments for different versions of the model in comparison with the baseline environment. Moments are computed by approximating the distribution of liquid and illiquid assets. The baseline model is the model with full convex adjustment costs, exit, leverage constraint, and submodular liquidity constraint. Liquid assets and illiquid assets are in units of deposits.

propensity to lend and the baseline model goes closer in fitting jointly the moments of the marginal propensity to lend, the price of loans, liquid assets and the kurtosis of illiquid assets, I take the baseline model as a benchmark to study assortative matching properties in the banking sector and the marginal propensity to lend.

6.3 Quantitative Results

Informed by the theoretical results and disciplined by the calibration and the micro empirical evidence, this Section examines the results of the model. It points out the need of bank heterogeneity in banks' portfolio choices to give rise to the share of constrained banks as a primary source that reconciles the redistribution of assets between assets and across banks. The Section then shows negative assortative matching in the banking sector, quantifies the contribution of

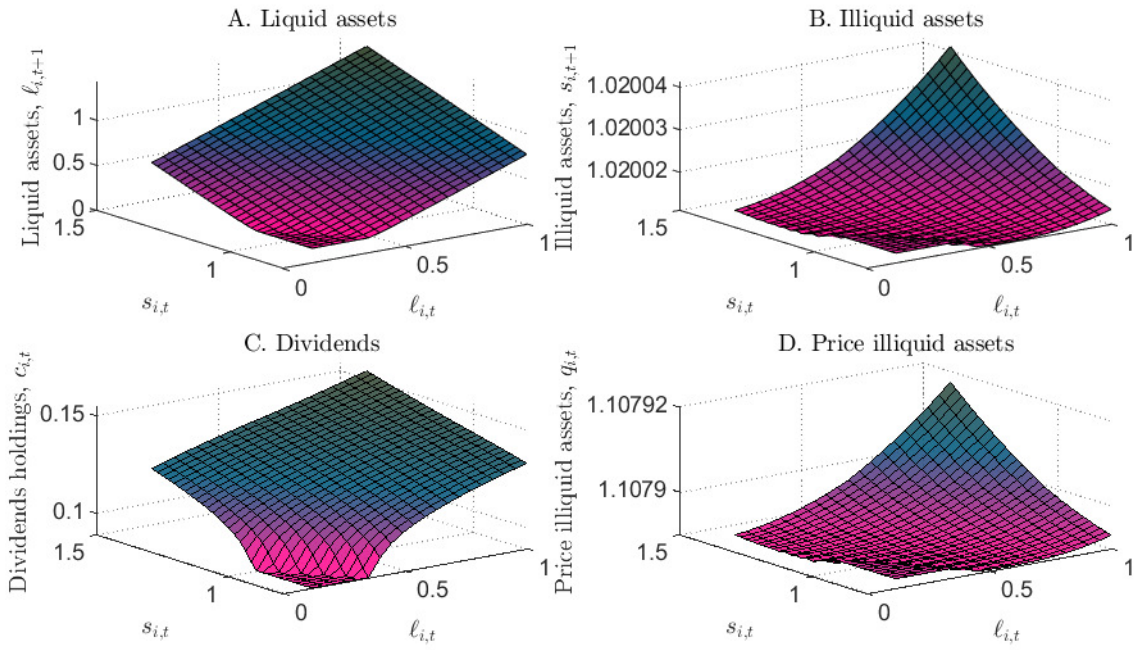


FIG. 8.—Bank decision rules.

NOTE. This figure shows decision rules for liquid assets, illiquid assets, dividends and the price of illiquid assets for the model with partial adjustment costs and no leverage constraint, and banks' entry and exit.

the marginal propensity to lend and assesses the quantitative importance of adjustment costs.

6.3.1 Illiquid Banks' Heterogeneity and Banks' Size Expansion

In this Section I solve the model with the partial adjustment cost function in which assets are infinitely illiquid. Moreover, to disentangle the difference and the mechanism surrounding the withdrawal of deposits, it is instructive to proceed by first solving the model with only the idiosyncratic component of deposits, without entry and exit and the leverage constraint. In this way the model can shed light on the liquidity frictions characterising banks' liquidity without overlapping effects from other sources of constraint. The purpose of the Section is to highlight the fraction of banks acting at the constraint and the redistribution of assets.

Figure 8 illustrates banks' decision rules with the partial adjustment costs function when assets are infinitely illiquid. There are two takeaways from the figure. First, Panel A and C show decision rules for liquidity and dividends and confirm that there is a fraction of banks acting at the liquidity and leverage constraint on liquid and illiquid assets' states, while dividends' decision rule becomes increasing and concave as liquid and illiquid assets increase. Banks expand in sizes through liquid assets, in particular for large banks at higher liquid assets holdings.

The illiquid assets decision rule in Panel B shows that, while increasing with only few banks redistributing their assets from illiquid to liquid assets, it does not generate a lot of variation given that the adjustment costs function is partially constant. In this model adjustment costs

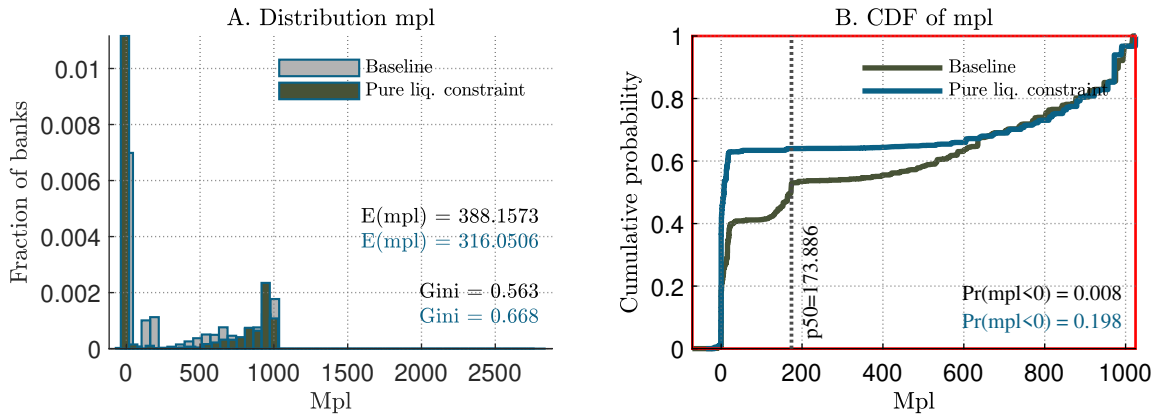


FIG. 9.—Distribution banks' marginal propensity to lend.

NOTE. Distribution of the banks' marginal propensity to lend in panel A and its CDF in panel B. In panel A light gray distribution reports the result for the baseline model and the green distribution presents the result for the model with the pure liquidity constraint. In panel B the vertical line reports the 50th percentile of the marginal propensity to lend. The first line of reported quantities in black refers to the baseline model and the green lines to the model with a pure liquidity constraint. The marginal propensity to lend is computed with finite difference as the change in loans given a change in the return on loans.

adjust relative to a fixed amount of illiquid assets. Consequently, the price of illiquid assets in Panel D changes upward as a function of liquid assets by a very small amount, since it is dependent upon the variation of illiquid assets decision rule. Therefore, a full convex adjustment costs functional form is needed to reproduce the distribution of liquid and illiquid assets and the marginal propensity to lend observed in the data.

Second, the model perfectly nests the environment when banks have only liquid assets in their budget constraint given that the change of illiquid assets along the grids of current liquid assets and illiquid assets is very tiny. This feature arises because the adjustment cost function offsets the desire of banks of holding illiquid assets regardless of the value of the adjustment cost parameter. By penalising fast changes in illiquid assets, through the full convex adjustment costs functional form and the leverage constraint the model can generate the path of illiquid assets and the marginal propensity to lend according to banks' size differentials.

6.3.2 Endogenous Market Structure and the Elasticity of Illiquid Assets in the Matching Regulatory Environment

I now examine the results of the baseline model with the full convex adjustment cost function, entry and exit, leverage constraint, and submodular liquidity constraint that motivated the premises of the paper. This paper started with the observation that convex adjustment costs in illiquid assets give rise to a redistribution across banks and the redistribution mechanism between liquid and illiquid assets.

Figure 9 Panel A shows the distribution of the marginal propensity to lend with the baseline model and the model that embeds the pure liquidity constraint which is not in submodular

TABLE 7.—

Elasticity of illiquid assets to return on loans

	Elasticity, %	p10	p25	p50	p75	p90
Baseline	0.050	-12.40	0.00	8.50	257.90	1,591.30
Pure constraint	0.051	-11.80	0.00	9.70	257.80	1,588.10
Submodularity, $\delta = 0.15$	0.051	-12.70	0.00	9.00	257.80	1,606.90
Submodularity, $\delta = -0.15$	0.049	-13.60	0.00	6.80	256.60	1,587.00

NOTE. This table reports the elasticity of illiquid assets to change in the interest rate on loans and its moments for the baseline model, the model with a pure liquidity constraint and the models in which the degree of submodularity and state dependency is modified to 0.15 and -0.15. The elasticity is computed by weighting the values with the ergodic distribution of banks.

functional form. The distribution of the marginal propensity to lend in the baseline model is very unequal along the distribution of banks with a Gini coefficient of 56.3%. The distribution is fat and approximately 0.7% of banks are concentrated in the bin with a low (16.21) marginal propensity to lend, whereas only 0.2% of banks show a very large (1006.51) marginal propensity to lend, that is 2.6 times the average marginal propensity to lend in the economy. Moreover, in practical terms given the model computation, the figure highlights that on average if the interest rate on loans was to increase by 1 percentage point, then illiquid assets would increase by 3.88 units, which is $\partial s_{i,t+1} = \mathbb{E}(\text{mpl}_{i,t}) \times \partial r_{s,t} = 388.1573 \times 0.01 \approx 3.88$.

The distribution of the marginal propensity to lend becomes even more unequal when considering a pure liquidity constraint that leads to a 66.8% inequality measured through the Gini coefficient. This is because the distribution becomes fatter where 1.12% of banks are concentrated in the lower bound of the marginal propensity to lend, while about 0.23% of banks have an average marginal propensity to lend of 944.963, about 3 times the average marginal propensity to lend. Under submodularity, the distribution moves towards large banks with high illiquid assets. This occurs since δ within the submodular liquidity constraint, tightens the constraint more for banks with high liquid assets, thereby changing the marginal value and the shape of the matching surplus between banks. The marginal value of liquidity for large banks becomes connected to that of small banks. While large banks become valuable because of the low marginal propensity to lend, small banks can stabilise by matching with large banks.

This pattern is denoted under the baseline economy by the reduction in the fraction of banks in the lower bound and a fatter distribution at the tails of the marginal propensity to lend, particularly towards the middle of the distribution. This is the essence of the negative assortative matching in the banking sector. The strength of submodularity in the submodular liquidity constraint, makes matching between large and small banks optimal since by matching both types of banks can benefit from expanding illiquid assets more strongly, relaxing at the margin the liquidity constraint.

Panel B shows the cumulative distribution function (hereafter, CDF) of the marginal propensity to lend. Under the baseline economy with a submodular liquidity constraint, the

distribution moves faster at lower values of the marginal propensity to lend and increases more quickly for high values of the marginal propensity. The median of the distribution is associated with a 173.886 marginal propensity to lend. This indicates that the distribution of the marginal propensity to lend is centered on large banks, resembling the pattern of the distribution of the marginal propensity to lend in 2010. The probability of observing banks with low marginal propensity to lend is also very small and equal to 0.8%. By contrast, the CDF in the economy with a pure liquidity constraint shows a slower increase at lower values of the marginal propensity to lend where the median bank is associated with a 7.251 marginal propensity to lend, that is remarkably 24 times lower than the median bank in the baseline economy. Indeed, the probability of observing a negative marginal propensity to lend is 19.8%. Effectively, fewer banks remain in the medium state under the baseline economy. The gap between the two CDFs closes at the 67th percentile of the distribution where banks of this size in both economies have 621.81 marginal propensity to lend.

These results imply that a back-of-the-envelope calculation for the elasticity of illiquid assets to change in the return on loans at each point of the bank distribution can be computed as follows:

$$\varepsilon_{s,r_s} = \left(\underbrace{\frac{\partial s_{t+1}}{\partial r_{s,t+1}}}_{\text{mpl}} \frac{r_{s,t+1}}{s_{t+1}} \right) \sum_{d^t \in \mathcal{D}^t} \mu_{t+1}(\ell_{t+1}, s_{t+1}, d_{t+1}, d_t), \tag{38}$$

where equation (38) is weighted by the distribution of banks.

Table 7 shows the elasticity implied by the model which indicates a very skewed distribution. The baseline economy shows an elasticity of 0.05% which implies that a change of 1% in the interest rate on loans, increases illiquid assets by 0.05%. Analysing the percentiles of the elasticity, they show that 25% of banks have an elasticity of zero while 10% of the mass is located with a negative elasticity of -12.4. Median large banks contribute to the economy with a moderate elasticity of 8.5%. The upper tail of the distribution appears to be very sensitive to changes in the interest rate on loans. The 90th percentile of banks has a very high elasticity of $1,591.30$, thus a 1% change in the interest rate on loans generates a remarkably high upward change in illiquid assets, since holding high deposits, these banks increases illiquid assets quickly.

Since banks in the 90th percentile are small banks, these are the banks with low liquid assets, high deposits, very unstable funding inflow, for which the liquidity and leverage constraint are binding almost surely and therefore, any small changes in the return of loans causes a very large change in illiquid assets. Moreover, because the overall elasticity decreases to 0.05%, small banks effectively match with large banks spending less time in the extreme states with high elasticity. Large banks that have more mass have an elasticity close to zero because these are banks with high liquid and illiquid assets, less deposits, and therefore, more stable funding inflow.

The elasticity of the model with a pure liquidity constraint is 0.051% slightly higher than the baseline model and shows that when banks are not penalised for the additional liquidity

they hold, the elasticity of small banks located in the 90th percentile is smaller and redistributes toward the median of the distribution. Medium large banks with high liquid and illiquid assets have an elasticity of 9.7, which is 14.12% higher than the baseline model. This means that the submodular liquidity constraint penalises more strongly liquidity holdings of large banks, generates more mass at the lower end of the distribution and makes the negative assortative matching stronger because it stabilises the system with a lower elasticity. The high elasticity states tend to be rare, and while the distribution has a very fat tail, the unstable states disappear once small banks match with large banks. By matching with large banks, small banks receive liquid assets and hence the liquidity constraint relaxes.

The pattern is more pronounced when the strength of submodularity and state dependency, δ, is modified to above and below the baseline value. A higher value of δ than the baseline leads to a higher elasticity concentrated in the upper tail of the distribution and in the median banks in the economy whose elasticity is 0.50 percentage points higher than the baseline economy. With a negative value of state dependency, the overall elasticity falls to 0.049% since δ relaxes the constraint of large banks. Consequently, the elasticity of medium large banks falls and an increase in the interest rate on loans generates a 6.80% rise in illiquid assets.

6.3.3 Distributional Consequences of Submodular Liquidity Constraint and Convex Adjustment Costs

Figure 10 shows the distribution of the price of illiquid assets for small banks in panel A and large banks in panel B. The Figure reveals that the price of illiquid assets is decreasing in the current value of illiquid assets holdings and increasing in current liquid assets, regardless of the size of banks. With low illiquid assets, banks are liquidity and leverage constrained and the marginal value of additional illiquid assets is high. As banks grow in size the value of holding additional illiquid assets falls, hence the price falls. Holding illiquid assets constant, banks with low liquid assets cannot expand illiquid assets because they are liquidity constrained. Therefore, the marginal value of illiquid assets is the largest for banks with high current liquid assets.

The price of illiquid assets for small banks in Panel A shows that the price of illiquid assets is valuable in the low current states of current liquid assets, and because they are liquidity constrained, relaxing the constraint makes the marginal value high and the price of illiquid assets rises at high values of current liquid assets. For large banks, the price of illiquid assets shown in Panel B, reveals that additional units of liquidity are less optimal and the path of the price of illiquid assets is almost flat along current liquid assets, whereas a smooth decreasing path is observed along the illiquid assets grid. Note that the state dependency of the liquidity constraint, δ, makes the constraint tighter for large banks in the present but relaxes the constraint in the future, and hence the price of illiquid assets reflects this marginal increase at high current liquid assets and is high for large banks. Moreover, the price of illiquid assets represents the shadow value of relaxing the constraints, and therefore it is high when banks are liquidity constrained

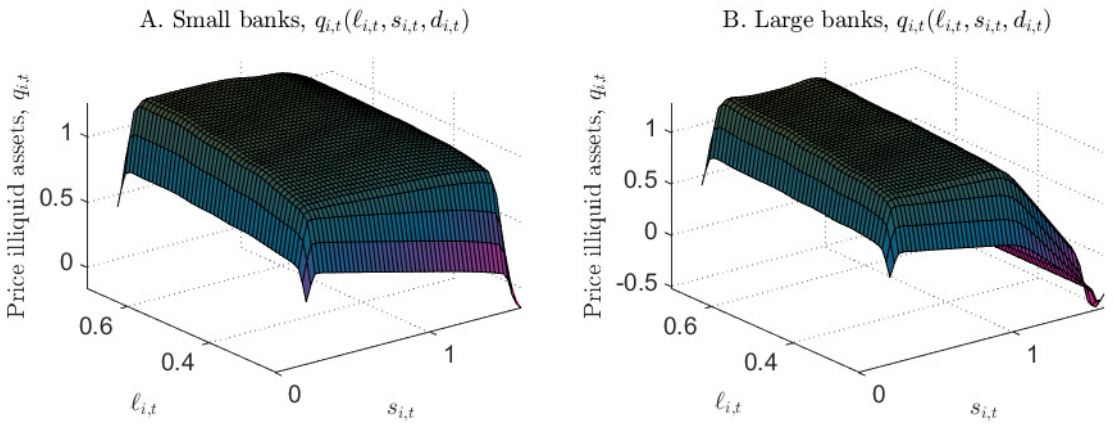


FIG. 10.—Distribution of the price of illiquid assets.

NOTE. The figure shows the cross-sectional distribution of the price of illiquid assets for small banks in panel A and large banks in panel B. Liquid assets are computed in units of deposits.

since they hold less liquid assets and the constraint is binding.

Appendix J.1 reports the distribution of liquidity and illiquid assets, decision rules for liquid assets, illiquid assets, and dividends by bank size. It also shows the distribution of the marginal propensity to lend and its CDF, for different values of the degree of submodularity.

6.3.4 Assortative Matching and Liquidity Regulations

In this Section I report the results for the negative assortative matching in the banking sector. Figure 11 shows the cross-sectional mean values in the deposit space for all banks in the economy and highlights the negative assortative matching. While liquid assets are increasing in deposits; illiquid assets, dividends, and the marginal propensity to lend are strongly decreasing in deposits. More precisely, because of the submodularity in the liquidity constraint, the surplus between banks submodular in their size and of the same type is costly. The negative path in illiquid assets and the marginal propensity to lend is driven mainly by large banks.

Large banks with high illiquid assets have a tight liquidity constraint, which makes them increase liquid assets along the deposit distribution to buffer the interaction between deposits and liquid assets. When liquid assets of large banks are paired with deposits of small banks, the interaction is optimal because large banks can expand liquid assets by relaxing the constraint and use the additional deposits to increase illiquid assets and pay dividends, thus their match generates a strictly dominant surplus. The marginal propensity to lend of large banks is very low and negative in the deposits space, because of the submodularity term in the liquidity constraint which penalizes additional holdings of deposits by decreasing the shadow value of liquidity when large banks match with small ones. Their decision rule for the marginal propensity to lend along the deposit distribution shows a kink toward the middle of deposit holdings, before increasing. The kink demonstrates the submodularity of the constraint, especially for large banks who are more strongly penalised by the submodularity parameter.

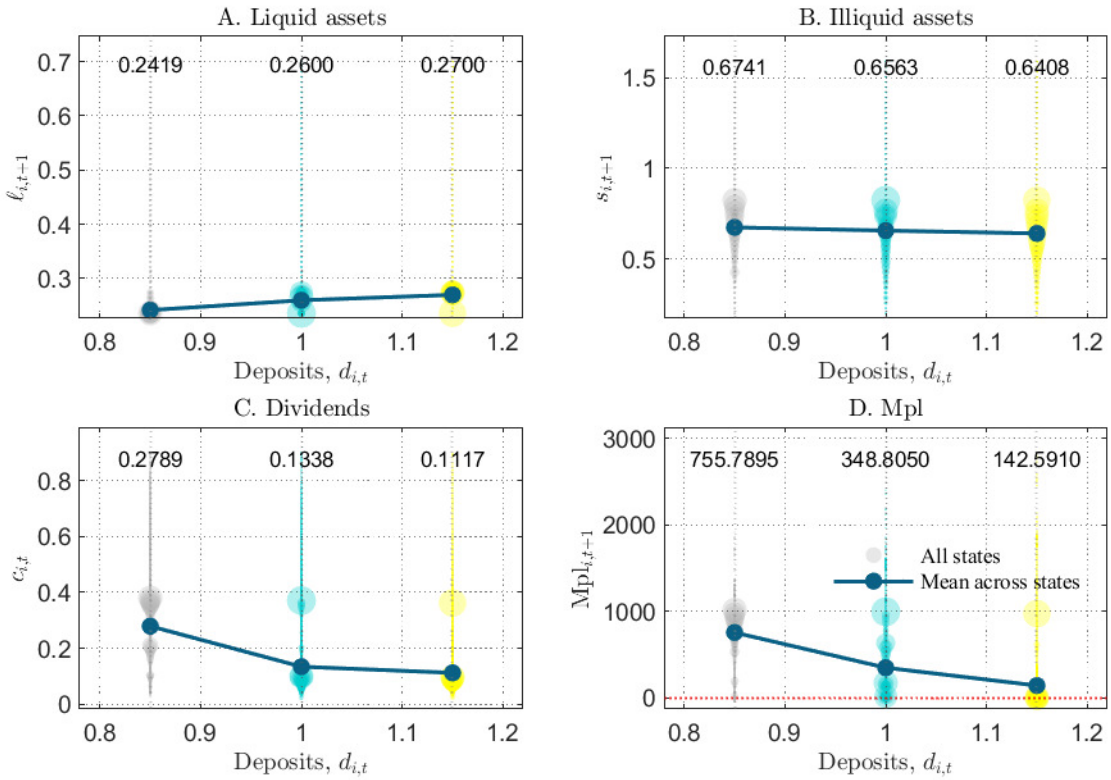


FIG. 11.—Negative assortative matching.

NOTE. The figure shows ergodic means across states for liquid assets, illiquid assets, dividends and the marginal propensity to lend in the space of deposits. Circles shows the states dimensions for each decision rule over the three states of deposits. Numbers on each state denotes mean values of decision rules across states.

Small banks are liquidity constrained along the deposit distribution. This implies that while they increase liquid assets because of self-insurance motive, they decrease dividends. As deposits grow for small banks, the liquid constraint is looser at the margin, hence, they can expand illiquid assets more strongly than large banks because they benefit from the match with large banks, while leading to a positive but decreasing marginal propensity to lend. In the space of low marginal propensity to lend, deposits are mainly used to cover dividends, as the marginal propensity rises, deposits are turned into investment in illiquid assets.

On average, high (low) deposit states are associated with high (low) liquid assets, a low (high) marginal propensity to lend, small (large) expansion in illiquid assets and lower (higher) dividends. The market structure is endogenously modified by the primitives of the model, whereby the submodular term leads large banks to interact with small banks generating a matching surplus that is negative assortative. Appendix J.2 shows the negative assortative matching by banks' size revealing the V-shaped marginal propensity to lend for large banks, while it is almost decreasing and flat for small banks.

The key message of this analysis is that liquidity and leverage regulations in an incomplete

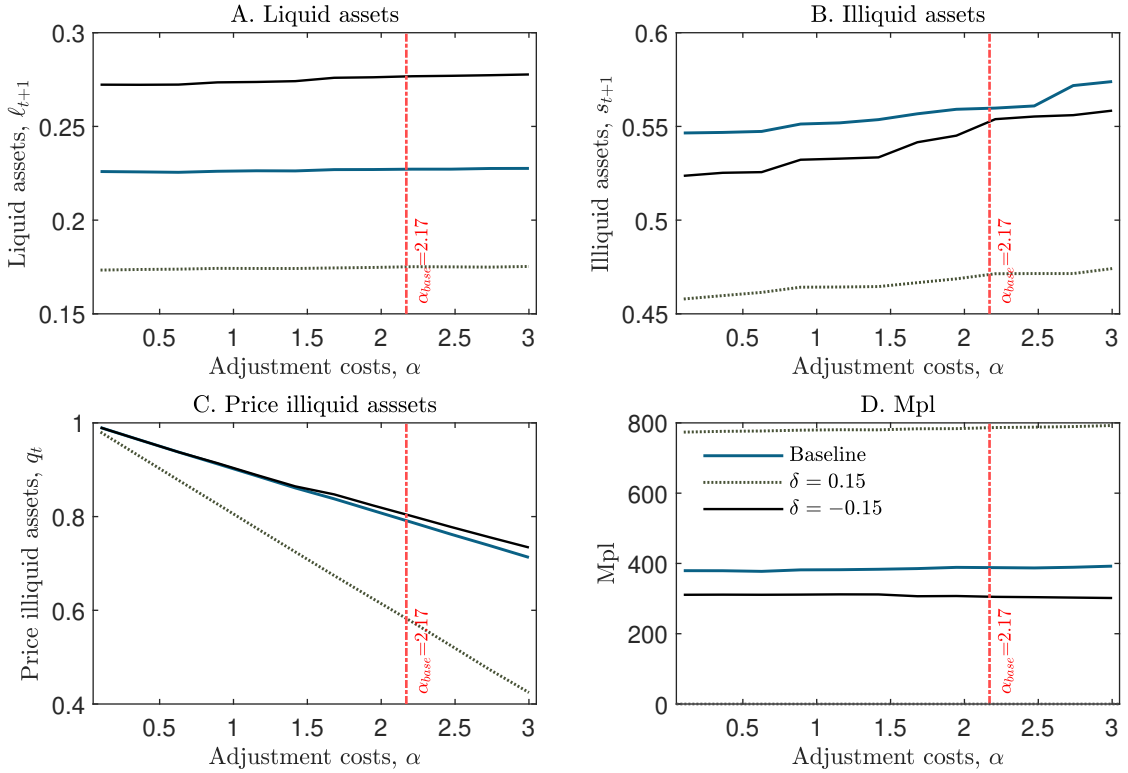


FIG. 12.—Adjustment costs sensitivity.

NOTE. The figure shows the path of liquid assets, illiquid assets, the price of illiquid assets and the marginal propensity to lend along different values of adjustment costs, for the baseline economy (solid blue lines), for the economy with strength of submodularity, $\delta = 0.15$ (dotted black line) and the model with $\delta = -0.15$ (solid black line). Vertical red lines denote the adjustment costs parameter under the baseline calibration.

market economy endogenously determine the market structure of banks, determining the optimal size and the matching between banks.

Appendix J.2 provides additional results of the negative assortative matching by varying the submodularity in liquidity regulations, the degree of leverage constraint and further corroborates the findings investigated in this Section.

6.3.5 Sensitivity to Adjustment Costs in Illiquid Assets

In this Section I study the sensitivity of the model to the adjustment costs parameter, α, since the literature provides a range of values for this parameter.

Figure 12 shows the path of liquid assets (Panel A), illiquid assets (Panel B), the price of illiquid assets (Panel C), and the marginal propensity to lend (Panel D) as a function of adjustment costs. The Figure compares these paths for the baseline economy with the economies that vary the strength of submodularity to above the baseline value, $\delta = 0.15$, in dotted black lines and below the baseline value, $\delta = -0.15$, in solid black lines. The adjustment costs parameter changes the price of illiquid assets and therefore, indicates how costly is to expand illiquid assets

for banks. Although liquid assets are almost invariant to adjustment costs in illiquid assets, the price of illiquid assets is sensitive to adjustment costs and continuously decreases as adjustment costs rise. Because the price of illiquid assets falls, banks expand illiquid assets more intensively, while maintaining a marginal propensity to lend that is minimally affected by adjustment costs.

Decreasing the strength of submodularity makes the liquidity constraint for large banks less tighter, as a result the response of liquid assets is larger in magnitude than the baseline economy, while illiquid assets and the marginal propensity to lend both fall below the baseline since banks must satisfy the liquidity constraint. More precisely, the lower the strength of submodularity, the more liquid assets and illiquid assets become complement each other.

By contrast, with a higher value of the strength of submodularity, the liquidity constraint strongly tightens for large banks because liquid assets are costly in high deposit states, while they are less responsive for small banks; consequently, banks find it difficult to increase liquid assets as adjustment costs rise, whereas they excessively increase their marginal propensity to lend. Effectively, liquid and illiquid assets become substitute.

Appendix J.3 shows additional sensitivity analysis on the distributional responses of adjustment costs to changes in the entry and exit fraction of banks, and the leverage constraint.

6.4 MIT Redistribution of Assets and The Bank Term Funding Program

This Section quantifies the impact of a liquidity programme−The Bank Term Funding Program−on banks' marginal propensity to lend. I envisage a counterfactual experiment in which I examine what the impact of this program would have been on the marginal propensity to lend and other banks' variables had it been introduced in 2010 following the turmoil of the financial crises.

The program is directed at eligible depository institutions that can claim funding in the form of loans in exchange for collateral. Loans are available for one year with the aim of supporting firms and households and assuring them that depository institutions have enough liquidity to satisfy depositor requests. The program was introduced in March 2023 and officially closed on March 11, 2024.[14]

I simulate the Bank Term Funding Program as an unanticipated relaxation of the liquidity constraint where banks have perfect foresight along the transition path that widens the liquidity capacity of banks. In particular, I assume that ϕ falls 15% below its steady state value at time, $t = 1$, and returns to its steady state value after 40 years with a speed of recovery, $\rho_\phi = 0.80$, thus the effects of the shock are meant to stay. Along the transition path, banks know the future path of the price of illiquid assets, leverage and liquidity constraints and therefore, they optimally choose liquid and illiquid assets, dividends and the marginal propensity to lend taking into account these paths. I use this experiment to study the redistribution of assets between banks and across liquid and illiquid assets in banks' portfolios, and the impact on the marginal propensity to lend. Therefore, this redistribution must come through frictions that make liquidity

[14]Further details on the program are available in Appendix K.

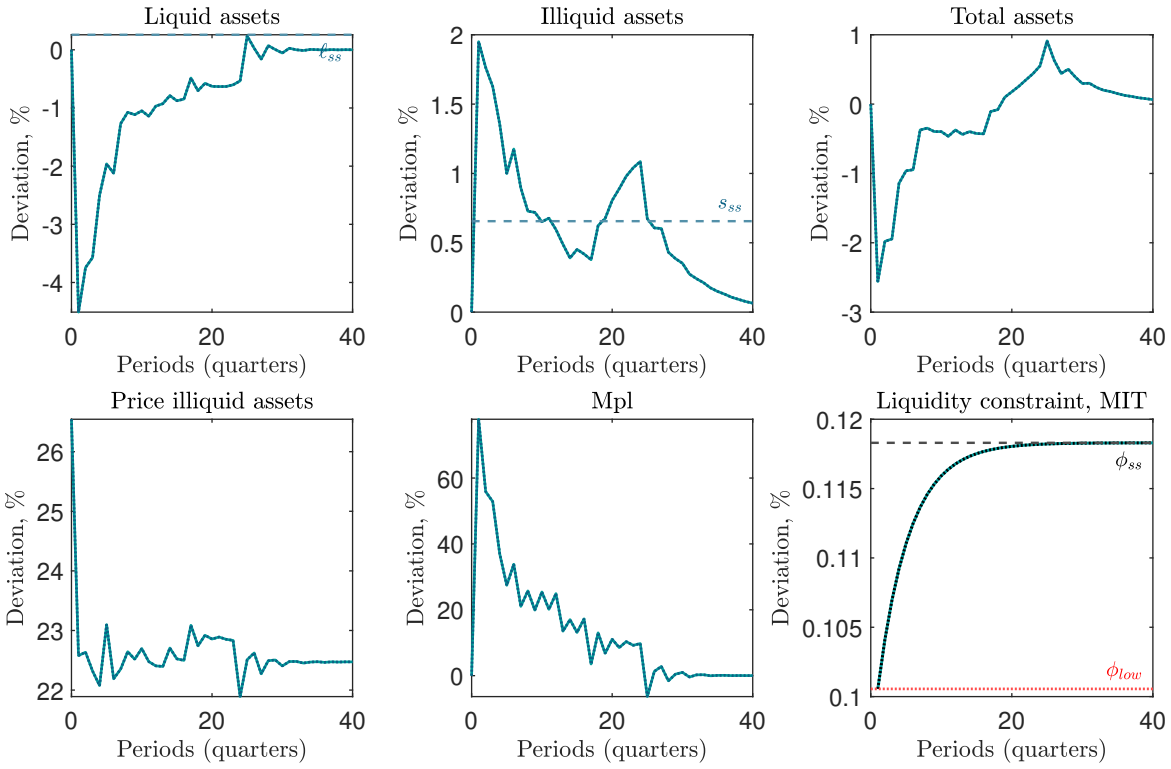


FIG. 13.—MIT shock and the Bank Term Funding Program.

NOTE. The figure shows transition paths to a MIT shock to the liquidity constraint that falls 15% below its steady state value and it is solved along the transition path with geometrically speed of recovery set to $\rho_\phi = 0.80$ in a horizon of 40 periods. The y-axis shows responses in percentage deviation from their steady state values. Blue lines indicate the steady state values of liquid and illiquid assets, while the boundaries of the liquidity constraint are denoted in dashed black line, ϕ_{ss}, and dotted red line, ϕ_{low}.

and illiquid assets to be not perfect substitutes.

Figure 13 shows transition paths for the baseline model economy. Essentially, with a relaxation of the liquidity constraint banks have more liquidity available in their balance sheet for use. Consequently, a fall in the liquidity constraint leads to 4.51% decay in liquid assets on impact, and a simultaneous 1.95% increase in illiquid asset at time $t = 1$ above its steady state value. As more banks invest in illiquid assets, the aggregate amount of illiquid assets in the economy rises. The price of illiquid assets increases on impact, but falls as banks accumulate more illiquid assets because of the inverse demand schedule between illiquid assets and the price of illiquid assets, whereby adjustment costs are incurred and penalise large deviations from the current value of illiquid assets. More importantly, the Bank Term Funding Program envisaged as a 15% relaxation of the liquidity constraint generates a 77.85% expansion in banks' lending by raising the banks' marginal propensity to lend on impact. Because the relaxation of the constraint changes the types of banks constrained along the transition, and the amount of assets that is redistributed between large and small banks, then it delivers kinks in the marginal propensity to lend and a low smooth path for the price of illiquid assets.

As banks become more leverage constrained, the effect of the liquidity constraint slowly

fades away. Banks need to rebuild their liquidity buffer, therefore illiquid assets and the marginal propensity to lend fall and banks reallocate illiquid assets back to liquid assets. Since banks know that the liquidity constraint will return to its steady state value, the contraction in liquid assets drives the contraction in total assets. Banks with the highest marginal propensity to lend are the banks that adjust illiquid assets aggressively which occurs for banks with the highest shadow value of relaxing the liquidity constraint that are small and more constrained banks. Large banks that are more leverage constrained, also increase illiquid assets but less than small banks. Therefore, the shock creates a redistribution of assets between banks. Along the transition path, the liquidity constraint is binding for small banks and less binding for large banks, then it relaxes. This change in the liquidity constraint generates kinks along the transition, importantly for the price of illiquid assets and the marginal propensity to lend which is computed as finite difference derivative.

Appendix J.4 shows robustness results of the MIT experiment to the degree of liquidity regulations, changes in adjustment costs and, entry and exit variations of the fraction of banks.

7 Conclusion

This paper developed a theoretical model of heterogeneous banks with banks' liquid assets, illiquid assets, adjustment costs in illiquid assets, entry and exit and idiosyncratic shocks to short-term debt. I embed an assignment matching game within a dynamic incomplete market economy. The paper places the observed cross-sectional micro banks' heterogeneity at a macro level to jointly study the relation between liquidity regulations and the market structure of banks, the banks' marginal propensity to lend, the redistribution of assets between banks and across banks' assets.

The novel feature of the theory is the combination of the assignment matching game with adjustment costs in illiquid assets and submodular liquidity constraints in the banking sector, based on the idea that a submodular liquidity constraint generates imperfect substitution of banks' assets and renders the match between small banks and large banks the strictly dominant marginal surplus of the match along the size distribution of banks. I show how the model delivers sizable fractions of banks who are constrained along the liquidity or leverage constraint or both, as well as liquid and illiquid banks. Moreover, I show how adjustment costs in illiquid assets affect the equilibrium of the model, by changing the price of illiquid assets, and therefore, banks' responses to liquid and illiquid assets, which are dependent upon the choice of the type of adjustment costs. I characterise the endogeneous marginal propensity to lend which is heterogeneous across banks and I prove that the market structure implied by the primitives of the model is negative assortative matching.

Using microdata on US commercial banks, I document cross-sectional patterns in the banking sector that motivated the distinctive features of the theoretical environment and highlight a

market structure characterised by an equilibrium with unequal distributions between large and small banks. I provide empirical evidence on the marginal propensity to lend that corroborates the theoretical predictions of the model. My estimated results on the marginal propensity to lend shows that this quantity is positive for all banks given a rise in illiquid assets and tend to be larger for banks located in the highest quantiles of the distribution of banks. Furthermore, they show how size heterogeneity and banks' technology drive the assignment matching game in the banking sector, that is shown to be negative assortative matching.

Quantitatively, I find that submodular liquidity constraints and adjustment costs in illiquid assets generate banks' responses to liquid and illiquid assets, and the marginal propensity to lend that are significantly different from absolute liquidity regulations. I then provide estimates for the elasticity of the marginal propensity to lend. Therefore, the model emphasises the importance of accounting for the endogenous interaction between non linear state dependent balance sheet constraints, liquid and illiquid assets in determining the market structure of banks. The paper then examines the impact of the Bank Term Funding Program aimed at providing additional sources of funding to banks. Evaluating the impact of this program I find an important effect on illiquid assets and the marginal propensity to lend, the latter increases with a 77.85% expansion in banks' illiquid assets.

This study is a first examination on how banks' regulations determine the market structure of banks in a context of matching through pair assignments with ex-post heterogeneous banks. While in my model banks pair according to the shape of the surplus, I leave for future research the characterisation of a frictional matching market where banks choose the counterpart they want to match with. Future work that quantifies the cross-sectional banking patterns documented in this paper in macroeconomic models will be valuable to study a number of questions on economic growth and monetary stabilisation along the full distribution of banks.

References

Adrian, T. and Shin, H. S. (2010). Liquidity and Leverage. *Journal of Financial Intermediation*, 19(3):418–437.

Adrian, T. and Shin, H. S. (2014). Procyclical Leverage and Value-at-Risk. *The Review of Financial Studies*, 27(2):373–403.

Agarwal, S., Chomsisengphet, S., Mahoney, N., and Stroebel, J. (2018). Do Banks Pass through Credit Expansions to Consumers Who Want to Borrow? *The Quarterly Journal of Economics*, 133(1):129–190.

Becker, G. S. (1973). A Theory of Marriage: Part I. *Journal of Political Economy*, 81(4):813–846.

Begenau, J. and Landvoigt, T. (2021). Financial Regulation in a Quantitative Model of the Modern Banking System. *Review of Economic Studies*, 0:1–37.

Berger, A. N., DeYoung, R., Flannery, M. J., Lee, D., and Öztekin, Ö. (2008). How Do Large Banking Organizations Manage their Capital Ratios? *Journal of Financial Services Research*, 34(2):123–149.

Berger, A. N. and Hannan, T. H. (1998). The Efficiency Cost of Market Power in the Banking Industry: A Test of the "Quiet Life" and Related Hypotheses. *Review of Economics and Statistics*, 80(3):454–465.

Berger, A. N. and Mester, L. J. (1997). Inside the Black Box: What Explains Differences in the Efficiencies of Financial Institutions? *Journal of Banking & Finance*, 21(7):895–947.

Bernanke, B. S. and Blinder, A. S. (1988). Credit, Money, and Aggregate Demand. *American Economic Review*, pages 435–439.

Bernanke, B. S. and Gertler, M. (1995). Inside the Black Box: the Credit Channel of Monetary Policy Transmission. *Journal of Economic Perspectives*, 9(4):27–48.

Bianchi, J. (2016). Efficient Bailouts? *American Economic Review*, 106(12):3607–59.

Bianchi, J. and Bigio, S. (2022). Banks, Liquidity Management, and Monetary Policy'. *Econometrica*, 90(1):391–454.

Bocola, L. (2016). The Pass-Through of Sovereign Risk. *Journal of Political Economy*, 124(4):879–926.

Boppart, T., Krusell, P., and Mitman, K. (2018). Exploiting MIT Shocks in Heterogeneous-Agent Economies: The Impulse Response as a Numerical Derivative. *Journal of Economic Dynamics and Control*, 89:68–92.

Brunnermeier, M. K. and Sannikov, Y. (2014). A Macroeconomic Model with a Financial Sector. *American Economic Review*, 104(2):379–421.

Cameron, A. and Trivedi, P. K. (2005). Microeconometrics: Methods and Applications. *Cambridge University Press*.

Chiappori, P.-A., Salanié, B., and Weiss, Y. (2017). Partner Choice, Investment in Children, and the Marital College Premium. *American Economic Review*, 107(8):2109–2167.

Coimbra, N. and Rey, H. (2023). Financial cycles with heterogeneous intermediaries. *The Review of Economic Studies*, pages 1–41.

Corbae, D. and D'Erasmo, P. (2021). Capital Buffers in a Quantitative Model of Banking Industry Dynamics. *Econometrica*, 89(6):2975–3023.

Corbae, D. and D'Erasmo, P. (2022). Banking Industry Dynamics Across Time and Space. Working Paper.

Corbae, D. and D'Erasmo, P. (2020). Rising Bank Concentration. *Journal of Economic Dynamics and Control*, 115:103877.

Curdia, V. and Woodford, M. (2010). Credit Spreads and Monetary Policy. *Journal of Money, Credit and Banking*, 42:3–35.

Cúrdia, V. and Woodford, M. (2016). Credit Frictions and Optimal Monetary Policy. *Journal of Monetary Economics*, 84:30–65.

Dempsey, K. (2025). Capital Requirements with Non-Bank Finance. *Review of Economic Studies*, 00:1–36.

Dempsey, K. and Faria-e Castro, M. (2025). A Quantitative Analysis of Bank Lending Relationships. *Journal of Financial Economics*, 170:104083.

Diamond, D. W. and Dybvig, P. H. (1983). Bank Runs, Deposit Insurance, and Liquidity. *Journal of Political Economy*, 91(3):401–419.

Diamond, D. W. and Rajan, R. G. (2001). Liquidity Risk, Liquidity Creation, and Financial Fragility: A Theory of Banking. *Journal of Political Economy*, 109(2):287–327.

Diamond, D. W. and Rajan, R. G. (2005). Liquidity Shortages and Banking Crises. *The Journal of Finance*, 60(2):615–647.

Diamond, D. W. and Rajan, R. G. (2012). Illiquid Banks, Financial Stability, and Interest Rate Policy. *Journal of Political Economy*, 120(3):552–591.

Drechsler, I., Savov, A., and Schnabl, P. (2017). The Deposits Channel of Monetary Policy. *The Quarterly Journal of Economics*, 132(4):1819–1876.

Farhi, E., Golosov, M., and Tsyvinski, A. (2009). A Theory of Liquidity and Regulation of Financial Intermediation. *The Review of Economic Studies*, 76(3):973–992.

Farhi, E. and Tirole, J. (2012). Collective Moral Hazard, Maturity Mismatch, and Systemic Bailouts. *American Economic Review*, 102(1):60–93.

Farhi, E. and Werning, I. (2016). A Theory of Macroprudential Policies in the Presence of Nnominal Rigidities. *Econometrica*, 84(5):1645–1704.

Fernández-Villaverde, J., Hurtado, S., and Nuno, G. (2023). Financial Frictions and the Wealth Distribution. *Econometrica*, 91(3):869–901.

Flannery, M. J. and Rangan, K. P. (2008). What Caused the Bank Capital Build-Up of the 1990s? *Review of Finance*, 12(2):391–429.

Gertler, M. and Karadi, P. (2011). A Model of Unconventional Monetary Policy. *Journal of Monetary Economics*, 58(1):17–34.

Gertler, M. and Kiyotaki, N. (2010). Financial Intermediation and Credit Policy in Business Cycle Analysis. *Handbook of Monetary Economics*, 3:547–599.

Gertler, M., Kiyotaki, N., and Prestipino, A. (2020). A Macroeconomic Model with Financial Panics. *The Review of Economic Studies*, 87(1):240–288.

Gertler, M., Kiyotaki, N., and Queralto, A. (2012). Financial Crises, Bank Risk Exposure and Government Financial Policy. *Journal of Monetary Economics*, 59:S17–S34.

Gigante, C. (2026). Heterogeneous Banks, Liquidity Risk and the Distribution of Banks' Liquidity. Working Paper.

Gowrisankaran, G. (1999). A Dynamic Model of Endogenous Horizontal Mergers. *The RAND Journal of Economics*, pages 56–83.

Gropp, R., Mosk, T., Ongena, S., and Wix, C. (2019). Banks Response to Higher Capital Requirements: Evidence from a Quasi-Natural Experiment. *The Review of Financial Studies*, 32(1):266–299.

He, Z. and Krishnamurthy, A. (2019). A Macroeconomic Framework for Quantifying Systemic Risk. *American Economic Journal: Macroeconomics*, 11(4):1–37.

Heckman, J. (2013). Sample Selection Bias as a Specification Error. *Applied Econometrics*, 31(3):129–137.

Holmström, B. and Tirole, J. (1998). Private and Public Supply of Liquidity. *Journal of Political Economy*, 106(1):1–40.

Jamilov, R. (2020). A Macroeconomic Model with Heterogeneous Banks. Working paper.

Jamilov, R. and Monacelli, T. (2025). Bewley Banks. *Review of Economic Studies*, 00:1–37.

Jiménez, G., Ongena, S., Peydró, J.-L., and Saurina, J. (2012). Credit Supply and Monetary Policy: Identifying the Bank Balance-Sheet Channel with Loan Applications. *American Economic Review*, 102(5):2301–26.

Jokipii, T. and Milne, A. (2011). Bank Capital Buffer and Risk Adjustment Decisions. *Journal of Financial Stability*, 7(3):165–178.

Judd, K. L. and Guu, S.-M. (2001). Asymptotic Methods for Asset Market Equilibrium Analysis. *Economic Theory*, 18(1):127–157.

Justiniano, A., Primiceri, G. E., and Tambalotti, A. (2019). Credit Supply and the Housing Boom. *Journal of Political Economy*, 127(3):1317–1350.

Kashyap, A. K., Rajan, R., and Stein, J. C. (2002). Banks as Liquidity Providers: An Explanation for the Coexistence of Lending and Deposit-Taking. *The Journal of Finance*, 57(1):33–73.

Kashyap, A. K. and Stein, J. C. (1994). Monetary Policy and Bank Lending. In *Monetary Policy*, pages 221–261. The University of Chicago Press.

Kashyap, A. K. and Stein, J. C. (1995). The Impact of Monetary Policy on Bank Balance Sheets. *Carnegie-Rochester Conference Series on Public Policy*, 42:151–195.

Kiyotaki, N. and Moore, J. (1997). Credit Cycles. *Journal of Political Economy*, 105(2):211–248.

Koenker, R. (2005). *Quantile Regression*, volume 38. Cambridge university press.

Koenker, R. and Bassett Jr, G. (1978). Regression Quantiles. *Econometrica*, pages 33–50.

Koenker, R. and Hallock, K. F. (2001). Quantile Regression. *Journal of Economic Perspectives*, 15(4):143–156.

Koenker, R. and Machado, J. A. (1999). Goodness of Fit and Related Inference Processes for Quantile Regression. *Journal of the American Statistical Association*, 94(448):1296–1310.

Mendoza, E. G. (2010). Sudden Stops, Financial Crises, and Leverage. *American Economic Review*, 100(5):1941–66.

Oberfield, E., Rossi-Hansberg, E., Trachter, N., and Wenning, D. (2024). Banks in Space. *NBER Working Paper Series, No. 32256*.

Rıos-Rull, J.-V., Takamura, T., and Terajima, Y. (2020). Banking Dynamics, Market Discipline and Capital Regulations. Working Paper.

Schmitt-Grohé, S. and Uribe, M. (2003). Closing Small Open Economy Models. *Journal of International Economics*, 61(1):163–185.

Shapley, L. S. and Shubik, M. (1971). The Assignment Game I: The Core. *International Journal of Game Theory*, 1(1):111–130.

Tauchen, G. and Hussey, R. (1991). Quadrature-Based Methods for Obtaining Approximate Solutions to Nonlinear Asset Pricing Models. *Econometrica*, pages 371–396.

Tobin, J. (1958). Estimation of Relationships for Limited Dependent Variables. *Econometrica*, pages 24–36.

Wharton Research Data Services (2024). WRDS Bank Regulatory Database. *wrds.wharton.upenn.edu accessed in 2024.*

Appendix for
Illiquid Heterogeneous Banks and Endogenous Market Structure

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A Research Gaps and Relation to the Literature

Table A.1 explains the contribution of the paper to the literature on bank heterogeneity, the research gaps, and the importance of addressing the gaps in my paper.

B Proofs of Propositions

B.1 Proof of Proposition 1

Proof. To prove the proposition I consider the case in which the liquidity constraint is binding and the leverage constraint is slack and there is no entry and exit. I consider this case, since I think of a worst case scenario for bank liquid assets that leads banks towards the liquidity constraint, thereby undertaking precautionary liquidity motive. Therefore in any non stochastic steady state $\beta r_\ell < 1$.

Given that the steady state is one single point along the stationary equilibrium, I derive it by assuming that all banks in the economy are symmetric. As a result, the deposit withdrawal risk that made banks to be heterogeneous is taken to be the average state in any steady state. Therefore, the steady state solves for the aggregate quantities in the economy and constant deposits guarantee existence of a constrained steady state.

In my model, the bank balance sheet contains second order terms that might give rise to risk into the model. It is known that in economies with portfolio allocations, the non stochastic steady state results into being indeterminate since the second order term for illiquid assets disappears. Nonetheless, the non stochastic steady state of this economy is still well defined.[15]

Therefore, I follow the insight of the debt elastic interest rate rule to prove the steady state. In particular, I redefine the interest rate on loans and adopt the following rule specification, $r_{s,t+1} = r_s + \psi(\exp(\overline{s} - s_{t+1}) - 1)$. The steady state model equations dropping time subscript can be written as follows:

$$c + (1 - r_\ell)\ell + (1 - r_s)s - 1 + r_d) = 0, \tag{B.1}$$

$$1 \geq \beta r_\ell - \beta\mu, \tag{B.2}$$

$$-1 = [\beta r_s - \beta\omega + \beta\psi(\exp(\overline{s} - s^*) - 1)], \tag{B.3}$$

$$\ell = \phi, \quad s < \kappa; \tag{B.4}$$

where, equation (B.1) is the steady state budget constraint, equations (B.2) and (B.3) represent the steady state bank optimality conditions for liquid and illiquid assets, respectively, and equation (B.4) contains the liquidity and leverage constraints, respectively. Moreover, $\overline{s}$, in equation

[15] The non stochastic steady state represents the state in which agents are aware that no shocks will happen in the future, in contrast to the stochastic steady state that is the environment in which agents know that a shock may occur, but it never materialises. Therefore, a sequence of null shocks occurs before all model outcomes have converged to their solutions.

TABLE A.1.—Contribution to the literature

Article	Framework	Differences	Research Gap	Importance of Research Gap
Corbae and D'Erasmo (2021)	Develop a model of imperfect competition between big and small banks to study regulation of the 2010 Dodd-Frank Act. The market structure is extrinsically embedded through the sequential Stackelberg competition between large and small banks.	My model is framed in a competitive environment and shows how the market structure between banks arises endogenously generating assortative matching between banks, through submodular regulatory constraints and convex adjustment costs in illiquid assets.	Research to date lacks how banks match between them, how the market structure evolves because of the identified matching pattern and how the distribution of the marginal propensity to lend changes with the market structure. This gap is embedded into the analysis of liquidity regulations.	The regulatory environment defines the industry and the market structure in which banks operate, understanding how banks match between them is important to understand the determination of the aggregate amount of banks' lending.
Bianchi and Bigio (2022)	Develop a general equilibrium framework to study the bank liquidity management in the US. They place particular importance to the liquidity frictions arising in the interbank market for the monetary policy transmission and show that the model delivers ex-post a homogeneous banking sector.	My model contains liquidity frictions in the commercial banking sector, rather than in the interbank market where banks are ex-post heterogeneous.	There is a need to understand the redistribution of assets between banks and reallocation across assets generated by regulatory environment and their consequences to the marginal propensity to lend.	The redistribution and reallocation of assets between banks is the mechanism through which the market structure evolves, that is important to examine the interconnection with financial stability.

NOTE. This table highlights the relationship to the literature of heterogeneous banks, the research gaps and the importance of addressing these gaps.

Contribution to the literature – continued.

Article	Framework	Differences	Research Gap	Importance of Research Gap
Dempsey (2025)	Develop a heterogeneous bank model with two-sided heterogeneity between firms and banks to analyse the rise in banks' capital requirements. The model includes a random matching between firms and banks intrinsic in the model, where banks are of a single type in each period.	My model studies liquidity requirements and includes an extrinsic assignment matching game between banks that influences the market structure.	Research to date has not investigated how the assignment matching game between banks occurs and what does it implies for the redistribution of assets in the system. As banks do not randomly match, it enables me to disentangle the type of the match and how regulation changes the technology banks operate.	In making the market structure endogenous, the assignment matching game preserves the distribution of banks as equilibrium object, and allows to identify which bank moves along the distribution.
Dempsey and Faria-e-Castro (2025)	Develop a heterogeneous banks model to study lending relationships between banks and borrowers where these relationships are modelled with deep habits.	My model contains adjustment costs in illiquid assets conceived as a convex functional form. They are essential for the assignment matching game between banks and the entire equilibrium of liquid and illiquid assets in the evolution of the market structure.	There is a need to analyse why large banks tend to end up with more illiquid assets, how this changes the stability of the market structure, and ultimately how illiquid assets are redistributed across the size distribution of banks.	Examining where illiquid assets are positioned in the distribution of banks, helps to understand which bank drives heterogeneity in the market, and therefore the responsiveness to shock which can be slow or fast. In addition to regulatory constraints, these embed the mechanism where the assignment matching game alters the fixed point of the market structure.
Jamilov and Monacelli (2025)	Develop a general equilibrium model of heterogeneous banks, where banks are heterogeneous in the return on risk to study aggregate fluctuations in the macroeconomy due to a technology shock.	My model features banks that are heterogeneous in their deposits and contains ex-post heterogeneity in deposit holdings. Moreover, banks can trade liquid and illiquid assets.	Research to date has not examined how the marginal propensity to lend distribution varies with illiquid assets and its role in the assignment matching game for the market structure.	The assignment matching game is the driver of the redistribution mechanism of banks' assets and is important to study how individual banks' choices aggregate into the market structure.

NOTE. This table highlights the relationship to the literature of heterogeneous banks, the research gaps and the importance of addressing these gaps.

(B.3) represents the long run mean value of illiquid assets, whereas, s^*, is the steady state level of illiquid assets. Since the leverage constraint holds with an inequality, then $\omega = 0$.

Taking the log in equation (B.3) gives $-\log(\beta r_s) = \log(\beta\psi) + \log(\exp(\overline{s} - s^*) - 1)$. Then, taking the exponential and further taking the log in a second round, leads to a steady state analytical solution for $s_{t+1}(d^t; v_0)$ equals to:

$$s^* = \overline{s} - \log(-\exp[\log(\beta r_s) + \log(\beta\psi)] + 1). \tag{B.5}$$

In a liquidity constrained steady state, $\ell^* = \phi$, and dividends consumption attains the minimum feasible dividends level, $c^* = 1 - r_d - (1 - r_\ell)\ell^* - (1 - r_s)s^*$. ■

B.2 Proof of Proposition 2

Proof. To prove the Proposition the optimal condition for illiquid assets under partial adjutsment costs solves the following maximisation problem:

$$s^*_{i,t+1}(d^t; v_0) = \underset{s_{i,t+1}\in S}{\arg\max} \left\{ u(c_{i,t}(d^{t-1}; v_0)) + \beta(1-\delta_e) \sum_{d^t\in\mathcal{D}^t} \mu^t(v_0, d^t)\mathcal{V}(\tilde{\ell}_{i,t+1}, s_{i,t+1}, d_{i,t+1}, d_{i,t}) \right\} \tag{B.6}$$

that yields:

$$u'(c_{i,t}(d^{t-1}; v_0)) + \alpha u'(c_{i,t}(d^{t-1}; v_0))(s_{i,t+1} - \overline{s}) = \beta(1-\delta_e) \sum_{d^t\in\mathcal{D}^t} \mu^t(v_0, d^t) \left[r_{s,t+1}(d_{i,t+1}) u'(c_{i,t+1}(d^t; v_0)) - \frac{\omega_{i,t}(d^{t-1}; v_0)}{d_{i,t+1}} \right]. \tag{B.7}$$

The optimality condition (B.7) makes clear that, when banks invest one more unit of illiquid assets then, while current dividends increase by one unit plus the additional marginal cost, then $\beta(1-\delta_e)\sum_{d^t\in\mathcal{D}^t}\mu^t(v_0, d^t)\left[u'(c_{i,t+1}(d^t; v_0))r_{s,t+1}(d_{i,t+1}) - \frac{\omega_{i,t}(d^{t-1};v_0)}{d_{i,t+1}}\right]$, that is the expected marginal benefit of one additional unit of illiquid assets, is now independent on the future adjustment cost term. This increases the quantity of illiquid assets, all else equal.

Solving equation (B.7) for illiquid assets $s_{i,t+1}$, it yields:

$$s^*_{i,t+1}(d^t; v_0) = \frac{1}{\alpha} \left\{ \beta(1-\delta_e) \frac{1}{u'(c_{i,t}(d^{t-1}; v_0))} \sum_{d^t\in\mathcal{D}^t} \mu^t(v_0, d^t) \left[r_{s,t+1}(d_{i,t+1}) u'(c_{i,t+1}(d^t; v_0)) - \frac{\omega_{i,t}(d^{t-1}; v_0)}{d_{i,t+1}} \right] - 1 \right\} + \overline{s}. \tag{B.8}$$

Therefore, when $\alpha \Longrightarrow \infty$ there is a closed form solution for banks' illiquid assets choice such that $s^*_{i,t+1}(d^t; v_0) = \overline{s}$. This solution provides a lower bound for the illiquid assets choice set. By contrast, when $\alpha = 0$, the optimal illiquid assets solution $s^*_{i,t+1}(d^t; v_0)$ becomes unbounded and the system cannot admit an equilibrium. ■

B.3 Proof of Proposition 3

Proof. From proposition 2 it was possible to derive the lower bound for the illiquid assets set. This equals $s_{min} = \overline{s}$. To derive the upper bound in the set for illiquid assets, it needs to be ruled out the possibility of holding no liquidity in banks' balance sheet. Therefore, the maximum solution for the upper bound of illiquid assets solves the following bank plan:

$$s_{max} = \underset{s_{i,t+1}}{\arg\max} \left\{ u(c_{i,t}(d^{t-1}, v_0)) + \beta(1-\delta_e) \sum_{d^t \in \mathcal{D}^t} \mu^t(v_0, d^t) \mathcal{V}(\ell_{i,t+1}(d^t, v_0) = 0, s_{i,t+1}, d_{i,t+1}, d_{i,t}) \right\}. \tag{B.9}$$

The solution for s_{max} in equation (B.9) guarantees the existence of the maximum attainable level of banks' illiquid assets if banks had no other assets in their balance sheet.

There cannot be an equilibrium in which $s_{i,t+1}(d^t, v_0) = 0$ because banks make negative profits since illiquid assets is the asset with the highest return. Therefore, the illiquid assets set is non empty and is bounded from below and from above, $S \subseteq [s_{min}, s_{max}]$. ■

B.4 Proof of Proposition 4

Proof. The optimality condition for illiquid assets according to the adjustment cost function can be written as follows:

$$\begin{aligned} &-u'(c_{i,t}(d^{t-1}, v_0)) - \alpha u'(c_{i,t}(d^{t-1}, v_0))(s_{i,t+1} - s_{i,t}) \\ &+ \beta(1-\delta_e) \sum_{d^t \in \mathcal{D}^t} \mu^t(v_0, d^t) \left\{ u'(c_{i,t+1}(d^t, v_0))[r_{s,t+1}(d_{i,t+1}) + \alpha(s_{i,t+2} - s_{i,t+1})] - \frac{\omega_{i,t}(d^{t-1}, v_0)}{d_{i,t+1}} \right\} = 0. \end{aligned} \tag{B.10}$$

Furthermore, I characterise equation (B.10) in terms of the price of illiquid assets that yields:

$$\begin{aligned} &q_{i,t}(s_{i,t+1}, s_{i,t}) = \\ &\beta(1-\delta_e) \sum_{d^t \in \mathcal{D}^t} \mu^t(v_0, d^t) \left\{ \frac{u'(c_{i,t+1}(d^t, v_0))}{u'(c_{i,t}(d^{t-1}, v_0))} (r_{s,t+1}(d_{i,t+1}) + q_{i,t+1}(s_{i,t+2}, s_{i,t+1}) - 1) \right. \\ &\left. - \frac{\omega_{i,t}(d^{t-1}, v_0)}{d_{i,t+1}} \frac{1}{u'(c_{i,t}(d^{t-1}, v_0))} \right\}. \end{aligned} \tag{B.11}$$

I define the marginal propensity to lend at the individual bank level as the change in illiquid assets given a change in the interest rate on loans. This quantity can be computed by considering first the change in the price of illiquid assets given a change in the interest rate on loans, all else equal. That is:

$$\frac{\partial q_{i,t+1}}{\partial r_{s,t+1}} = \beta(1-\delta_e) \sum_{d^t \in \mathcal{D}^t} \mu^t(v_0, d^t) \left[\underbrace{\frac{u''(c_{i,t+1})}{u''(c_{i,t})} \frac{\partial s_{i,t+1}}{\partial r_{s,t+1}}}_{\text{redistribution effect}} \left(1 + \underbrace{\frac{\partial q_{i,t+1}}{\partial q_{i,t}} \frac{\partial q_{i,t}}{\partial s_{i,t+1}}}_{\text{price effect}} \underbrace{\frac{\partial s_{i,t+1}}{\partial r_{s,t+1}}}_{\text{mpl}} \right) - \underbrace{u''(c_{i,t}) \frac{\partial \left(\frac{\omega_{i,t}}{d_{i,t+1}} \right)}{\partial s_{i,t+1}} \frac{\partial s_{i,t+1}}{\partial r_{s,t+1}}}_{\text{shadow price effect}} \right], \tag{B.12}$$

where the first term is the redistribution effect whereby increasing the interest rate on loans by one unit raises the loan supply of banks, and substitute away dividends and $r_{s,t+1}$ is evaluated at $r_{s,t+1}(d_{i,t+1})$. The effect is positive because of concavity of bankers' utility function and is larger for large banks. Moreover, illiquid assets redistribute from large banks to small banks. Therefore, there is a redistribution across assets and between banks. The second term is the price effect that is the marginal change in price from extending one extra unit of loans. This effect is positive for lending banks, but it is negative for borrowing banks.

The third term is the marginal propensity to lend, where a higher interest rate on loans increases loans supply. The last term is the shadow price effect that captures the financial frictions of the leverage constraint. This is the unitary change in the Lagrangian multiplier given a unitary change in loans supply multiplied by the individual marginal propensity to lend. The shadow price effect increases banks' net worth through the increase of the price of a loan.

Therefore, as demonstrated in equation (B.12) the marginal propensity to lend is endogenous and since the equation holds for each bank, then the marginal propensity to lend is heterogeneous across banks and varies with banks' size and the price of loans.

The second part of the proof follows by inspection of the leverage constraint, $\frac{s_{i,t+1}}{d_{i,t+1}} \leq \kappa$, and the liquidity constraint. With high deposits the leverage constraint is slack, $\omega_{i,t} = 0$. An increase in deposits makes small banks more responsive to change in $r_{s,t+1}$ with an incentive to expand illiquid assets. However, with more deposits the liquidity constraint is tighter, $\mu_{i,t} > 0$, because of the submodularity term. Thus, while small banks can fund more illiquid assets with more deposits, their incentive to expand illiquid assets following a contemporaneous change in the interest rate on loans falls.

For large banks, the constraint set changes. When deposits are low, the leverage constraint is tight, $\omega_{i,t} > 0$ and banks have incentives to reduce illiquid assets in response to a change in the interest rate on loans. When the constraint stop binding, the marginal propensity to lend can be negative and falls with deposits. At this point the marginal propensity to lend changes direction of movement. As deposits rise, the constraint does not bind anymore, $\omega_{i,t} = 0$, and high deposits make large banks more responsive to changes in the interest rate on loans. Thus, for large banks a V-shaped marginal propensity to lend occurs. ■

B.5 Proof of Proposition 5

Proof. To prove the Proposition consider an ordered set of realised deposits for small banks $d_{1i,t+1} \leq d_{2i,t+1} \leq ... \leq d_{mm,t+1}$ and large banks $d_{1j,t+1} \leq d_{2j,t+1} \leq ... \leq d_{pp,t+1}$. Taking the first order condition with respect to liquid assets yields for small and large banks, respectively:

$$-u'(c_{i,t}(d^{t-1}; v_0)) + \beta(1-\delta_e) \sum_{d^t \in \mathcal{D}^t} \mu^t(v_0, d^t)[r_\ell u'(c_{i,t+1}(d^t; v_0))] - \mu_{i,t}(d^{t-1}; v_0)\frac{1}{d_{i,t}} = 0, \quad \text{(B.13)}$$

$$-u'(c_{j,t}(d^{t-1};v_0))+\beta(1-\delta_e)\sum_{d^t\in\mathcal{D}^t}\mu^t(v_0,d^t)[r_\ell u'(c_{j,t+1}(d^t;v_0))]-\mu_{j,t}(d^{t-1};v_0)\frac{1}{d_{j,t}}=0. \quad \text{(B.14)}$$

Consider now the envelope terms for both types of banks: $\frac{\partial v_{\ell^*,s_{min},\overline{d}}(\tilde{\ell}_{i,t},s_{i,t},d_{i,t},d_{i,t-1})}{\partial \ell_{i,t+1}}=-\mu_{i,t}(d^{t-1};v_0)\frac{1}{d_{i,t}}<0$ and $\frac{\partial v_{\overline{\ell},s_{max},\underline{d}}(\ell_{j,t},s_{j,t},d_{j,t},d_{j,t-1})}{\partial \ell_{j,t+1}}=-\mu_{j,t}(d^{t-1};v_0)\frac{1}{d_{j,t}}<0$. Regardless of banks' size, one extra unit of liquid assets relaxes the constraint, and its marginal value is greater the lower the deposits. In particular, take any two banks such that $d_{j,t}<d_{i,t}$, then in absolute values it holds:

$$\left|\frac{\partial v_{\overline{\ell},s_{max},\underline{d}}(\tilde{\ell}_{j,t},s_{j,t},d_{j,t},d_{j,t-1})}{\partial \ell_{j,t+1}}\right|=\mu_{j,t}(d^{t-1};v_0)\frac{1}{d_{j,t}}>\mu_{i,t}(d^{t-1};v_0)\frac{1}{d_{i,t}}=\left|\frac{\partial v_{\ell^*,s_{min},\overline{d}}(\tilde{\ell}_{i,t},s_{i,t},d_{i,t},d_{i,t-1})}{\partial \ell_{i,t+1}}\right|, \quad \text{(B.15)}$$

therefore, large banks with small deposits have high marginal value of liquidity and viceversa.

Lemma 1 *Within each type of banks, the marginal value of liquidity is decreasing in deposits.*

The proof of Lemma 1 follows from the proof of Proposition 5 by taking any two small banks $(i,i')\in M$, and any two large banks $(j,j')\in P$ such that $d_{i',t}<d_{i,t}$ and $d_{j,t}<d_{j',t}$. Therefore, the following holds:

$$\left|\frac{\partial v(\tilde{\ell}_{i',t},s_{i',t},d_{i',t},d_{i',t-1})}{\partial \ell_{i',t+1}}\right|=\mu_{i',t}(d^{t-1};v_0)\frac{1}{d_{i',t}}>\mu_{i,t}(d^{t-1};v_0)\frac{1}{d_{i,t}}=\left|\frac{\partial v(\tilde{\ell}_{i,t},s_{i,t},d_{i,t},d_{i,t-1})}{\partial \ell_{i,t+1}}\right|, \quad \text{(B.16)}$$

$$\left|\frac{\partial v(\tilde{\ell}_{j,t},s_{j,t},d_{j,t},d_{j,t-1})}{\partial \ell_{j,t+1}}\right|=\mu_{j,t}(d^{t-1};v_0)\frac{1}{d_{j,t}}>\mu_{j',t}(d^{t-1};v_0)\frac{1}{d_{j',t}}=\left|\frac{\partial v(\tilde{\ell}_{j',t},s_{j',t},d_{j',t},d_{j',t-1})}{\partial \ell_{j',t+1}}\right|, \quad \text{(B.17)}$$

where equation (B.16) and (B.17) apply within small banks and large banks, respectively.

Consequently, the marginal value of liquidity is decreasing in deposits not only between banks but also within each type of banks. ■

B.6 Proof of Proposition 6

Proof. Assume an interior optimum exists. Let $\ell^{**}\in L$ be the choice of amount of liquidity that is transferred from one bank to another bank. Consider the value of a small and large bank in the presence of this additional liquidity $v_{\ell^*,s_{min},\overline{d}}(\tilde{\ell}_{i,t+1}-\ell^{**},s_{i,t+1},d_{i,t+1},d_i)$ and $v_{\overline{\ell},s_{max},\underline{d}}(\tilde{\ell}_{j,t+1}+\ell^{**},s_{j,t+1},d_{j,t+1},d_{j,t})$, respectively. Define the joint value of a bank pair within a period as $w_{ij,t}(\ell^{**})=v_{\ell^*,s_{min},\overline{d}}(\tilde{\ell}_{i,t+1}-\ell^{**},s_{i,t+1},d_{i,t+1},d_i)+v_{\overline{\ell},s_{max},\underline{d}}(\tilde{\ell}_{j,t+1}+\ell^{**},s_{j,t+1},d_{j,t+1},d_{j,t})$. Taking the first order condition with respect to ℓ^{**} one obtains: $\frac{\partial w_{ij,t}}{\partial \ell^{**}}=-\frac{\partial v_{\ell^*,s_{min},\overline{d}}(\tilde{\ell}_{i,t+1}-\ell^{**},s_{i,t+1},d_{i,t+1},d_i)}{\partial \ell_{i,t+1}}+\frac{\partial v_{\overline{\ell},s_{max},\underline{d}}(\tilde{\ell}_{j,t+1}+\ell^{**},s_{j,t+1},d_{j,t+1},d_{j,t})}{\partial \ell_{j,t+1}}=0$. Intuitively, consider a small increase in ℓ^{**}, such that $\Delta\ell^{**}>0$. Then $\Delta w_{ij,t}(\ell^{**})\approx\left(\frac{\partial v_{\overline{\ell},s_{max},\underline{d}}(\tilde{\ell}_{j,t+1}+\ell^{**},s_{j,t+1},d_{j,t+1},d_{j,t})}{\partial \ell_{j,t+1}}-\frac{\partial v_{\ell^*,s_{min},\overline{d}}(\tilde{\ell}_{i,t+1}-\ell^{**},s_{i,t+1},d_{i,t+1},d_i)}{\partial \ell_{i,t+1}}\right)\Delta\ell^{**}$. When $\frac{\partial v_{\overline{\ell},s_{max},\underline{d}}(\tilde{\ell}_{j,t+1}+\ell^{**},s_{j,t+1},d_{j,t+1},d_{j,t})}{\partial \ell_{j,t+1}}>\frac{\partial v_{\ell^*,s_{min},\overline{d}}(\tilde{\ell}_{i,t+1}-\ell^{**},s_{i,t+1},d_{i,t+1},d_i)}{\partial \ell_{i,t+1}}$ then $\Delta w_{ij,t}(\ell^{**})>0$ and the optimal response is to increase ℓ^{**}, that is sending liquid assets to large banks, and viceversa. Because any small change in ℓ^{**} can effectively improve upon the coalition,

equalising the marginal values blocks any further coalition improvements. Therefore, denoting $v_i^{\ell}(d_{i,t}) = \frac{\partial v_{\ell^*,s_{min},\overline{d}}(\tilde{\ell}_{i,t+1}-\ell^{**},s_{i,t+1},d_{i,t+1},d_i)}{\partial \ell_{i,t+1}}$ and $v_j^{\ell}(d_{j,t}) = \frac{\partial v_{\overline{\ell},s_{max},\underline{d}}(\tilde{\ell}_{j,t+1}+\ell^{**},s_{j,t+1},d_{j,t+1},d_{j,t})}{\partial \ell_{j,t+1}}$, then $v_i^{\ell}(d_{i,t}) = v_j^{\ell}(d_{j,t})$. Banks trade liquid assets ℓ^{**} within a period in the interbank market until the marginal values are equalised and $W_t(\sigma_{i,t},\sigma_{j,t}) \geq (v_{\ell^*,s_{min},\overline{d}}(\sigma_{i,t}) + v_{\overline{\ell},s_{max},\underline{d}}(\sigma_{i,t}))$. ■

B.7 Proof of Proposition 7

Proof. The proof of the Proposition follows by considering the characteristics of the model environment. Recall that small banks have higher deposits than large banks who hold a higher marginal value of liquidity $v_j^{\ell}(d_{j,t})$. Consider any two banks $(i,i') \in M$ and $(j,j') \in P$ such that $d_{i',t} < d_{i,t}$ and $d_{j,t} < d_{j',t}$ where the following cross match pairs can occur (i,j), (i',j'), (i,j'), (i',j). I then use condition of submodularity in Becker (1973), however on banks' surplus. Then, the following holds:

$$S_{ij}(d_{i,t},d_{j,t}) + S_{i'j'}(d_{i',t},d_{j',t}) \leq S_{ij'}(d_{i,t},d_{j',t}) + S_{i'j}(d_{i',t},d_{j,t}). \tag{B.18}$$

Since as shown from Proposition 5 and Lemma 1 banks have decreasing marginal value of liquidity, then the joint value of the match $W_t(\sigma_{i,t},\sigma_{j,t})$ retains the same property across the four pairs of matches. Therefore, the cross partial derivative in deposits of the joint surplus is also negative for any of the cross match pairs considered, $\frac{\partial^2 S_{ij,t}(d_{i,t},d_{j,t})}{\partial d_{i,t}\partial d_{j,t}} \leq 0$. ■

B.8 Proof of Proposition 8

Proof. To prove the Proposition consider any banks $(i,i') \in M$ and $(j,j') \in P$ with deposits such that $i < i'$, and $j < j'$ where $x_{ij=1}$, $x_{i'j'=1}$. Then, consider a new banks' matching x' such that $x'_{ij} = 0$, $x'_{i'j'} = 0$, $x'_{ij'} = 1$ and $x'_{i'j} = 0$, where I have dropped time subscript to simplify notation. Intuitively, under the x matching small banks (i,i') are matched with one large bank, and conversely large banks (j,j') with one small bank. Under the x' matching a small bank i is matched with the least largest bank j' and, the least smallest bank i' is matched with the largest bank j. By computing the difference in total surpluses between banks' matching I obtain:

$$\begin{aligned} v(x') - v(x) = &\sum_{fk} S_{fk}(x'_{fk} - x_{fk}) = \\ &S_{ij}(x'_{ij} - x_{ij}) + S_{i'j'}(x'_{i'j'-x_{i'j'}}) + S_{ij'}(x'_{ij'} - x_{ij'}) + S_{i'j}(x'_{i'j} - x_{i'j}) + \\ &\sum_{(fk)\notin\{(ij),(i'j'),(ij'),(i'j)\}} S_{fk}(x'_{fk} - x_{fk}). \end{aligned} \tag{B.19}$$

Since it holds that $x'_{fk} = x_{fk}$, then the pairs (fk) do not contribute to total difference in surplus. Consequently, equation (B.19) can be simplified to:

$$\begin{aligned} v(x') - v(x) &= S_{ij}(-1) + S_{i'j'}(-1) + S_{ij'}(1) + S_{i'j}(1) \\ &= (S_{ij'} + S_{i'j}) - (S_{ij} + S_{i'j'}). \end{aligned} \tag{B.20}$$

Because the surplus is submodular in deposits from Proposition 7, then $S_{ij'} + S_{i'j} \geq S_{ij} + S_{i'j'}$, therefore $v(x') - v(x) \geq 0$. The banks pairs (i, j') and (i', j) generate a greater surplus than the pairs (i, j) and (i', j').

I now need to prove the existence of a matching that is negative assortative. Recall that a matching is negative assortative if there does not exist a coalition that improves upon the cross partial, namely, there must not exist matches where $x_{ij} = 1$ and $x_{i'j'} = 1$. Suppose there exists x^*_{ij} that maximises total surplus as negative assortative matching. Take any matching $\tilde{x}$ such that $v(\tilde{x}) = \max_x v(x)$. We need to verify that $\tilde{x}$ is negative assortative. If not, there exist $i < i'$ and $j < j'$ such that $\tilde{x}_{ij} = 1$, $\tilde{x}_{i'j'} = 1$ at the matching $\tilde{x}'$, for which $v(\tilde{x}') \geq v(\tilde{x})$. Because $\tilde{x}$ is surplus maximising then $v(\tilde{x}') = v(\tilde{x})$. However, if $\tilde{x}'$ contains matching coalition improvements then a new matching $\tilde{x}''$ can be found such that $v(\tilde{x}'') \geq v(\tilde{x}') = v(\tilde{x})$. Because the number of banks' coalitions is finite at N, this process will stop at a matching x^{**} where $v(x^{**}) = v(x^*)$ and no matching $x^{**}_{ij} = x^{**}_{i'j'} = 1$ exists. Therefore, x^{**}_{ij} is negative assortative and attains the maximal surplus. From Proposition (7) the total surplus, that is submodular in deposits, is optimal. Therefore, a match with banks of different types brings more to the surplus than a match with banks of the same type. The gains from matching with banks of different types are larger because the marginal value of increasing liquidity within the same type are smaller than the marginal value between different types of banks. ■

B.9 Proof of Proposition 9

Proof. Consider the minimum and maximum deposit realisations $\{\overline{d}, \underline{d}\} \in D$ with the deposit set being an ordered set $d_{1,t+1} \leq d_{2,t+1} \leq ... \leq d_{m,t+1}$. Recall banks' budget constraint:

$$\begin{aligned} &c_{i,t}(d^t; v_0) + \ell_{i,t+1}(d^t; v_0) + s_{i,t+1}(d^t; v_0) + \eta(s_{i,t+1}, s_{i,t}) - d_{i,t+1} = \\ &\quad r_\ell \ell_{i,t}(d^{t-1}; v_0) + r_{s,t}(d_{i,t}) s_{i,t}(d^{t-1}; v_0) - r_d d_{i,t} + (r_\ell - r_d)(d_{i,t+1} - d_{i,t}). \end{aligned} \tag{B.21}$$

We want to show that the model contains an endogenous matching market of liquidity that arises from the transfer of $(d_{i,t+1} - d_{i,t}) > 0$ units of deposits. This means we need to show that the model delivers the optimal assignment matching game as an endogenous outcome that is negative assortative and delivers $(S_{ij'}(d_i, d_{j'}) + S_{i'j}(d_{i'}, d_j)) \geq (S_{ij}(d_i, d_j) + S_{i'j'}(d_{i'}, d_{j'}))$.

The proof follows a similar reasoning in Gigante (2026) in Proposition 9, with the difference that now banks are subject to a leverage constraint and the choice of illiquid assets is influenced by adjustment costs and entry and exit. Suppose that the smallest banks are defined

as $\mathcal{V}_{\ell^*,\underline{s},\overline{d},d_{i,t+1}}(\ell_{i,t},s_{i,t},d_{i,t+1},d_{i,t})$, the least small banks as $\mathcal{V}_{\ell^*,\underline{s},\underline{d},d_{i,t+1}}(\ell_{i,t},s_{i,t},d_{i,t+1},d_{i,t})$, the largest banks as $\mathcal{V}_{\overline{\ell},\overline{s},\underline{d},d_{i,t+1}}(\ell_{i,t},s_{i,t},d_{i,t+1},d_{i,t})$ and the least large banks as $\mathcal{V}_{\overline{\ell},\overline{s},\overline{d},d_{i,t+1}}(\ell_{i,t},s_{i,t},d_{i,t+1},d_{i,t})$.

Case 1. Take the smallest banks $\mathcal{V}_{\ell^*,\underline{s},\overline{d},d_{i,t+1}}(\ell_{i,t},s_{i,t},d_{i,t+1},d_{i,t})$ and assume that $d_{i,t+1}=\underline{d}$. Then, by evaluating banks' budget constraint gives $c_{i,t}+\ell_{i,t+1}+s_{i,t+1}(d^t;v_0)+\eta(s_{i,t+1},s_{i,t})-d_{i,t+1}=r_\ell\ell^*+r_{s,t}(d_{i,t})\underline{s}-r_d\overline{d}+\underbrace{(r_\ell-r_d)}_{<0}\underbrace{(d_{i,t+1}-\overline{d})}_{<0}$. Banks experience an inflow of deposits, the liquidity constraint $\ell_{i,t+1}\geq\phi\overline{d}-(d_{i,t+1}-\overline{d})+\delta\ell^*\overline{d}$ tightens, and the leverage constraint $s_{i,t+1}\leq\kappa\underline{d}$ is more likely to bind. Now consider $d_{i,t+1}=\overline{d}$. The inflow/outflow effect in banks' budget constraint does not arise any more, the liquidity constraint reduces to $\ell_{i,t+1}\geq\phi\overline{d}+\delta\ell^*\overline{d}$ while the leverage constraint $s_{i,t+1}\leq\kappa\overline{d}$ is less likely to bind.

Case 2. Take the largest banks $\mathcal{V}_{\overline{\ell},\overline{s},\underline{d},d_{i,t+1}}(\ell_{i,t},s_{i,t},d_{i,t+1},d_{i,t})$ and assume that $d_{i,t+1}=\overline{d}$. Then, their budget constraint $c_{i,t}+\ell_{i,t+1}+s_{i,t+1}(d^t;v_0)+\eta(s_{i,t+1},s_{i,t})-d_{i,t+1}=r_\ell\overline{\ell}+r_{s,t}(d_{i,t})\overline{s}-r_d\underline{d}+\underbrace{(r_\ell-r_d)}_{<0}\underbrace{(d_{i,t+1}-\underline{d})}_{>0}$ obtains an outflow of deposits, where the liquidity constraint $\ell_{i,t+1}\geq\phi\underline{d}-(d_{i,t+1}+\underline{d})-\delta\overline{\ell}\underline{d}$ and the leverage constraint $\frac{s_{i,t+1}}{\overline{d}}\leq\kappa$ are less binding. Assume now that $d_{i,t+1}=\underline{d}$. In this case, the inefficiency arising from the inflow/outflow effect disappears in the banks' budget constraint, $c_{i,t}+\ell_{i,t+1}+s_{i,t+1}(d^t;v_0)+\eta(s_{i,t+1},s_{i,t})-d_{i,t+1}=r_\ell\overline{\ell}+r_{s,t}(d_{i,t})\overline{s}-r_d\underline{d}$ and in the liquidity constraint, $\ell_{i,t+1}\geq\phi\underline{d}-\delta\overline{\ell}\underline{d}$, whereas the leverage constraint is tighter.

Case 3. Take the least large banks $\mathcal{V}_{\overline{\ell},\overline{s},\overline{d},d_{i,t+1}}(\ell_{i,t},s_{i,t},d_{i,t+1},d_{i,t})$ and assume that $d_{i,t+1}=\underline{d}$. In this case the banks' budget constraint becomes $c_{i,t}+\ell_{i,t+1}+s_{i,t+1}(d^t;v_0)+\eta(s_{i,t+1},s_{i,t})-d_{i,t+1}=r_\ell\overline{\ell}+r_{s,t}(d_{i,t})\overline{s}-r_d\overline{d}+\underbrace{(r_\ell-r_d)}_{<0}\underbrace{(d_{i,t+1}-\overline{d})}_{<0}$. Banks experience an inflow of deposits, where the liquidity constraint $\ell_{i,t+1}\geq\phi\overline{d}-(d_{i,t+1}-\overline{d})+\delta\overline{\ell d}$ and the leverage constraint $\frac{s_{i,t+1}}{d_{i,t+1}}\leq\kappa$ are tight. In the event $d_{i,t+1}=\overline{d}$, the inflow/outflow effect does not arise in the banks' budget constraint and the liquidity constraint, and effectively banks cannot expand in size through liquid assets.

Case 4. Take the least small banks $\mathcal{V}_{\ell^*,\underline{s},\underline{d},d_{i,t+1}}(\ell_{i,t},s_{i,t},d_{i,t+1},d_{i,t})$ and assume that $d_{i,t+1}=\overline{d}$. The bank budget constraint becomes $c_{i,t}+\ell_{i,t+1}+s_{i,t+1}(d^t;v_0)+\eta(s_{i,t+1},s_{i,t})-d_{i,t+1}=r_\ell\ell^*+r_{s,t}(d_{i,t})\underline{s}-r_d\underline{d}+\underbrace{(r_\ell-r_d)}_{<0}\underbrace{(d_{i,t+1}-\underline{d})}_{>0}$ by which banks experience an outflow of deposits. The liquidity constraint $\ell_{i,t+1}\geq\phi\underline{d}-(d_{i,t+1}-\underline{d})+\delta\ell^*\underline{d}$ is less likely to bind, while the leverage constraint $\frac{s_{i,t+1}}{d_{i,t+1}}\leq\kappa$ is also less binding. In the opposite case in which $d_{i,t+1}=\underline{d}$, the inefficiency of the inflow/outflow effect in banks' budget constraint and liquidity constraint vanishes, while the leverage constraint continues is tighter.

Focus on the cases that give rise to an inflow or outflow of deposits, since the other cases do not change the incentive banks have towards liquidity from the inflow/outflow of deposits term. At the highest deposits realisation $d_{i,t+1}=\overline{d}$ banks are outflow banks, while at the lowest deposits realisation $d_{i,t+1}=\underline{d}$ banks are inflow banks. In particular, the smallest banks $\mathcal{V}_{\ell^*,\underline{s},\overline{d},d_{i,t+1}=\underline{d}}(\ell_{i,t},s_{i,t},d_{i,t+1},d_{i,t})$, and the least large banks $\mathcal{V}_{\overline{\ell},\overline{s},\overline{d},d_{i,t+1}=\underline{d}}(\ell_{i,t},s_{i,t},d_{i,t+1},d_{i,t})$ are inflow banks, while the least small banks $\mathcal{V}_{\ell^*,\underline{s},\underline{d},d_{i,t+1}=\overline{d}}(\ell_{i,t},s_{i,t},d_{i,t+1},d_{i,t})$ and the largest banks

$\mathcal{V}_{\overline{\ell},\overline{s},\underline{d},d_{i,t+1}=\overline{d}}(\ell_{i,t}, s_{i,t}, d_{i,t+1}, d_{i,t})$ are outflow banks. If we identify $\mathcal{V}_{\ell^*,\underline{s},\overline{d},d_{i,t+1}=\underline{d}}(\ell_{i,t}, s_{i,t}, d_{i,t+1}, d_{i,t})$ with i, $\mathcal{V}_{\ell^*,\underline{s},\underline{d},d_{i,t+1}=\overline{d}}(\ell_{i,t}, s_{i,t}, d_{i,t+1}, d_{i,t})$ with i', $\mathcal{V}_{\overline{\ell},\overline{s},\underline{d},d_{i,t+1}=\overline{d}}(\ell_{i,t}, s_{i,t}, d_{i,t+1}, d_{i,t})$ with j' and $\mathcal{V}_{\overline{\ell},\overline{s},\overline{d},d_{i,t+1}=\underline{d}}(\ell_{i,t}, s_{i,t}, d_{i,t+1}, d_{i,t})$ with j, then the surplus from the matching of banks of dissimilar types is greater than the surplus from the matching of banks of the same type. This is simply because banks within their type have the same incentive towards liquid assets and because of the decreasing marginal value of liquidity. The optimal banks' matching is between i and j', and between i' and j, which are between small inflow banks and largest outflow banks; and between the least large inflow banks and the least small outflow banks, respectively.

Therefore, banks have an incentive to match instead of competing alone because:

$$W_{ij',t} \geq \mathcal{V}_{\ell^*,\underline{s},\overline{d},d_{i,t+1}=\underline{d}}(\ell_{i,t}, s_{i,t}, d_{i,t+1}, d_{i,t}) + \mathcal{V}_{\overline{\ell},\overline{s},\underline{d},d_{i,t+1}=\overline{d}}(\ell_{i,t}, s_{i,t}, d_{i,t+1}, d_{i,t}); \tag{B.22}$$

$$W_{i'j,t} \geq \mathcal{V}_{\ell^*,\underline{s},\underline{d},d_{i,t+1}=\overline{d}}(\ell_{i,t}, s_{i,t}, d_{i,t+1}, d_{i,t}) + \mathcal{V}_{\overline{\ell},\overline{s},\overline{d},d_{i,t+1}=\underline{d}}(\ell_{i,t}, s_{i,t}, d_{i,t+1}, d_{i,t}), \tag{B.23}$$

and they match with banks of dissimilar types because:

$$S_{ij'}(d_i, d_{j'}) + S_{i'j}(d_{i'}, d_j) \geq S_{ij}(d_i, d_j) + S_{i',j'}(d_{i'}, d_{j'}). \tag{B.24}$$

Thus, the optimal assignment matching game arises endogenously in the federal funds market and this respects negative assortative matching. Large banks are then the buyers of liquidity, while small banks are the sellers of liquidity, encapsulating the properties of the assignment game presented in Section 4.3.3. ■

C Omitted Proofs and Derivations

C.1 Constrained and Unconstrained Banks

Given the specificity of the model of liquidity and leverage constraints and idiosyncratic deposit withdrawal shock, I characterise four types of heterogeneous banks. These are unconstrained banks, constrained liquidity medium banks, constrained leverage medium banks, constrained small banks.

Unconstrained banks. These types of banks have enough liquidity to face the deposit withdrawal shock and never hit the leverage constraint. Their programme is to maximise the value function in equation (11) subject to the budget constraint (12), the intraperiod liquidity constraint (13) and leverage constraint (14). The programme generates optimal decision rules for dividends, $c^*_{i,t}(d^t; v_0)$, liquidity, $\ell^*_{i,t+1}(d^t; v_0)$, in equation (15), illiquid assets, $s^*_{i,t+1}(d^t; v_0)$, in equation (16) when adjusting them upward, with null multipliers, $\mu_{i,t}(d^{t-1}; v_0) = 0$ and $\omega_{i,t}(d^{t-1}; v_0) = 0$.

Since unconstrained banks are indifferent between issuing dividends and investing in liquid assets, the rationale for the redistribution mechanism of assets owns to the slope of the expected adjustment costs that drives the expected marginal return. When the discounted marginal

return is expanding, banks invest in illiquid assets.

Banks entering the economy with a high level of deposits are more likely to be unconstrained in the next period if the liquidity and leverage constraints do not bind. These banks are large in size. The accumulation of liquidity of this type of banks is enough to guarantee they will remain unconstrained in the future.

Constrained liquidity medium banks. These types of banks are liquidity constrained but not leverage constrained. They are illiquid banks and medium small banks. The optimal liquidity decision rule is at the corner solution and it is constrained by the liquidity constraint that yields:

$$\ell^*_{i,t+1}(d^t; v_0) = \phi - (d_{i,t+1} - d_{i,t}), \tag{C.1}$$

and in case of the submodular liquidity constraint, equation (C.1) becomes $\ell^*_{i,t+1}(d^t; v_0) = \phi d_{i,t} - (d_{i,t+1} - d_{i,t}) + \delta \ell_{i,t} d_{i,t}$; whereas the optimal decision rule for illiquid assets, $s^*_{i,t+1}(d^t; v_0)$, solves equation (16), with its multiplier $\omega_{i,t}(d^{t-1}; v_0) = 0$. This type of banks captures those banks that whilst without enough liquidity, they are solid since the constrained amount of liquidity ensures that the unconstrained amount of illiquid assets is covered by any possible withdrawal. Since they face the unconstrained level of illiquid assets and make enough profits because of the high return on illiquid assets, these liquidity constrained banks can still grow in size. However, an unexpected withdrawal of deposits can turn these banks to be leverage constrained, thereby losing their dimensionality as they may need to disinvest in illiquid assets. In this instance, the bank can become a constrained small bank.

Moreover, when the value of liquidity is constrained by the liquidity constraint, banks can rebalance their illiquid assets towards more liquidity. The support of illiquid assets is then bounded below and above by $\underline{s}$ and $\overline{s}$, respectively.

Constrained leverage medium banks. These types of banks are leverage constrained but not liquidity constrained, they are medium large banks. The maturity mismatch imposes banks to stay on their leverage constraint. Therefore, the optimal decision rule for loans is:

$$s^*_{i,t+1}(d^t; v_0) = \kappa d_{i,t+1} \quad \text{or} \quad s^*_{i,t+1}(d^t; v_0) = \max\{\kappa d_{i,t+1}, 0\}, \tag{C.2}$$

where a fraction of illiquid assets needs to be covered by deposits, which is in turn connected to the accumulated liquidity in banks' balance sheets. Similarly to the constrained liquidity medium banks, constrained leverage medium banks can grow in size through the accumulation of liquidity by holding a nonnegative amount of dividends. Available dividends will be consumed inasmuch banks will not be liquidity constrained in the future.

Constrained small banks. Lastly, constrained small banks are banks that are both liquidity and leverage constrained banks. The multipliers on both constraints for this type of banks hold with strict inequality, $\mu_{i,t}(d^{t-1}; v_0) > 0$ and $\omega_{i,t}(d^{t-1}; v_0) > 0$. Therefore, these types of banks find it difficult to expand their liquid and illiquid assets. However, they are the banks that in

response to an increase in illiquid assets, strongly increase the marginal propensity to lend.

The precautionary liquidity motive for these types of banks is high since they have to build their buffer of liquidity. They extend loans inasmuch they are allowed by the leverage constraint and consume the remaining amount by the end of the period. Banks entering the economy with a permanent low level of deposits will be less likely to expand in size in the future.

C.2 Illiquid Assets Redistribution Mechanism

This section analyses the redistribution effects from large banks to small banks within assets and between banks. Recall that any increase in deposits that allows banks to expand in size, has to be counterbalanced by a tighter liquidity constraint and leverage constraint. The model gives rise to a banks' size dependent minimum liquidity requirement and inherits initial level of banks' illiquid assets of each individual bank. Proposition 10 demonstrates the formal analysis.

Proposition 10 *The minimum liquidity requirement, ℓ_{min}, is bank size dependent and depends on individual bank illiquid assets characteristics.*

Proof. In the Appendix C.2. ■

Proof. To show the Proposition recall the balance sheet identity in this model, $\ell_{i,t} + s_{i,t} = d_{i,t} + \mathcal{P}_{i,t}$. Moreover, assume that at the beginning of the period profits equal zero to give a clear role to the assets side of banks' balance sheet. Solving the balance sheet identity for deposits and substituting the corresponding quantity in the liquidity constraint at the beginning of period at the equality sign, $\ell_{i,t} + (d_{i,t+1} - d_{i,t}) = \phi$, I obtain $\ell_{i,t} + (\ell_{i,t+1} + s_{i,t+1} - \ell_{i,t} - s_{i,t}) = \phi$. That is, $\ell_{i,t+1} + s_{i,t+1} - s_{i,t} = \phi$. This implies that at the lower bound, $\ell_{min} = \phi + s_{i,t} - s_{min}$. Therefore, the minimum level of liquidity is banks' size dependent and inherits the initial conditions of each bank liquid assets. ■

Therefore, the level of illiquid assets with which banks enter the economy is sensitive to their position in the market, *i.e.*, whether they will be large banks or small banks.

The inflow/outflow effect in the banks' budget constraint creates an inefficiency that vanishes through the redistribution of assets between banks. In particular, as Section 4.3.3 will demonstrate, the largest banks are outflow banks while the smallest banks are inflow banks. As outflow banks need to pay the interest wedge on the liquidity outflow to inflow banks, a match arises between these two types of banks. The submodular liquidity constraint together with the leverage constraint have opposite effects between large and small banks, following an outflow of deposits, making the match between these banks feasible. Indeed the liquidity and leverage constraint are binding for inflow banks, while they are loose for outflow banks.

As liquidity is transferred to large banks, a redistribution of assets manifests as the marginal difference in assets between large and small banks. This redistribution has, therefore, important

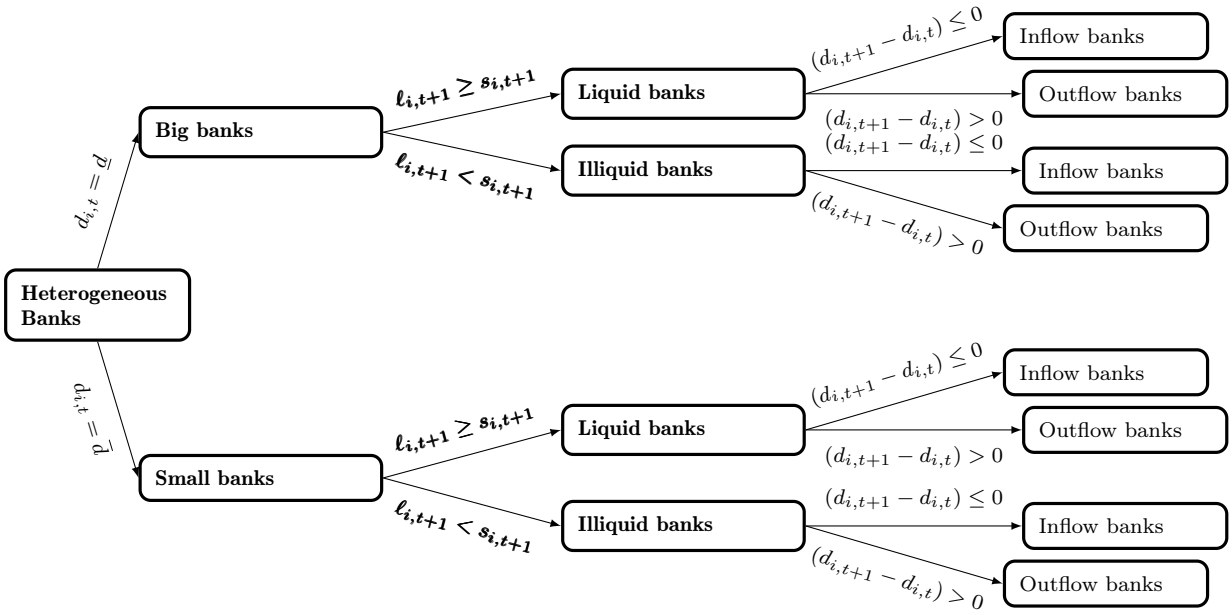


FIG. D.1.—Taxonomy of banks

efficiency properties because it removes the inefficiency arising from the inflow/outflow effect in banks' budget constraint.

By being outflow banks, large banks have a lower incentive toward adjusting illiquid assets the higher the adjustment costs since they have to mitigate the outflow of liquidity, consequently their supply of illiquid assets increases less than the supply of illiquid assets of small banks. The price of illiquid assets makes the redistribution of illiquid assets between banks to occur because of the adjustment costs in illiquid assets that restrict fast deviations in illiquid assets and because of the redistribution effect in the equation of the marginal propensity to lend in Proposition 4.

D Taxonomy of Banks

The model contains different types of banks which are described in Figure D.1. In particular, depending on the initial conditions of deposits and assets a bank can be big when $d_{i,t} = \underline{d}$ and $s_{i,t} = \overline{s}$ or small when $d_{i,t} = \overline{d}$ and $s_{i,t} = s^*$, where $\overline{d}$ ($\underline{d}$) and $\overline{s}$ (s^*) are the maximum (minimum) level of deposits and illiquid assets.

The liquidity structure of banks relies on the pledging collateral in banks' balance sheet to satisfy short term liquidity needs. Therefore, they can be liquid (illiquid) when $\ell_{i,t+1} \geq s_{i,t+1}$ ($\ell_{i,t+1} < s_{i,t+1}$). Lastly, since banks may experience a withdrawal of deposits the recipient banks is experiencing an inflow of deposits. Therefore, the outflow of deposits which occurs when $(d_{i,t+1} - d_{i,t}) > 0$ determines the type of banks as inflow or outflow banks. This type, however, is idiosyncratic since it depends on the future realisation of deposits.

E Computational and Numerical Appendix

In this Appendix I describe the numerical solutions I used to solve the models.

The recursive formulation of the model is given below:

$$1 = \beta r_\ell \mathbb{E}_{d'|d}\left[\frac{g(\ell,s,d,d_{-1})}{g(h(\ell,s,d,d_{-1}),e(\ell,s,d,d_{-1}),d',d)}\right]^\gamma; \tag{E.1}$$

$$1 = \beta \mathbb{E}_{d'|d}\left[\left(\frac{g(\ell,s,d,d_{-1})}{g(h(\ell,s,d,d_{-1}),e(\ell,s,d,d_{-1}),d',d)}\right)^\gamma \right.$$
$$\left.\left[\alpha(v(h(\ell,s,d,d_{-1}),e(\ell,s,d,d_{-1}),d',d) - e(\ell,s,d,d_{-1})) + r_s] + \frac{\alpha}{\beta}(e(\ell,s,d,d_{-1}) - s)\right] - \frac{\omega(\ell,s,d,d_{-1})g(\ell,s,d,d_{-1})}{d}\right]; \tag{E.2}$$

in which $s'' = v(h(\ell,s,d,d_{-1}),e(\ell,s,d,d_{-1}),d',d)$,

$$\begin{aligned} r_s s + r_\ell \ell - r_d d_{-1} + (r_\ell - r_d)(d - d_{-1}) - g(\ell,s,d,d_{-1}) - e(\ell,s,d,d_{-1}) + \\ -\frac{\alpha}{2}(e(\ell,s,d,d_{-1}) - s)^2 - h(\ell,s,d,d_{-1}) + d = 0; \end{aligned} \tag{E.3}$$

$$h(\ell,s,d,d_{-1}) + (d - d_{-1}) \geq \phi, \quad \frac{e(\ell,s,d,d_{-1})}{d} \leq \kappa; \tag{E.4}$$

$$\mu(\ell,s,d,d_{-1})(h(\ell,s,d,d_{-1}) + (d - d_{-1}) - \phi) = 0, \quad \mu(\ell,s,d,d_{-1}) \geq 0; \tag{E.5}$$

$$\omega(\ell,s,d,d_{-1})\left(\frac{e(\ell,s,d,d_{-1})}{d} - \kappa\right) = 0, \quad \omega(\ell,s,d,d_{-1}) \geq 0. \tag{E.6}$$

Equation (E.5) with submodular liquidity constraint becomes: $\mu(\ell,s,d,d_{-1})(h(\ell,s,d,d_{-1}) + (d - d_{-1}) - \delta\ell d - \phi d) = 0, \quad \mu(\ell,s,d,d_{-1}) \geq 0.$

E.1 Numerical Solution Algorithm Model Partial Adjustment Costs

This Section describes the solution procedure used to solve the model with convex adjustment costs. The state space consists of two endogenous state variables and one exogenous state, (ℓ,s,d,d_{-1}). In this environment, the leverage constraint does not hold. It is discretised such that the grid for liquid assets contains as a lowest value the corresponding liquidity constraint and the highest upper bound set such that a bank is never going to hit the constraint. The grid for ℓ is a polynomial grid with more nodes located at the end of the liquidity grid from below.

The algorithm proceeds as follows. The model is solved with fixed point iteration.

1. Specify the grid for ℓ such that $\ell_{min} = \phi + d_{max} - d_{min}$;
2. Specify the grid for s;
3. Guess $\ell' = h^0(\ell,s,d,d_{-1})$;
4. Guess $s' = e^0(\ell,s,d,d_{-1})$;
5. Enforce the liquidity constraint: for each element (i,j,k), if $\ell'_{i,j,k} < \phi + d_k - d_{min}$ replace with $\ell'_{i,j,k} = \phi + d_k - d_{min}$; assume the leverage constraint is not binding;

6. Calculate the guess for consumption at iteration $q = 0$, from the budget constraint in equation (E.3), and get $c = g(\ell, s, d, d_{-1})$;
7. Compute the marginal utility of consumption and obtain future values of marginal utility of consumption through an interpolation method and get $c' = g(h^0(\ell, s, d, d_{-1}), e^0(\ell, s, d, d_{-1}), d', d)$;
8. Compute the expected value of the marginal utility of consumption, $\mathbb{E}_{d'|d}\left[(g(h^0(\ell, s, d, d_{-1}), e^0(\ell, s, d, d_{-1}), d', d))^{-\gamma}\right]$;
9. Update current consumption, $g^{next}(\ell, s, d, d_{-1})$, through the Euler equation, in equation (E.1) and get $c' = g(h^0(\ell, s, d, d_{-1}), e^0(\ell, s, d, d_{-1}), d', d)$ for each node;
10. Calculate the updated $\ell'_{next} = h_{next}(\ell', s', d', d_{-1})$ through the budget constraint;
11. Check whether the liquidity constraint is binding, if so replace elements with $\ell'_{next,i,j,k} = \phi + d_k - d_{-1}$, otherwise skip this step and move to the next grid node;
12. Interpolate linearly to find future illiquid assets, $s'' = v^0(h^0(\ell, s, d, d_{-1}), e^0(\ell, s, d, d_{-1}), d', d)$;
13. Given computed policies $g^{next}(.)$ and $h^{next}(.)$, compute future illiquid assets through equation (E.2); in particular,
    (a) Interpolate $g(.)^{-\gamma}$ to obtain future the value of the marginal utility of consumption and calculate its expected value;
    (b) Compute the expected term on the rhs of the Euler equation of illiquid assets;

    $$\mathbb{E}_{d'|d}\left[g(h^0(\ell, s, d, d_{-1}), e^0(\ell, s, d, d_{-1}), d', d)[r_s - \eta(e^0(h^0(\ell, s, d, d_{-1}), e^0(\ell, s, d, d_{-1}), d', d)]\right];$$

    (c) Calculate the updated $s'_{next} = e^{next}(.)$ through the Euler equation for illiquid assets in equation (E.2);
14. Check criterion of convergence of the guessed policies, $\ell' = h^0(\ell, s, d, d_{-1})$ and $s' = e^0(\ell, s, d, d_{-1})$. Hence, define the tolerance for convergence equal to ϵ and if,

    $$\text{supnorm}_{\{\ell,s,d,d_{-1}\}}(\|h_{next}(\ell, s, d, d_{-1}) - h^0(\ell, s, d, d_{-1})\|, \|e_{next}(\ell, s, d, d_{-1}) - e^0(\ell, s, d, d_{-1})\|) < \epsilon,$$

    stop. Else, update the guesses $h^0(\ell, s, d, d_{-1})$ and $e^0(\ell, s, d, d_{-1})$ according to the damping rule, go to the next iteration $q = q + 1$, and return to step 3.

E.2 Numerical Solution Algorithm Baseline Model

To solve the baseline model with full convex adjustment costs, banks' exit, submodular liquidity constraint and leverage constraint, I solve the model with value function iteration with Howard improvement step and Nelder-Meade optimisation for the outer loop, since the computation of the marginal propensity to lend involves the computation of a derivative of one of the policy functions on the grid of illiquid assets and their interest rate. The algorithm is solved using the effective discount factor $\beta(1 - \delta_e)$. The state space consists of two endogenous state variables and one exogenous state, (ℓ, s, d, d_{-1}), with adjustment costs on s illiquid assets. The algorithm

proceeds as follows.

1. Set up the grids for ℓ and s;
2. Compute the lower bounds for ℓ satisfying a worst case deposit outflow for each bank type, for each element (i,j,k), if $\ell'_{i,j,k} < \phi + d_k - d_{min}$ replace with $\ell'_{i,j,k} = \phi + d_k - d_{min}$;
3. Compute the entry state;
4. Determine the maximum level of illiquid assets s;
5. Initiliase the value function $\mathcal{V}^0(\ell, s, d, d_{-1})$;

   For each iteration j,
6. Calculate the guess for current consumption from the budget constraint in equation (E.3), and get $c = g(\ell, s, d, d_{-1})$;
7. Compute the expected continuation value $\mathbb{E}_{d'|d}\left[(\mathcal{V}^j(h^0(\ell, s, d, d_{-1}), e^0(\ell, s, d, d_{-1}), d', d))\right]$;
8. Check the leverage constraint in equation (E.6) is satisfied;
9. Ensure feasibility of the liquidity constraint in equation (E.5) and non-negativity of consumption;
10. Store argmax of ℓ and s for Howard improvement step;
11. Every 10 iterations, apply Howard improvement step over $n = 50$ iterations, and set $\mathcal{V}^{j+1}(\ell, s, d, d_{-1}) = \mathcal{V}^{j,n}(\ell, s, d, d_{-1})$;
12. Given $(h^0(\ell, s, d, d_{-1}), e^0(\ell, s, d, d_{-1}))$, evaluate analytically the value function $\mathcal{V}^j(\ell, s, d, d_{-1})$ by linear indexation of its expected value:

$$\mathcal{V}^j(\ell, s, d, d_{-1}) = u(g(\ell, s, d, d_{-1})) + \mathbb{E}_{d'|d}\left[(\mathcal{V}^j(h^0(\ell, s, d, d_{-1}), e^0(\ell, s, d, d_{-1}), d', d))\right]; \tag{E.7}$$

13. Check criterion of convergence supnorm$(\|\mathcal{V}^{j+1}(\ell, s, d, d_{-1}) - \mathcal{V}^j(\ell, s, d, d_{-1})\|) < \epsilon$, if the condition is satisfied stop, else update the guess for $\mathcal{V}^0(\ell, s, d, d_{-1})$, move to the next iteration and return to step 6;
14. For each grid point element (i,j,k) compute Nelder-Meade optimisation (fminsearch). In particular, set the outer grid over the deposit state, initialise policies $\ell' = h^0(\ell, s, d, d_{-1})$ and $s' = e^0(\ell, s, d, d_{-1})$, bilinearly interpolate $\mathbb{E}_{d'|d}\left[(\mathcal{V}^j(h^0(\ell, s, d, d_{-1}), e^0(\ell, s, d, d_{-1}), d', d))\right]$; for each (i,j,k) construct lower bounds for ℓ' and s' so that the feasible set satisfies the intersection of endogenous states with constraints:
   $\Theta(X) = \left\{(\ell, s) \in \mathbb{R} \times \mathbb{R}_+ : \frac{e(\ell,s,d,d_{-1})}{d} \leq \kappa, \quad \frac{(h(\ell,s,d,d_{-1})+(d-d_{-1})-\delta\ell d)}{d} \geq \phi, \right\}$,
   set the guess for Nelder-Mead starting point to $\ell^0(\ell, s, d, d_{-1})$ and $e^0(\ell, s, d, d_{-1})$, upper and lower bounds for ℓ and s and leverage constraint, compute consumption through the budget constraint, evaluate the expected value, build the initial simplex and apply Nelder-Meade algorithm with reflection, expansion, contraction and shrink over the objective function without derivatives, apply penalisation for infeasible points returned from the

minimisation, stop when optimisation is reached and store policies. To reduce numerical noise, for each k apply Gaussian smoothing over the policies ℓ' and s';

15. Compute the price of illiquid assets and apply Gaussian smoothing;
16. Compute the marginal propensity to lend as $\frac{\partial(e^0(\ell,s,d,d_{-1}))}{\partial r_s}$, using Nelder-Mead optimisation twice applied to r_s and $(r_s + \partial r_s)$ separately, using initial conditions for guess and interpolant as defined in step 14 to avoid convergence to different local optima; then apply Gaussian smoothing;
17. Compute the stationary distribution $\mu_{k+1}(h(\ell, s, d, d_{-1}), e(\ell, s, d, d_{-1}), d, d_{-1})$ with forward iteration by applying the transition matrix over $\mu_k(h(\ell, s, d, d_{-1}), e(\ell, s, d, d_{-1}), d, d_{-1})$; adjust stationary distribution with entry mass according to:

$$\mu_{k+1}(h(\ell, s, d, d_{-1}), e(\ell, s, d, d_{-1}), d, d_{-1}) = (1 - \delta_e)\mu_{k+1}(h(\cdot), e(\cdot) + \delta_e \mu_e(\ell, s, d, d_{-1}); \tag{E.8}$$

where $\mu_e(\ell, s, d, d_{-1})$ is the entry mass; normalise the distribution, assess convergence based on the supnorm of
supnorm$(\|\mu_{k+1}(h(\ell, s, d, d_{-1}), e(\ell, s, d, d_{-1}), d, d_{-1}) - \mu_k(h(\ell, s, d, d_{-1}), e(\ell, s, d, d_{-1}), d, d_{-1})\|)$, if less than tolerance stop; else iterate using the transition matrix to convergence on $\mu_{k+1}(h(\ell, s, d, d_{-1}), e(\ell, s, d, d_{-1}), d, d_{-1})$.

E.3 MIT Model Algorithm

To solve for the unanticipated shock along the transition path I assume ϕ drops at $t = 1$. I set the time horizon $T = 40$ and compute the path for ϕ_t as follows, $\phi_t = \phi_{ss} - (\phi_{ss} - \phi_{low})\rho_\phi^{t-1}$, $\forall t$, where $\phi_{low} = \phi_{ss}(1 - \phi_{drop})$ and ϕ_{drop} is the relaxation of the constraint.

The algorithm consists of two loops: a backward loop and a forward loop over the distribution of banks. The backward loop proceeds as follows:

1. Set $\mathcal{V}_t = \mathcal{V}^0(\ell, s, d, d_{-1})$;
   For each $t = T, T-1, ..., 1$,
2. Compute $\mathbb{E}_{d'|d}\left[(\mathcal{V}_{t+1}(h(\ell, s, d, d_{-1}), e(\ell, s, d, d_{-1}), d', d))\right]$;
3. Enforce time varying lower bounds for the liquidity constraint and compute t bounds;
4. Enforce leverage constraint;
5. Guess $\ell_t^0(\ell, s, d, d_{-1})$, $e_t^0(\ell, s, d, d_{-1})$ and $\mathcal{V}_{t+1}$;
6. For each element (i, j, k) compute feasible consumption dividends;
7. Compute argmax over value function, store $\mathcal{V}_{t+1}$, store policies $\ell_t(\ell, s, d, d_{-1})$, $e_t(\ell, s, d, d_{-1})$;
8. Check criterion of convergence supnorm$(\|\mathcal{V}_{t+1} - \mathcal{V}^0(\ell, s, d, d_{-1})\|) < \epsilon$ and move to the previous period.

The forward loop proceeds as follows:

1. Set $\phi_t = \mu(h(\ell, s, d, d_{-1}), e(\ell, s, d, d_{-1}), d, d_{-1})$ to the ergodic steady state distribution of

banks;

For each $t = 1, 2, ..., T + 1$,

2. Compute aggregate moments for distributions of $(c, \ell, s, q, d, \text{mpl})$ using distribution ϕ_t;
3. If $t < T$ update ϕ_t forward using $\ell_t(\ell, s, d, d_{-1})$, $e_t(\ell, s, d, d_{-1})$, according to $\phi_{new}(h(\ell, s, d, d_{-1}), e(\ell, s, d, d_{-1}), d, d_{-1}) = \phi_{new}(\cdot) + \mathbb{E}_{d'|d}\left[\phi_t(\ell, s, d, d_{-1})\right]$;
4. Update $\phi_{new}(h(\ell, s, d, d_{-1}), e(\ell, s, d, d_{-1}), d, d_{-1}) = (1 - \delta_e)\phi_{new}(\cdot) + \delta_e \mu_e$;
5. Set $\phi_t(\ell, s, d, d_{-1}) = \phi_{new}(h(\ell, s, d, d_{-1}), e(\ell, s, d, d_{-1}), d, d_{-1})$ and proceed to the next iteration.
6. At iteration $T + 1$ use steady state policies for ℓ' and s'.

F Additional Empirical Evidence

This Section provides further descriptive evidence along the lines of the ones presented in the main text in Section 3.

In the main text banks' liquidity consists of cash assets, amortised cost of held to maturity treasury securities, government securities and mortgage passthrough securities, fair value available for sale treasuries securities, government securities and mortgage passthrough securities, federal funds sold and securities purchased under agreements to resell in domestic offices of banks, held to maturity securities, total trading assets, mortgage backed securities held to maturity. Illiquid assets consist of commercial and industrial loans, total loans and leases, loans secured by real estate, total assets, other borrowed money, commercial real estate construction and consumer loans.[16]

Table F.1 provides summary statistics of the main variables of interest for banks. It shows substantial heterogeneity across banks. The median bank in the sample in 1998 has illiquid assets to deposits of 2.567 that increased to 2.765 in 2010. Banks located in the top decile of the distribution hold more illiquid assets than banks located in the bottom decile in both years. By contrast, liquidity in the top percentile of banks is six times larger than the bottom percentile in 1998, however by 2010 liquidity in the top percentile of banks decreased considerably becoming only three times higher than banks in the bottom percentile.

There is substantial cross-sectional heterogeneity in the return on loans with the median value remaining almost invariant across the years considered. Yet, the loan return of the top percentile of banks experienced a 40% fall in 2010 and turning eight times higher than banks in the bottom percentile, in comparison to the return on loans in 1998 when for the top percentile of banks was 169 times higher than the bottom percentile. There is heterogeneity in the price of loans and while the marginal propensity to lend shows a lot of variation across years, the difference between the bottom and top percentile of banks is substantially large. The negative values of the marginal propensity to lend represent a fall in banks' intensity to lend which are due either to a drop in loans from the previous year of a bank operation or to a drop in the interest

[16] Specific variables are given in Appendix L.

TABLE F.1.—

Descriptive statistics

	1998					2010				
	Mean	Median	Std	10th	90th	Mean	Median	Std	10th	90th
Illiquid assets	2.697	2.567	0.900	2.089	3.368	2.792	2.765	0.355	2.410	3.239
Liquidity	0.496	0.375	0.686	0.194	0.797	0.236	0.217	0.109	0.126	0.371
Price of loans	0.018	0.016	0.0139	0.006	0.030	0.019	0.017	0.010	0.007	0.034
Return loans	0.373	0.032	1.860	0.003	0.508	0.036	0.032	0.028	0.009	0.072
Adjust costs	0.017	0.001	0.093	0.001	0.017	0.043	0.011	0.186	0.006	0.081
Mpl	0.010	0.00001	0.417	-0.001	0.004	0.0004	-0.0001	0.009	-0.0003	0.0004
No. banks	1,193	1,193	1,193	1,193	1,193	904	904	904	904	904

NOTE. Summary statistics of the main variables of interest of the model. Illiquid assets, liquidity, adjustment costs and the marginal propensity to lend are accounted relative to deposits, whereas the return on loans is relative to illiquid assets. The terms 10th and 90th indicate percentiles. Data source is provided in Appendix L.

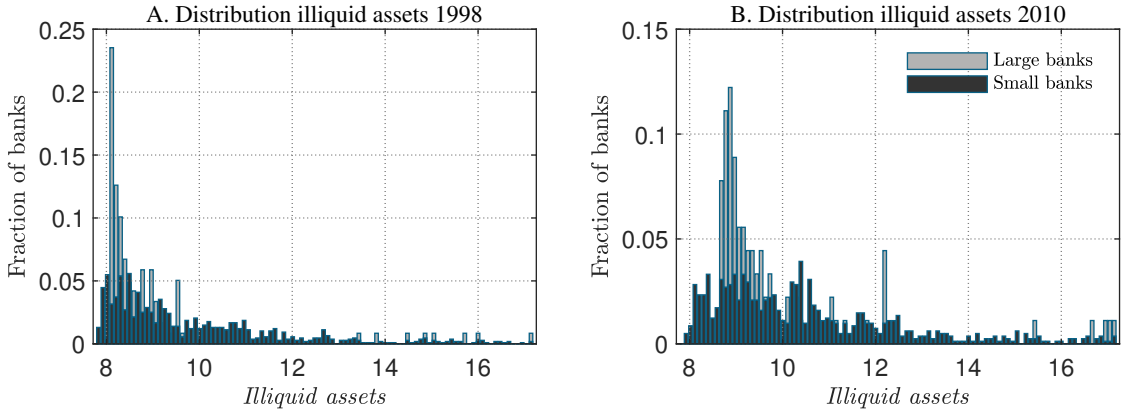


FIG. F.1.—Distribution banks' illiquid assets.

NOTE. Distribution of banks' illiquid assets in 1998 in panel A and in 2010 in panel B by bank size. Data source is provided in Appendix L.

rate on loans from the previous year. I consider the change in loans from the previous year to elicit the marginal propensity to lend, since in the model I focus on the heterogeneity in banks' loans, which is the primary research investigation of the paper and abstract from heterogeneity in interest rates.

Figure F.1 confirms the predictions made in Figure 1 and 2 in the main text. It shows the histogram distributions of large and small banks' illiquid assets. I split the distribution between large and small banks to identify any redistribution of illiquid assets between banks and between assets. It emerges a significant shift to the right of the distribution of large banks' illiquid assets. The shift was accompanied by a reduction in the fraction of large banks and a simultaneous fatter left tail of the distribution of small banks' illiquid assets. Similarly, Figure F.2 graphs the distribution of the price of loans. While the fraction of large banks charging higher prices halved from 1998 to 2010, by the same time it is more difficult to find large banks

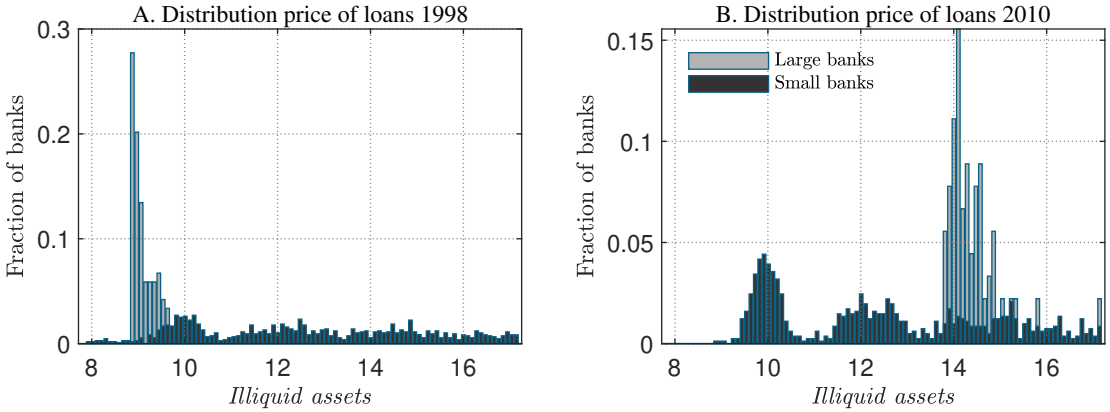


FIG. F.2.—Distribution banks' price of loans.

NOTE. Distribution of banks' price of loans in 1998 in panel A and in 2010 in panel B by bank size. The price of loans is the per unit of loans interest income loans to total illiquid assets. Data source is provided in Appendix L.

charging low loan prices. The high loan price may be the reason for the fall in loan supply driven by a fall in borrowers' loan demand. Large banks exhibited an increase in size with the ability to charge a higher price, thereby leading to a less concentrated marginal propensity distribution which is shown in Figure 4 of the main text.

Table F.2 presents further summary statistics based on banks' size in 1998 and 2010, respectively. Variables are in units of deposits expect the price of loans. The Table points out that in 1998 on average illiquid assets of large banks is almost double the illiquid assets of small banks, with a volatility of small banks (0.360) that is significantly lower than the volatility of large banks (2.068) in the same year. In 2010, the pattern for the volatility reverts with illiquid assets of small banks (0.273) being more volatile than large banks (0.268). While large banks have an average value of liquidity larger than small banks in 1998, their volatility in liquidity is significantly larger than small banks volatility in both years.

Figure F.3 shows the evolution of the average distribution of the loan-to-deposits ratio and the marginal propensity to lend from 1998 to 2011 for large banks, small banks and all banks. The first distribution assesses the standing of liquidity in banks' balance sheet.[17] A high loan-to-deposit ratio means that banks may not have enough liquidity to cover any possible loan losses. This translates into a larger marginal propensity to lend and viceversa with a low loan-to-deposits ratio. This pattern is consistent throughout the majority of the sample period considered, but shows a reverse by the end of the sample. In particular, beginning 2007 until 2011 at a low (high) loan-to-deposits ratio for large (small) banks corresponds a high and increasing (low and decreasing) marginal propensity to lend. The pattern is more pronounced for large banks. This fact suggests that the additional liquidity in banks' balance sheets arising from a low loan-to-deposits ratio at the time of the financial crisis was used for additional lending.

[17]The loan-to-deposits ratio is disciplined in Section 109 of the Riegle-Neal Interstate Banking and Branching Efficiency Act. The larger the deposits, the larger the loan demand that can be financed with them.

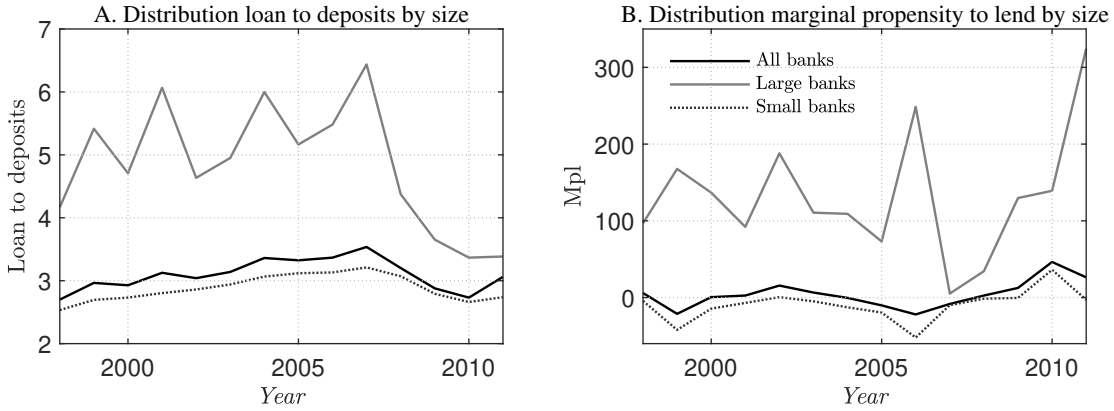


FIG. F.3.—Loans to deposits and marginal propensity to lend distributions by bank size over time.

NOTE. Distribution of loans to deposits in panel A is computed as the ratio of banks' illiquid assets to deposits. The marginal propensity to lend distribution in panel B is computed as the change in illiquid assets given a unit change in the return on loans. The evolution of both variables are from 1998 to 2011. Large banks correspond to the top 10 percentile of banks' distribution, small banks correspond to the rest of banks. Data source and details are provided in Appendix L.

Additionally, the decline in the loan-to-deposits ratio emphaises that banks were not transforming the additional deposits in lending, making the market more reliant on liquidity. Therefore, in holding more illiquid assets, large banks faced different regulatory constraints than small banks. In this environment characterised by decreasing loan-to-deposits ratio and opposite movement in the marginal propensity to lend, the market structure becomes characterised by an upper tail dominated equilibrium skewed to the right, whereby small banks become more equally distributed in the illiquid assets distribution and large banks are the main driver of heterogeneity in illiquid assets. I further inspect this joint interaction with the endogenous market structure in the theoretical model.

The figure highlights important heterogeneity in the marginal propensity to lend and the loan-to-deposits ratio. The loan-to-deposit ratio and the marginal propensity to lend of large banks have been higher than those of small banks in all considered years. Post financial crisis both variables have been decreasing over time and the fall has been driven by large banks whose distribution approached the average distribution of the loan-to-deposits and the marginal propensity to lend of all banks in the sample. This fact indicates that banks redistributed their assets from illiquid assets to liquid assets. As a result the distribution of banks' liquid assets became more widely spread.

The average marginal propensity to lend for large banks is positive and large in 1998 reflecting the expansion of illiquid assets which is then made available by increasing the amount of loans lent. A significant reduction in the marginal propensity to lend is experienced by large banks in 2010 in light of the stricter capital and liquidity requirements put forward after the financial crisis. By contrast, the average marginal propensity of small banks is negative pointing out a reduction in the amount of loans lent in 1998, which increases in 2010 while remaining

TABLE F.2.—

Summary statistics

	1998									
	Large banks					Small banks				
	Mean	Median	Std	10th	90th	Mean	Median	Std	10th	90th
Illiquid assets	4.261	3.691	2.068	3.405	4.791	2.524	2.523	0.360	2.062	2.993
Liquidity	1.636	1.123	1.761	0.846	2.167	0.370	0.352	0.151	0.185	0.585
Price of loans	0.042	0.035	0.028	0.030	0.051	0.015	0.014	0.007	0.006	0.026
Return loans	3.154	1.321	5.104	0.592	7.760	0.063	0.025	0.091	0.003	0.183
Adjust costs	0.151	0.048	0.261	0.019	0.475	0.002	0.001	0.003	0.001	0.005
Mpl	0.190	0.012	1.095	0.005	0.083	-0.009	0.00001	0.238	-0.002	0.0001
Observations	119	119	119	119	119	1,074	1,074	1,074	1,074	1,074
	2010									
	Large banks					Small banks				
	Mean	Median	Std	10th	90th	Mean	Median	Std	10th	90th
Illiquid assets	3.473	3.374	0.268	3.266	3.813	2.716	2.726	0.273	2.362	3.068
Liquidity	0.468	0.434	0.121	0.377	0.573	0.210	0.206	0.070	0.123	0.311
Price of loans	0.038	0.037	0.003	0.035	0.042	0.017	0.016	0.008	0.007	0.029
Return loans	0.089	0.084	0.015	0.074	0.111	0.031	0.029	0.023	0.008	0.059
Adjust costs	0.287	0.155	0.531	0.091	0.407	0.016	0.008	0.018	0.001	0.045
Mpl	0.007	0.002	0.029	0.0001	0.009	-0.0001	-0.000	0.002	-0.0001	0.0001
Observations	90	90	90	90	90	814	814	814	814	814

NOTE. Summary statistics by banks' size. Illiquid assets, liquidity, adjustment costs and the marginal propensity to lend are accounted relative to deposits, whereas the return on loans is relative to illiquid assets. Data source is provided in Appendix L.

negative highlighting the stricter liquidity requirements to banks.

Figures F.4 and F.5 present cross-sectional heterogeneity in the return of loans and the liquidity to assets ratio. While in 1998 banks with more deposits received a high return on loans, following the financial crisis income for the same type of banks dropped. Moreover, the distribution of the return on loans shifted to the right by 2010 indicating that banks with little deposits and therefore, large banks exited or merged with other banks. Similarly, I show the liquidity to assets ratio distribution to control for the size of banks. In particular, in 1998 this ratio experienced a steady decline as deposits increase in size. This pattern continues to hold in 2010 despite the ratio stabilises around its mean value.

Figure F.6 illustrates the distribution of the loan-to-deposits ratio as a measure of banks' liquidity from the perspective of the loan side. Large (small) banks are represented by the histogram in light (dark) green. The figure shows that in 1998 more than 30% of large banks held low loan-to-deposits. In 2010 the distribution of large banks shifted to the right and the move was led by the fall in deposits simultaneously to the rise in deposits. In 2010 the fraction of small with a large loan-to-deposits also rises.

Figure F.7 depicts the evolution of the price of loans and adjustment costs from 1998 to

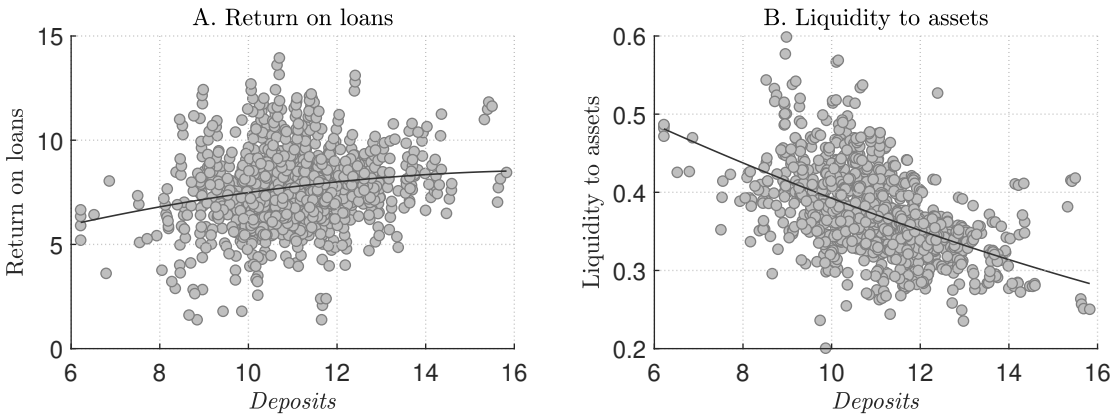


FIG. F.4.—Banks' cross-sectional patterns in 1998.

NOTE. Scatter plots of cross-sectional patterns of banks' return on loans and liquidity to illiquid assets in 1998. Data source and details are provided in Appendix L.

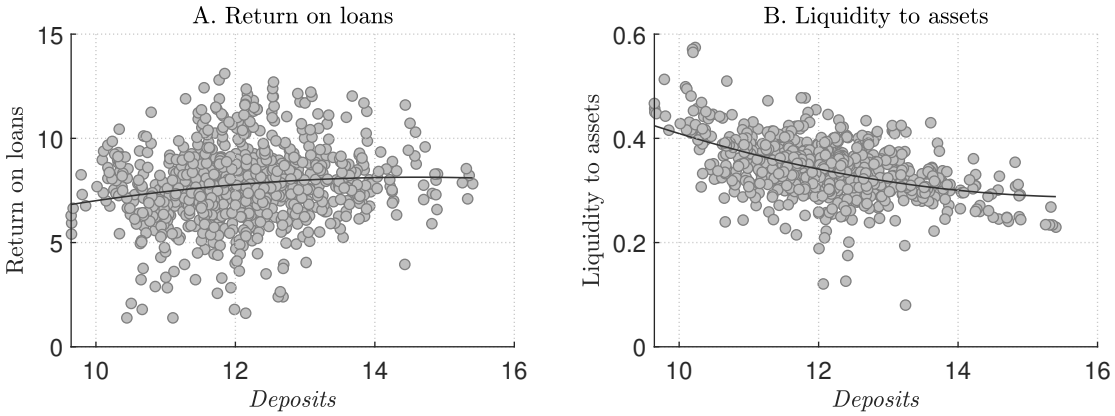


FIG. F.5.—Banks' cross-sectional patterns in 2010.

NOTE. Scatter plots of cross-sectional patterns of banks' return on loans and liquidity to illiquid assets in 2010. Data source and details are provided in Appendix L.

2011 for all banks, large banks and small banks with black lines, gray lines and dotted gray lines, respectively. The price of loans ranges between 2 and 6 percent across banks and while it rises around the 2008 financial crisis, it suddenly drops. More variation is highlighted in the adjustment costs that range between 0 and 5 percent. While adjustment costs surge pre-financial crisis, they drop immediately post-financial crisis. These facts document a considerable heterogeneous impact in banks' balance sheet which I explore in the theoretical model.

G Tobit Model and the Inverse Mills Ratio

I provide details on the derivation of the Inverse Mills ratio. These derivations follow the seminal articles by Tobin (1958) and Heckman (2013) with methodology also available in Cameron and Trivedi (2005). Assume that the density distribution function of the marginal propensity to lend is identified with density $\upsilon(\text{mpl}_{i,t})$ with $\text{mpl}_{i,t}$ truncated from below at 0. Then the truncated

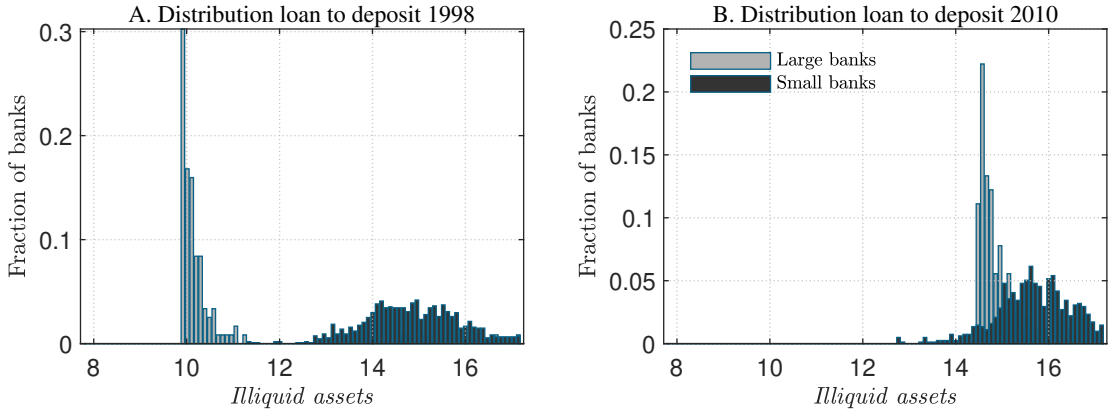


FIG. F.6.—Distribution loan-to-deposit.

NOTE. Distribution of loan-to-deposit in 1998 in panel A and in 2010 in panel B by bank size. Data source is in Appendix L.

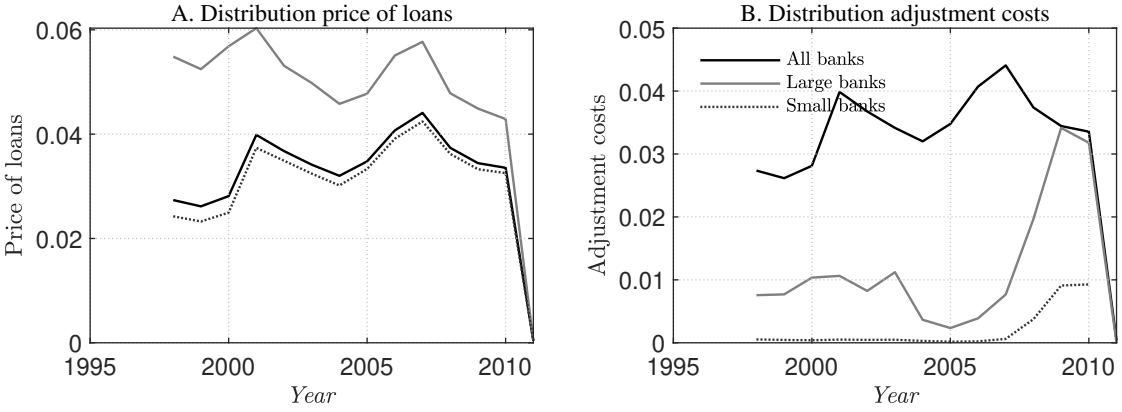


FIG. F.7.—Price of loans and adjustment costs in illiquid assets by bank size over time.

NOTE. Distribution of price of loans in panel A is computed as the ratio of interest income on loans to total loans. The adjustment costs in illiquid assets in panel B are computed as difference between charge off loans and recovery loans. The evolution of both variables are from 1998 to 2011. Large banks correspond to the top 10 percentile of banks' distribution, small banks correspond to the rest of banks. Further details are given in Appendix L.

density function $\upsilon(\text{mpl}_{i,t}|\text{mpl}_{i,t} > 0)$ is given by:

$$\upsilon(\text{mpl}_{i,t}|\text{mpl}_{i,t} > 0) = \frac{\upsilon(\text{mpl}_{i,t})}{P(\upsilon(\text{mpl}_{i,t} > 0)} = \frac{\upsilon(\text{mpl}_{i,t})}{1 - \Upsilon(0)}, \tag{G.1}$$

where, $\Upsilon(0)$ is the cumulative distribution density function at the truncation point 0. Therefore, the truncated density function with normally distributed dependent variable can be written as:

$$\upsilon(\text{mpl}_{i,t}|\text{mpl}_{i,t} > 0) = \frac{1}{\sigma}\phi\left(\frac{\text{mpl}_{i,t} - \mu}{\sigma}\right) / \left[1 - \Phi\left(\frac{0 - \mu}{\sigma}\right)\right], \tag{G.2}$$

where $\phi(\cdot)$ and $\Phi(\cdot)$ are the density and distribution function associated to the Normal distribution. It follows that the expected value of the truncated Normal distribution can be characterised

as:

$$\mathbb{E}(\epsilon_{i,t}|\epsilon_{i,t}>0)=\int_0^\infty \epsilon_{i,t}\upsilon(\epsilon_{i,t}|\epsilon_{i,t}>0)d\epsilon_{i,t}, \tag{G.3}$$

hence:

$$\mathbb{E}(\epsilon_{i,t}|\epsilon_{i,t}>0)=\frac{1}{1-\Phi(0)}\int_0^\infty \epsilon_{i,t}\phi(\epsilon_{i,t})d\epsilon_{i,t}=\frac{\phi(0)}{1-\Phi(0)}. \tag{G.4}$$

The characterisation of the truncated expectation in equation (G.4) as the ratio between the density function and the complement distribution follows from the normality assumption.

The Inverse Mills Ratio is best known for being the ratio between the density and the cumulative distribution of the standard normal distribution. In particular, for $\delta \in (-\infty,\infty)$, $\lambda(\delta)=\phi(\delta)/\Phi(\delta)$. Therefore, equation (G.4) can be written as $\frac{\phi(0)}{1-\Phi(0)}=\frac{\phi(-0)}{\Phi(-0)}$ that means:

$$\mathbb{E}(\epsilon_{i,t}|\epsilon_{i,t}>0)=\lambda(-0). \tag{G.5}$$

By construction the Inverse Mills Ratio is bounded below at zero and it is bounded at $-\delta$. The latter bound arises because of $\lambda(\delta)=\mathbb{E}(\epsilon_{i,t}|\epsilon_{i,t}>-\delta)$ which needs to be larger than its smallest value, $-\delta$. Hence, the lower bound for the Inverse Mills Ratio is $\lambda(\delta)\geq\max(0,\delta)$. By differentiating this expression for δ, it is possible to learn how much $\lambda(\delta)$ changes given a change in δ which results in $\lambda'(\delta)=-\lambda(\delta)[\lambda(\delta)+\delta]$ which is negative and therefore, the Inverse Mills Ratio is monotonically decreasing in δ if and only if $\log[\Upsilon(\delta)]$ is concave. Since $\delta=-0$ then $\mathbb{E}(\epsilon_{i,t}|\epsilon_{i,t}>0)$ is monotonically increasing in the truncation point. The result continues to hold for the standardised Normal random variable.

The latent model $\mathbb{E}(\text{mpl}^*_{i,t}|x_{i,t})$ is linear in γ and therefore, it does not deliver a plausible and meaningful interpretation. Recall that I am interested in analysing how deposits impact the probability of giving one unit of loan and the effect of the the return on loans on the marginal propensity to lend. Given the definition of the empirical model in the main text and including the variable $\tilde{s}_{i,t}$ and its coefficient, γ_1, in the matrix of variables $\mathbf{X}_{i,t}$ and the vector of coefficients γ_2', the Tobit expression equations are:

$$\text{P}(\text{mpl}_{i,t}>0|\mathbf{X}_{i,t})=\Phi\left(\frac{\gamma_2'\mathbf{X}_{i,t}}{\sigma}\right), \tag{G.6}$$

$$\mathbb{E}(\text{mpl}_{i,t}|\text{mpl}_{i,t}>0,\mathbf{X}_{i,t})=\gamma_2'\mathbf{X}_{i,t}+\sigma\lambda\left(\frac{\gamma_2'\mathbf{X}_{i,t}}{\sigma}\right), \tag{G.7}$$

$$\mathbb{E}(\text{mpl}_{i,t}|\mathbf{X}_{i,t})=\Phi\left(\frac{\gamma_2'\mathbf{X}_{i,t}}{\sigma}\right)\left[\gamma_2'\mathbf{X}_{i,t}+\sigma\lambda\left(\frac{\gamma_2'\mathbf{X}_{i,t}}{\sigma}\right)\right]. \tag{G.8}$$

Since $\mu=\gamma_2'\mathbf{X}_{i,t}$ then $\mathbb{E}(\text{mpl}_{i,t}|\text{mpl}_{i,t}>0,\mathbf{X}_{i,t})>\mathbb{E}(\text{mpl}_{i,t}|\mathbf{X}_{i,t})>\mathbb{E}(\text{mpl}^*_{i,t}|\mathbf{X}_{i,t})=\mu$.

Consider now a τ increase in μ then the marginal effect for the probability at the truncation point equals:

$$\frac{\partial\text{P}(\text{mpl}_{i,t}=0|\mathbf{X}_{i,t})}{\partial\mathbf{X}_{i,t,\tau}}=-\phi\left(\frac{\gamma_2'\mathbf{X}_{i,t}}{\sigma}\right)\gamma_{2,\tau}. \tag{G.9}$$

Using the derivative of the Inverse Mills Ratio, the conditional expectation of the Tobit model can be recovered as:

$$\frac{\partial \mathbb{E}(\text{mpl}_{i,t}|\text{mpl}_{i,t}>0,\mathbf{X}_{i,t})}{\partial \mathbf{X}_{i,t,\tau}}=\gamma_{2,\tau}\left\{1-\lambda\left(\frac{\gamma_2'\mathbf{X}_{i,t}}{\sigma}\right)\left[\frac{\gamma_2'\mathbf{X}_{i,t}}{\sigma}+\lambda\left(\frac{\gamma_2'\mathbf{X}_{i,t}}{\sigma}\right)\right]\right\}. \tag{G.10}$$

Equation (G.10) implies that the total marginal effect is computed as:

$$\begin{aligned}\frac{\partial \mathbb{E}(\text{mpl}_{i,t}|\mathbf{X}_{i,t})}{\partial \mathbf{X}_{i,t,\tau}}=&\frac{\partial \text{P}(\text{mpl}_{i,t}>0|\mathbf{X}_{i,t})}{\partial \mathbf{X}_{i,t,\tau}}\mathbb{E}(\text{mpl}_{i,t}|\text{mpl}_{i,t}>0,\mathbf{X}_{i,t})+\\&\text{P}(\text{mpl}_{i,t}>0|\mathbf{X}_{i,t})\frac{\partial \mathbb{E}(\text{mpl}_{i,t}|\text{mpl}_{i,t}>0,\mathbf{X}_{i,t})}{\partial \mathbf{X}_{i,t,\tau}}.\end{aligned} \tag{G.11}$$

The two terms in equation (G.11) are the extensive and the intensive margin, respectively. I interpret the extensive margin as the probability that a bank will extend a line of loan as deposits increase times the expected marginal propensity to lend. By contrast, I interpret the intensive margin as the increase in the expected marginal propensity to lend given a deposit increase times the probability of extending a line of loans by banks. Therefore, the total marginal effect is scaled by $\gamma_{2,\tau}$. Furthermore, the relative marginal effects between any two variable changes are constant, because:

$$\frac{\partial \mathbb{E}(\text{mpl}_{i,t}|\mathbf{X}_{i,t})/\mathbf{X}_{i,t,\tau}}{\partial \mathbb{E}(\text{mpl}_{i,t}|\mathbf{X}_{i,t})/\mathbf{X}_{i,t,\kappa}}=\frac{\gamma_{2,\tau}}{\gamma_{2,\kappa}}, \tag{G.12}$$

which implies that a large value of $\gamma_2'\mathbf{X}_{i,t}$ leads to large $\Phi\left(\frac{\gamma_2'\mathbf{X}_{i,t}}{\sigma}\right)$ and therefore a small probability of obtaining zero, and the more likely is that the two models $\text{mpl}_{i,t}$ and $\text{mpl}_{i,t}^*$ with conditional expectations and marginal effects are similar.

H Supplement to Empirical Assortative Matching and Inequality in the Marginal Propensity to Lend

In this Section I report further empirical results on the Tobit estimates relating to the assortative matching in Section H.1 and the inequality in the marginal propensity to lend in Section H.2.

H.1 Further Empirical Results on Assortative Matching

I now report the results of the remaining coefficients for the first step regression in Table 1 of the main text. The important role of banks' regulation is further highlighted by banks' size within each size of banks. While a unit increase in banks' size for small banks leads to an increase in illiquid assets of 0.128 and 0.031 units in 1998 and 2010 increasing their illiquidity per unit of deposits, respectively, large banks reduce illiquid assets as they grow in size because of economies of scale, indicating a self-insurance motive in liquidity and more importantly, the role of adjustment costs in illiquid assets. While in 1998 small banks appear to be more constrained and reliant on deposits, by 2010 they do not increase illiquid assets substantially as they grow

in size. Likewise, by 2010 large banks do not reduce illiquid assets significantly as they grow in size. Both effects are evidence of tighter liquidity regulations put in place post-financial crisis.

The coefficients on the price of loans, return on loans and adjustment costs show a more equal effect across banks' sizes and years. The price of loans variable indicates that for large banks, an increase in the price of loans reduces illiquid assets by an average of 70.31 units between both years. Similarly, the negative effect on illiquid assets is recorded for banks with an average of 93.611 units. However, while the price of loans was related to the size and liquidity positions of banks in 1998, whereby small banks with more liquid assets tended to charge a higher loan price since they are strongly liquidity constrained, in 2010 small (-80.169) and large (-76.027) banks charge an almost equal price of loans regardless of their liquidity position consistent with more uniform liquidity requirements.

The return on loans highlights the constrained and unconstrained nature of banks and the tighter liquidity regulation after the financial crisis. The effect of the return on loans for large banks is almost identical in both years, whereas the effect is stronger for small banks. Small banks tend to hold 39.769 more illiquid assets given an increase in the return on loans in 1998, while it decreases to 28.996 in 2010. By this year, any bank size difference between large and small banks disappears, yielding a more homogeneous effect of the return on loans between banks. This result indicates tighter liquidity regulations, where small banks need to charge loans more aggressively than large banks to gain in profitability.

The relationship between adjustment costs and illiquid assets shows the redistribution mechanism of illiquid assets between and within banks. This relation is positive and strongly statistically significant for large and small banks in both years. In 1998 higher adjustment costs lead to a 36.012 and 39.521 rise in illiquid assets for large and small banks, respectively. By 2010 this relation weakens while being positive and strongly statistically significant. It becomes more homogeneous between banks, and decreases by 20.02% for large banks and 26.95% for small banks, which indicates a redistribution of illiquid assets stronger for small banks. These estimates suggest a reduction in heterogeneity of adjustment costs across bank size due to liquidity regulations and a more homogeneous redistribution of illiquid assets between banks.

The interest rate on deposits shows that for small banks a high interest rate on deposits is associated with an increase of illiquid assets by 19.069, that is more than three times larger than the 2010 coefficient. The relationship is statistically significant in 1998 highlighting that small banks are strongly constrained because of their funding structure and this continues to hold in 2010, but the estimates are not significant. An opposite pattern arises for large banks. In 1998 an increase in the interest rate on deposits reduces illiquid assets by 34.398, while it has a positive effect in 2010 rising by 130.32% relative to the pre-financial crisis year. This result is consistent with the fact that with a high interest rate on deposits banks tend to be riskier and therefore, banks must hold more liquid assets than illiquid assets. However, in 2010 high deposits rates lead large banks to have more illiquid balance sheets.

TABLE H.1.—

Second step Tobit regression estimates of the marginal propensity to lend

	1998			2010		
	Large banks	Small banks	All banks	Large banks	Small banks	All banks
Illiquid res., $\tilde{s}_{i,t}$	-0.0001	0.004*	0.005*	-0.0005	0.033	0.018
	(0.000)	(0.002)	(0.003)	(0.000)	(0.022)	(0.016)
Liquid assets	0.0002**	-0.002	0.002***	-0.0001*	-0.002	-0.0002
	(0.000)	(0.002)	(0.000)	(0.000)	(0.001)	(0.0005)
Bank size	-0.0003**	0.001	-0.004***	0.0001	-0.001	-0.0003
	(0.000)	(0.002)	(0.001)	(0.000)	(0.001)	(0.0006)
Price loans	-0.020**	-0.282	-0.363**	0.0004	0.706	0.312
	(0.009)	(0.211)	(0.171)	(0.004)	(0.459)	(0.271)
Return loans	0.005**	0.019	0.136**	-0.000	-0.171*	-0.061
	(0.002)	(0.081)	(0.061)	(0.001)	(0.100)	(0.050)
Adjust. cost	0.008	0.008	0.211**	-0.002	-0.164*	-0.073
	(0.007)	(0.165)	(0.101)	(0.002)	(0.093)	(0.053)
Interest deposits	0.029**	1.408***	0.744**	-0.006	-0.976	-0.092
	(0.014)	(0.377)	(0.338)	(0.017)	(1.177)	(0.415)
Constant	0.002**	0.003	0.011***	0.001**	0.021	0.001
	(0.001)	(0.010)	(0.004)	(0.000)	(0.017)	(0.006)
Controls	✓	✓	✓	✓	✓	✓
Observations	138	165	304	94	87	182
Inverse Mills ratio	0.798	0.749	0.762	0.798	0.799	0.798

NOTE. Results of the second step Tobit regression specification of the marginal propensity to lend in units of deposits on residualised illiquid assets and banks' controls. 'Adjust. cost' indicates adjustment costs, 'size' identifies the level of total assets, 'illiquid res.' is the residualised illiquid assets, 'controls' indicates the presence of other controls beyond the ones reported in the table. Parentheses show robust standard errors. Significance with ***, **, * indicates p-values at 1%, 5% and 10% critical level, respectively.

Table H.1 reports the estimates of the Tobit model. The Tobit estimates in Table H.1 show a negative effect on the marginal propensity to lend for large banks in 1998 and 2010 due to idiosyncratic illiquidity. By contrast, small banks that are more illiquid than predicted tend to rise their marginal propensity to lend with a statistically significant effect in 1998. However, the Tobit estimates cannot provide guidance on the magnitude of the coefficient estimates, for which conditional and unconditional marginal effects are computed.

Table H.2 shows conditional marginal effects. When large banks become unexpectedly more illiquid, their marginal propensity to lend decreases, in contrast to small banks who experience a rise in the marginal propensity to lend. This effect is stronger post-financial crisis. The rise in the marginal propensity to lend of small banks in 2010 is approximately 869.058% higher than the 1998 counterpart.

By interpreting this finding through the market structure of banks, it shows evidence of negative assortative matching. Small banks, who increase the marginal propensity to lend because of higher illiquidity, amplify the shock of illiquidity risk in the market. When their

TABLE H.2.—

Conditional marginal effects of Tobit estimates on the marginal propensity to lend

	1998			2010		
	Large banks	Small banks	All banks	Large banks	Small banks	All banks
Illiquid res., $\tilde{s}_{i,t}$	-0.066	0.223**	0.333***	-1.581	2.161**	1.634*
	(0.127)	(0.114)	(0.142)	(1.492)	(0.954)	(0.905)
Liquid assets	0.128***	-0.103	0.164***	-0.455***	-0.141*	-0.015
	(0.053)	(0.094)	(0.045)	(0.173)	(0.075)	(0.052)
Bank size	-0.232***	0.070	-0.235***	0.335**	-0.063	-0.031
	(0.083)	(0.124)	(0.045)	(0.150)	(0.078)	(0.057)
Price loans	-13.328**	-15.812	-23.611***	1.306	46.617***	28.989**
	(6.467)	(11.860)	(07.305)	(14.188)	(19.344)	(15.498)
Return loans	3.573***	1.064	8.837***	-0.110	-11.297***	-5.634**
	(1.394)	(4.582)	(2.383)	(3.972)	(4.176)	(3.007)
Adjust. cost	5.028	0.438	13.710***	-6.395	-10.868***	-6.783**
	(3.956)	(9.242)	(4.658)	(4.436)	(3.873)	(2.992)
Interest deposits	19.049***	78.927***	48.376***	-20.655	-64.485	-8.561
	(7.191)	(16.114)	(12.720)	(51.085)	(67.634)	(36.462)
Controls	✓	✓	✓	✓	✓	✓
Observations	138	165	304	94	87	182

NOTE. Results of conditional marginal effects of the Tobit regression specification of the marginal propensity to lend. 'Adjust. cost' indicates adjustment costs, 'size' identifies the level of total assets, 'illiquid res.' is the residualised illiquid assets, 'controls' indicates the presence of other controls beyond the ones reported in the table. Parentheses show robust standard errors. Significance with ***, **, * indicates p-values at 1%, 5% and 10% critical level, respectively.

counterparts are large banks, this risk is absorbed by large banks who tend to be more stable and do not expand the marginal propensity to lend. In the event their counterparts are other small banks, the risk amplifies within the small banks' market. This pattern is consistent with negative assortative matching where idiosyncratic illiquidity of small banks translates into stronger marginal propensity to lend and the heterogeneity in the marginal propensity to lend is the result of how banks of different types match with one another in the given market structure.

The price of loans further corroborates the result on the illiquidity coefficient, and has a negative effect on the marginal propensity to lend of banks regardless of their size in 1998, and this effect becomes positive in 2010. In particular, large banks experience a 13.328 decay in the marginal propensity to lend given a rise in the price of loans, with no significant effect in 2010. By this time, higher loan prices lead to an increase of the marginal propensity to lend of small banks of 46.617. This result shows negative assortative matching between small banks and it is highlighted by the high price of loans that leads to high marginal propensity to lend and asymmetrically interacts with large banks.

The return on loans highlights the constrained and unconstrained nature of banks' balance sheet. This coefficient is positive in the years pre-financial crisis and negative post-financial crisis regardless of banks' size. In 1998 large banks increase illiquid assets following an increase

in the return on loans with an 3.573 unit increase that is statistically significant at 1% critical value. However, this relationship no longer holds in 2010 indicating that large banks become less constrained and can hold illiquid assets regardless of the return on loans. By contrast, for small banks in 1998 a higher return on loans increase illiquid assets, but the estimate is not statically significant, whereas in 2010 they drastically reduce illiquid assets by 11.297 units showing small banks becoming more constrained due to tightening in banks' regulations where small banks now price loans aggressively to compensate for a more risky balance sheet. Therefore, while the liquidity return trade-off is very pronounced for large banks, it is almost absent for small banks. Furthermore, the estimates for all banks cannot account for the heterogeneity between banks, since they overestimate the effect of the return on loans on illiquid assets of large banks and underestimate it for small banks.

Likewise, adjustment costs have a positive effect on the marginal propensity to lend in 1998, while they have a negative effect in 2010. However, the effect is not significant in 1998, and they reduce illiquid assets for small banks by 10.868 in 2010. The market structure of banks exhibits negative assortative matching, whereby small banks become highly sensitive to adjustment costs matching with large banks within their type. The matching pattern is therefore type dependent since large banks, as more stable banks, remain less affected by adjustment costs.

The conditional marginal effect on liquid assets shows negative assortative matching particularly in 2010. During this year, large banks strongly reduce the marginal propensity to lend by 0.455 when becoming more liquid, similarly to small banks who decreases the marginal propensity to lend by 0.141. The coefficient on all banks reduces to 0.015 because large and small banks have very different behavior. In particular, while the fall in the marginal propensity to lend of large banks is mainly stabilising, the fall of small banks is a more aggressive response toward liquid assets dictated by stronger liquidity requirements. This asymmetric response in the magnitude of liquid assets, highlights negative assortative matching with large banks matching with small banks generating this asymmetric response.

The Inverse Mills ratio ranges between 0.749 and 0.799 across all considered specifications, indicating the presence of a moderate positive selection into the marginal propensity to lend making the Tobit model specification necessary. Bank who lend, have higher marginal propensity to lend, therefore ignoring this positive selection would bias the coefficient estimates. However, differences in these estimates are not entirely driven by selection, but by the underlying size heterogeneity between banks. Furthermore, the Inverse Mills ratio strongly corroborates negative assortative matching between banks. This is because the Inverse Mills ratio is very similar and stable between large and small banks which means that the matching between large and small banks drives the selection into the marginal propensity to lend. Therefore, the heterogeneity in the coefficient estimates reflects the fact that small banks match with large banks and not differences in sample selection.

In Table 2 of the main text, the unconditional marginal effects of the return on loans and

the adjustment costs in illiquid assets show a similar pattern across banks. Both effects on the marginal propensity to lend are positive in 1998, and turn negative in 2010. The response of small banks' marginal propensity to lend is negative and statistically significant at 10% critical value with an estimated coefficient of 0.073 and 0.071 for the return on loans and the adjustment cost in illiquid assets, respectively; while these responses are mainly zero for large banks. Since the response of all banks' marginal propensity to lend shows a difference in magnitude relative to large and small banks, it reflects the asymmetric interaction between large and small banks and their matching rather than a homogeneous within type behavioral pattern.

Likewise, an increase in the price of loans has a consistent negative effect on the marginal propensity to lend pre-financial crisis for large and small banks, while it turns to a positive effect post-financial crisis. This result has implications for the market structure of banks, that is absent in 1998 but strongly negative assortative in 2010. While the estimates in 2010 are insignificant, they show a clear pattern of negative assortative matching. In particular, an increase in the price of loans has a large effect on the marginal propensity to lend of small banks of about 0.304 units, whereas is mainly zero for large banks. Combining this effect with the coefficient on all banks that is driven by small banks' behavior, then the marginal propensity to lend is mainly influenced by the interactions between large and small banks and not within each group of banks, corroborating the presence of negative assortative matching. The coefficient of the price of loans for all banks allows to disentangle the assortative matching pattern. This is because, if negative assortative matching between banks did not hold, one would expect similar coefficients in sign and magnitude between these two types of banks, and the coefficient on all banks would not be driven entirely from one type of banks, in contrast to the evidence.

The direction of the relationship between liquid assets and the marginal propensity to lend is shown with the estimates of liquid assets, which although small they are economically meaningful for the market structure. In 1998 more liquid assets lead large banks to rise the marginal propensity to lend slightly, and the magnitude of the effect is unchanged in 2010, but the sign reverts and shows a statistically significant 0.0001 reduction in the marginal propensity to lend of large banks. The negative effect is evident also for small banks whose liquidity impact is more persistent. While this effect is insignificant for small banks, it is 10 times the liquid assets effect of large banks in both considered years, which is economically meaningful since scaled by the probability of being uncensored. The coefficient for all banks mainly converges to zero because the market is driven by matching between large and small banks, that shows a very different behavior toward liquid assets.

H.2 Further Empirical Results on the Inequality in the Marginal Propensity to Lend and its Connection to Assortative Matching

I now supplement the remaining coefficients in Table 3 and 4 of the paper and their connection to the assortative matching. In 1998, liquid assets strongly increase the marginal propensity to

lend of large banks in the 25th quantile of the distribution. Any rise in liquid assets leads to a 12.590 increase in the marginal propensity to lend. Since large banks hold liquid buffers, they expand lending when liquidity is large. Medium large banks and medium small banks in the 50th and 75th quantile of the distribution, respond similarly to large banks to rise in liquidity, but their effect is lower since they are less sensitive to change in liquidity. Banks in the upper tail of the marginal propensity to lend distribution (90th quantile) are small banks who reduce the marginal propensity to lend given an increase in liquid assets. Even though the coefficient is not significant, it represents the expected response from small banks that are liquidity constrained and use liquidity to reduce the risk instead of expanding lending and the marginal propensity to lend, which only occurs when small banks become more illiquid.

When size is measured by deposits, the marginal propensity to lend pattern highlights the negative assortative matching between banks. In particular, an increase in deposits leads to a 3.25 increase in the marginal propensity to lend of the 25th quantile of the distribution. Progressing along the distribution of quantiles, this effect decreases until it becomes negative for the 90th quantile of banks who experience a 11.905 fall in the marginal propensity to lend. This monotonic pattern emphasises the peculiar characteristic that underpins heterogeneity in banks' deposits. Large banks that are mainly in the 25th quantile of the distribution, hold less deposits than small banks, therefore any increase in deposits fosters lending and the marginal propensity to lend. By contrast, small banks are the holder of large deposits and as banks liquidity constrained, upward changes in deposits are used to build the liquid buffer as self-insurance motive, rather than expanding loans.

In 2010, liquidity plays a significant role in the middle of the distribution for medium large and medium small banks that increase the marginal propensity to lend by 24.721 and 47.237 given an upward change in liquid assets. Relative to 1998, the behavior of large and small banks is different. Post-financial crisis large banks strongly reduce the marginal propensity to lend, in contrast to small banks that aggressively expand the marginal propensity to lend by 96.711 since the liquidity constraint has relaxed, thereby depicting size heterogeneity in the marginal propensity to lend.

The effect of deposits on the marginal propensity to lend reverts only for large banks in the 25th quantile of the distribution, relative to the effect in 1998. In this case, upward changes in deposits result in a fall of 1.571 of the marginal propensity to lend as large banks try to stabilise their balance sheet following the turmoils of the financial crisis. Medium large banks in the 75th quantile continue to expand lending and the marginal propensity to lend, while medium small banks and small banks liquidity constrained, use deposits with a self-insurance motive to reduce the exposure to risk constraining the banks' ability to expand lending. This result further highlights a redistribution of assets and substitution of funding across banks.

TABLE I.1.—

Tobit regression estimates of the marginal propensity to lend with interaction terms

	1998			2010		
	Large banks	Small banks	All banks	Large banks	Small banks	All banks
Profitab, $s^*_{i,t} r_{s,i,t}$	0.0002	0.041	0.016**	-0.0001	-0.019	-0.006
	(0.001)	(0.028)	(0.008)	(0.0003)	(0.041)	(0.011)
Illiquid res., $\tilde{s}_{i,t}$	-0.0001	0.003	0.005*	-0.0005	0.032	0.017
	(0.0002)	(0.002)	(0.003)	(0.0006)	(0.021)	(0.014)
Liquid assets	0.0002**	-0.001	0.002***	-0.0001	-0.002	-0.0002
	(0.000)	(0.002)	(0.001)	(0.0001)	(0.001)	(0.001)
Bank size	-0.0004	-0.001	-0.005***	0.0001	0.0003	0.0001
	(0.0002)	(0.002)	(0.001)	(0.0001)	(0.002)	(0.001)
Price loans	-0.019**	-0.195	-0.287***	0.0002	0.689	0.306
	(0.009)	(0.187)	(0.144)	(55.003)	(0.445)	(0.261)
Return loans	0.003	-0.461	-0.109	0.002	0.062	0.021
	(0.021)	(0.324)	(0.106)	(0.005)	(0.496)	(0.119)
Adjust. cost	0.008	-0.033	0.167*	-0.002	-0.154*	-0.067
	(0.007)	(0.157)	(0.093)	(0.002)	(0.090)	(0.047)
Interest deposits	0.028**	1.374***	0.710**	-0.007	0.930	-0.084
	(0.013)	(0.359)	(0.316)	(0.017)	(1.160)	(0.407)
Constant	0.003	0.0269	0.024***	0.001	0.007	-0.004
	(0.002)	(0.018)	(0.009)	(0.000)	(0.028)	(0.012)
Controls	✓	✓	✓	✓	✓	✓
Observations	138	165	304	94	87	182
Inverse Mills ratio	0.798	0.749	0.761	0.798	0.799	0.798

NOTE. Results of the Tobit regression specification of the marginal propensity to lend on residualised illiquid assets, interaction terms and banks' controls. Parentheses show robust standard errors. Significance with ***, **, * indicates p-values at 1%, 5% and 10% critical level, respectively.

I Robustness Empirical Analysis

In this Section I implement a robustness analysis on the empirical study investigated in the main text for the Tobit and the quantile analysis.

I.1 Robustness Tobit Marginal Propensity to Lend

In this robustness experiment, I take into account interaction effects in the Tobit specification in Subsection 5.1 and implement the following model:

$$\begin{aligned} \text{mpl}_{i,t} &= \beta_0 + \beta_1 s^*_{i,t} r_{s,i,t} + \beta_2 \tilde{s}_{i,t} + \beta_3' \mathbf{X}_{i,t} + \epsilon_{i,t}, \quad \forall i, \forall \mathbb{1}(i \in [s_{min}, \overline{s}_{min}]), \\ \text{mpl}_{i,t} &= \max(0, \text{mpl}^*_{i,t}), \end{aligned} \tag{I.1}$$

where, in equation I.1 the marginal propensity to lend is identified as the change in loans induced by the change in the return on loans. The interaction term of interest is $s^*_{i,t} r_{s,i,t}$. This term is illiquid assets interacted with the return on loans, $s^*_{i,t} r_{s,i,t}$, which I interpret as a measure of

TABLE I.2.—

Conditional marginal effects with interaction terms

	1998			2010		
	Large banks	Small banks	All banks	Large banks	Small banks	All banks
Profitab, $s^*_{i,t} r_{s,i,t}$	0.125	2.315	1.092***	-0.431	-1.259	-0.578
	(0.911)	(1.596)	(0.465)	(1.169)	(2.626)	(0.857)
Illiquid res., $\tilde{s}_{i,t}$	-0.054	0.194*	0.367***	-1.632	2.133**	1.582*
	(0.161)	(0.112)	(0.153)	(1.484)	(0.935)	(0.862)
Liquid assets	0.131**	-0.088	0.161***	-0.425**	-0.154**	-0.024
	(0.056)	(0.093)	(0.044)	(0.195)	(0.079)	(0.054)
Bank size	-0.247*	-0.077	-0.313***	0.332**	0.022	0.013
	(0.141)	(0.143)	(0.053)	(0.151)	(0.158)	(0.086)
Price loans	-12.815**	-10.999	-18.887***	0.905	45.771***	28.499**
	(6.651)	(10.576)	(6.692)	(14.238)	(18.985)	(15.026)
Return loans	1.769	-25.989	-7.196	6.067	4.127	1.982
	(12.940)	(18.183)	(6.861)	(17.616)	(32.803)	(10.785)
Adjust. cost	5.039	-1.879	10.973**	6.251	-10.209***	-6.229**
	(3.957)	(8.813)	(4.921)	(4.475)	(4.101)	(2.882)
Interest deposits	18.747***	77.497***	46.739***	-21.757	-61.768	-7.786
	(6.877)	(15.605)	(12.045)	(51.167)	(67.757)	(36.125)
Controls	✓	✓	✓	✓	✓	✓
Observations	138	165	304	94	87	182

NOTE. Results of conditional marginal effects of the Tobit regression specification of the marginal propensity to lend with interaction terms. Parentheses show robust standard errors. Significance with ***, **, * indicates p-values at 1%, 5% and 10% critical level, respectively.

profitability. With a large interaction term, high return on loan amplifies the marginal propensity to lend since banks are willing to extend more loans, and viceversa. Consequently, banks may focus on improving the performance of loans when the coefficient is positive.

Tables I.1, I.2 and I.3 provide the results for the Tobit estimates, the conditional marginal effects, and the unconditional marginal effects. The results are consistent with the main analysis in the text even after controlling for the interaction term, which highlight a consistent negative assortative matching between banks.

In particular, the results on censored observations indicate that profitable loans increase the marginal propensity to lend for all banks in 1998 by 1.092 whereas the response is negative in 2010 and equals -0.578. For small banks profitable loans raise the marginal propensity to lend by 2.315 in 1998 and decrease it to -1.259 in 1998, that is a 154.384% fall relative to the 1998 response. The change in the sign and magnitude of profitable loans on the marginal propensity to lend is also highlighted for large banks, whose response is a fall of 444.7% which albeit insignificant, show a strong negative assortative matching in the market structure. In particular, the response of all banks' marginal propensity to lend is highly driven by small banks whose match with large banks makes very unlikely this estimate to be half the large and

TABLE I.3.—
Unconditional marginal effects with interaction terms

	1998			2010		
	Large banks	Small banks	All banks	Large banks	Small banks	All banks
Profitab, $s^*_{i,t} r_{s,i,t}$	0.0001	0.020	0.008**	-0.0001	-0.008	-0.003
	(0.0008)	(0.014)	(0.004)	(0.0002)	(0.018)	(0.004)
Illiquid res., $\tilde{s}_{i,t}$	-0.000	0.002	0.003*	-0.0003	0.014	0.007
	(0.000)	(0.001)	(0.001)	(0.0003)	(0.009)	(0.006)
Liquid assets	0.0001**	-0.001	-0.001***	-0.0001*	-0.001	-0.0001
	(0.000)	(0.001)	(0.000)	(0.0000)	(0.001)	(0.0002)
Bank size	-0.0002	-0.001	-0.002***	0.0001*	0.0001	0.0001
	(0.0001)	(0.001)	(0.000)	(0.0000)	(0.001)	(0.0004)
Price loans	-0.011**	-0.096	-0.136**	0.0001	0.296	0.130
	(0.005)	(0.092)	(0.066)	(0.002)	(0.189)	(0.113)
Return loans	0.001	-0.228	-0.052	0.001	0.027	0.009
	(0.011)	(0.158)	(0.050)	(0.002)	(0.213)	(0.051)
Adjust. cost	0.004	-0.016	0.079**	-0.001	-0.066*	-0.028
	(0.004)	(0.077)	(0.043)	(0.001)	(0.039)	(0.021)
Interest deposits	0.016**	0.680***	0.338**	-0.003	-0.399	-0.036
	(0.007)	(0.171)	(0.144)	(0.009)	(0.492)	(0.173)
Controls	✓	✓	✓	✓	✓	✓
Observations	138	165	304	94	87	182

NOTE. Results of unconditional marginal effects of the Tobit regression specification of the marginal propensity to lend allowing for interaction terms. Parentheses show robust standard errors. Significance with ***, **, * indicates p-values at 1%, 5% and 10% critical level, respectively.

small banks' responses.

The unconditional marginal effects maintain the same sign of response of the conditional marginal effects however in smaller magnitude. For all banks, profitable loans lead to a 0.008 (0.003) rise (decrease) in the marginal propensity to lend in 1998 (2010). The response of loans profitability for large banks is sign reverted with loans' profitability decreasing the marginal propensity to lend in 2010 by a small magnitude, whereas the response is positive but not significant in 1998. A rise in small banks loans' profitability leads to a fall in the marginal propensity to lend that is 80 times higher than the response of large banks. The rest of the coefficients leads to a response in the marginal propensity to lend that is largely consistent with the main specification.

I.2 Robustness Quantile Inequality Marginal Propensity to Lend

In the robustness experiments for the quantile regression, I perform the analysis for large and small banks separately to study the heterogeneity in the marginal propensity to lend within banks. Table I.4 and I.5 show the results for large banks in 1998 and 2010, respectively. The results underscore strong within heterogeneity. In 1998 large banks always consistently reduce

TABLE I.4.—

Inequality in the marginal propensity to lend of large banks in 1998

	1998			
	25th quantile	50th quantile	75th quantile	90th quantile
Illiquid assets	-10.132	-2.023	-1.083	-1.912
	(8.164)	(4.377)	(8.841)	(38.502)
Liquid assets	19.628*	7.294	4.387	2.216
	(10.889)	(7.282)	(25.118)	(109.162)
Size deposits	1.727	-2.936	-2.060	-16.831
	(4.243)	(2.696)	(3.767)	(17.610)
Constant	-0.397	55.539*	54.843	278.711
	(53.993)	(34.191)	(44.066)	(254.005)
Controls	✓	✓	✓	✓
Observations	138	138	138	138
Pseudo R^2	0.014	0.012	0.014	0.032

NOTE. Results of quantile regression analysis of the marginal propensity to lend for large banks in 1998. 'Adjust. cost' indicates adjustment costs, 'size' identifies the log of deposits, 'controls' indicates the presence of other controls beyond the ones reported in the table. Parentheses show robust standard errors computed with bootstrapping method in 5,000 resampling with replacement. Significance with ***, **, * indicates p-values at 1%, 5% and 10% critical level, respectively.

TABLE I.5.—

Inequality in the marginal propensity to lend of large banks in 2010

	2010			
	25th quantile	50th quantile	75th quantile	90th quantile
Illiquid assets	6.989	-3.249	-12.661	-38.122
	(14.888)	(7.800)	(13.783)	(37.797)
Liquid assets	31.006	31.195	63.912*	19.276
	(37.739)	(22.521)	(34.528)	(143.596)
Size deposits	-0.749	2.704	-1.623	20.810
	(5.961)	(3.264)	(5.887)	(15.470)
Constant	-19.763	-32.397	41.188	-155.847
	(80.382)	(48.167)	(91.414)	(208.173)
Controls	✓	✓	✓	✓
Observations	94	94	94	94
Pseudo R^2	0.017	0.041	0.055	0.108

NOTE. Results of quantile regression analysis of the marginal propensity to lend for large banks in 2010. 'Adjust. cost' indicates adjustment costs, 'size' identifies the log of deposits, 'controls' indicates the presence of other controls beyond the ones reported in the table. Parentheses show robust standard errors computed with bootstrapping method in 5,000 resampling with replacement. Significance with ***, **, * indicates p-values at 1%, 5% and 10% critical level, respectively.

the marginal propensity to lend given a rise in illiquid assets across quantiles. The higher the quantile, the larger the fall in magnitude of the marginal propensity to lend.

Likewise, the marginal propensity response to liquid assets shows a consistent increase in

the marginal propensity to lend of large banks across each quantile of large banks' distribution. The effect is declining as quantiles increase and large banks decrease in size. Since all sample estimates show a sign reversal at the 90th quantile of the distribution, they corroborate that the inequality in the marginal propensity to lend is mainly due to small banks' fundamentals because large banks show a more uniform behavior across quantiles.

Banks in the 25th quantile of the distribution slightly increase lending in light of deposit growth, mainly in those banks where deposits are used as stable funding source. The rest of large banks strongly reduces the marginal propensity to lend to redistribute their assets and reduce the liquidity risk.

In 2010 large banks' marginal propensity to lend response to illiquid assets shows a sign reversal at the tails of the quantile distribution. This response is characterised by a strong reduction in the marginal propensity to lend for banks in the 90th quantile of the distribution, while an upward movement is recorded for large banks in the 25th quantile. Liquid assets play an important role during this time, since large banks can expand lending faster and redistribute their liquidity towards loans, thereby increasing the marginal propensity to lend.

Within large banks, deposits drives the largest degree of heterogeneity in the marginal propensity to lend. The results show that large banks and medium small banks reduce the marginal propensity to lend when deposits rise, whereas medium large banks and small banks strongly increase the marginal propensity to lend to diversify their assets.

Table I.6 and I.7 show the quantile estimates within the distribution of small banks. In 1998 they show a response of the marginal propensity to lend due to increase in illiquid assets and liquid assets that is consistent with the pattern highlighted for large banks in this year. The difference in heterogeneity is shown by deposits. In this case the largest banks within small banks reduce the marginal propensity to lend given a rise in deposits, particularly as these are the banks that already hold less deposits and therefore the additional deposits are used to redistribute assets and self-insure against the liquidity risk. Banks located in the rest of the quantiles instead hold more deposits, are willing to increase their lending activity and therefore the marginal propensity to lend.

In 2010 any rise in illiquid assets and deposits strongly reduces the marginal propensity to lend of small banks. The magnitude of the fall decreases as small banks grow in size and the reach they 25th quantile. Following the financial crisis, small banks become mainly liquidity constrained and any additional source of short term debt is used to cover the liquidity risk. Liquid assets have a contractionary effect on the marginal propensity to lend of the largest and medium largest banks within small banks, since they use the additional liquidity to build stable funds. The medium smallest and smallest banks within small banks, increase the marginal propensity to lend since they are the banks more sensitive to liquidity shocks and therefore, they expand lending more quickly when liquidity rises.

These results indicate uniform heterogeneity within large banks and a more pronounced

TABLE I.6.—

Inequality in the marginal propensity to lend of small banks in 1998

	1998			
	25th quantile	50th quantile	75th quantile	90th quantile
Illiquid assets	-15.345***	-15.095***	-9.417	-3.838
	(5.511)	(4.813)	(6.761)	(46.057)
Liquid assets	12.784	10.713	7.071	8.061
	(11.723)	(10.581)	(14.456)	(61.005)
Size deposits	-3.238	2.112	3.316	21.6655
	(4.493)	(3.073)	(5.042)	(25.112)
Constant	-5.868	14.172	-14.637	-125.582
	(53.351)	(39.928)	(55.033)	(282.593)
Controls	✓	✓	✓	✓
Observations	165	165	165	165
Pseudo R^2	0.049	0.033	0.019	0.022

NOTE. Results of quantile regression analysis of the marginal propensity to lend for small banks in 1998. 'Adjust. cost' indicates adjustment costs, 'size' identifies the log of deposits, 'controls' indicates the presence of other controls beyond the ones reported in the table. Parentheses show robust standard errors computed with bootstrapping method in 5,000 resampling with replacement. Significance with ***, **, * indicates p-values at 1%, 5% and 10% critical level, respectively.

heterogeneity within small banks. Therefore, the size heterogeneity that underpins all banks' distribution is remarkably shaped by small banks who have a more heterogeneous marginal propensity to lend.

TABLE I.7.—

Inequality in the marginal propensity to lend of small banks in 2010

	2010			
	25th quantile	50th quantile	75th quantile	90th quantile
Illiquid assets	-2.032	-9.357	-10.989	-96.331
	(12.285)	(11.988)	(27.682)	(101.793)
Liquid assets	-47.472	-13.768	10.893	173.185
	(41.941)	(41.971)	(86.051)	(349.168)
Size deposits	-7.866	-0.702	-15.902	-112.174
	(5.110)	(6.520)	(27.068)	(115.502)
Constant	101.393	36.595	214.235	1,621.439
	(72.297)	(90.079)	(388.473)	(1,648.498)
Controls	✓	✓	✓	✓
Observations	87	87	87	87
Pseudo R^2	0.022	0.004	0.022	0.109

NOTE. Results of quantile regression analysis of the marginal propensity to lend for small banks in 2010. 'Adjust. cost' indicates adjustment costs, 'size' identifies the log of deposits, 'controls' indicates the presence of other controls beyond the ones reported in the table. Parentheses show robust standard errors computed with bootstrapping method in 5,000 resampling with replacement. Significance with ***, **, * indicates p-values at 1%, 5% and 10% critical level, respectively.

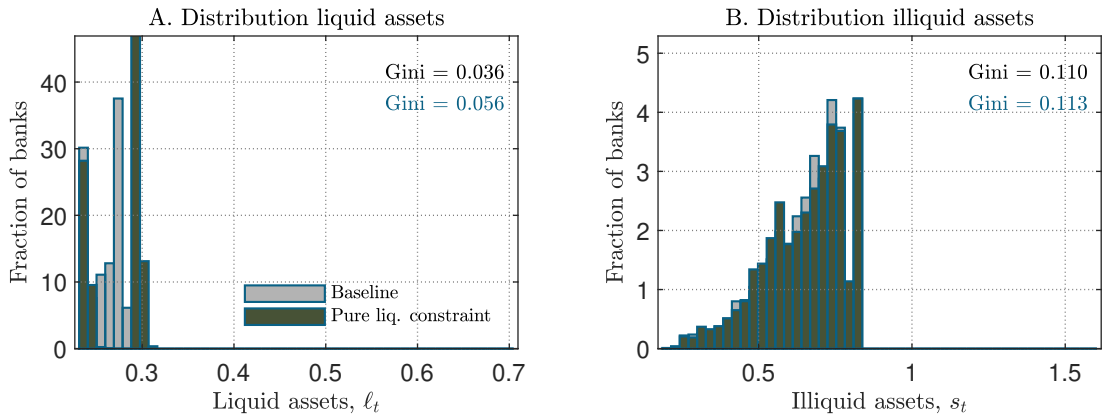


FIG. J.1.—Distribution banks' liquid and illiquid assets.

NOTE. Distribution of the banks' liquid assets in panel A and illiquid assets in panel B. Light gray distributions report the result for the baseline model and the green distributions present the result for the model with the pure liquidity constraint. The first line of reported numbers in black refers to the baseline model and the green lines to the model with a pure liquidity constraint. Liquid and illiquid assets are computed in units of deposits and total assets, respectively. Each point mass is divided by the width of the bin in the respective space of liquid and illiquid assets.

J Quantitative Appendix

In this Section I present additional quantitative results of the model.

J.1 Stationary Equilibrium Additional Analysis

Figure J.1 presents the cross-sectional distribution of liquid assets in Panel A and the distribution of illiquid assets in Panel B. The x-axis represents the histogram bins for the liquidity distribution and the illiquid assets distribution, whilst the y-axis represents the fraction of banks in each liquidity and illiquid assets bin, respectively. The model reveals that in the baseline economy the distribution of liquid assets shows denser mass at the lower end of the distribution where approximately 28.54% of banks hold a small amount of liquid assets and are liquidity constrained. The model with a pure liquidity constraint shows a more unequal distribution of liquid assets that is mainly concentrated in two mass points of the distribution corresponding to a fraction of 26.69% of banks holding less than 0.236 liquid assets in units of deposits, and a small fraction of 12.43% of banks holding large amount of liquid assets.

Panel B shows the cross-sectional distribution of illiquid assets. Under both model specifications, the model contains a mass concentrated toward the middle of the illiquid assets distribution. More precisely, in the baseline economy a fraction of 11.93% of banks hold high illiquid assets whereas 12% of banks are in the high values of the illiquid assets distribution in the economy with a pure liquidity constraint. The small Gini coefficient in both economies respectively at 11% and 11.3%, reflects the normalisation of illiquid assets to total assets in the calibration of the model. The distribution of illiquid assets obtained from the model is consistent with the distribution observed in the data in 2010 where there was a shift to the right of the

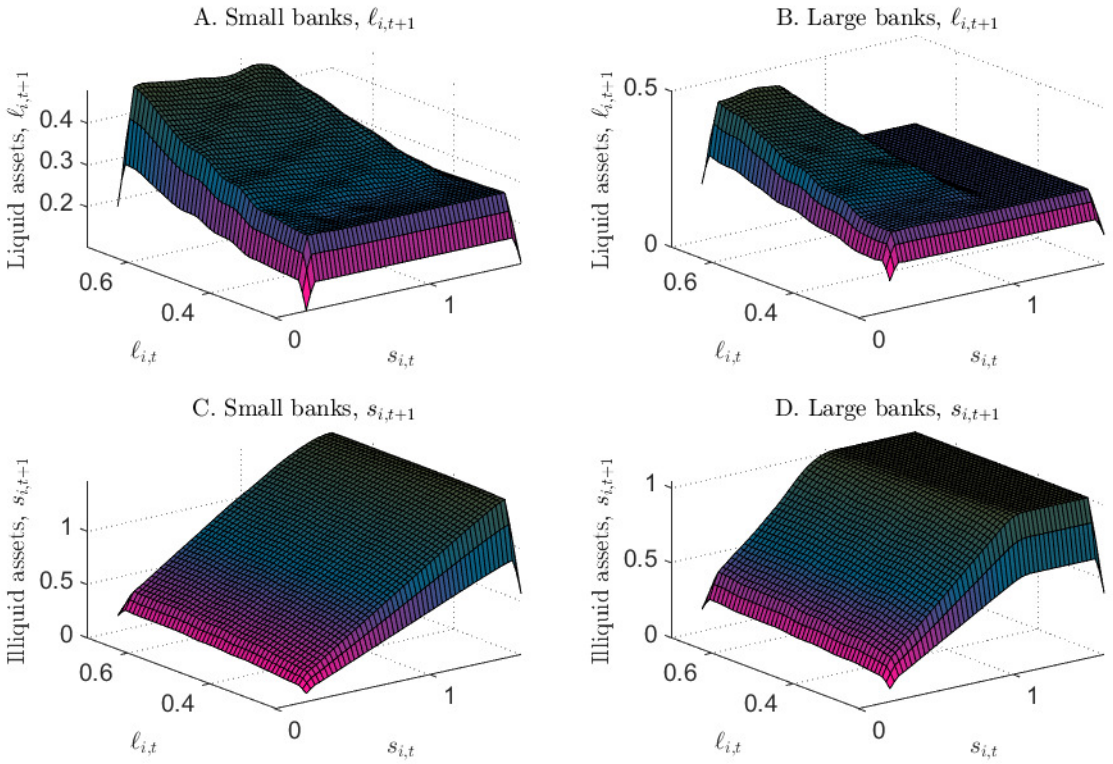


FIG. J.2.—Decision rules for liquid and illiquid assets by bank size.

NOTE. The figure shows decision rules for liquid and illiquid assets for small banks in panel A and C, and large banks in panel B and D, respectively.

distribution with a fatter left tail indicating the presence of some matching between large and small banks.

Figure J.2 shows decision rules for liquid assets and illiquid assets for small banks in panel A and C and large banks in panel B and D, respectively. For large banks, liquid and illiquid assets decision rules are monotonically increasing, and additional liquidity is used to expand illiquid assets. By contrast, small banks need to build the buffer of liquidity before expanding illiquid assets. Holding current illiquid assets constant, small banks show higher current liquid assets than large banks and higher future liquid assets by 0.72% at 0.6053 holding of liquid assets, consistent with the assumptions of the submodular liquidity constraint which make the constraint tighter for banks holding high current liquid assets and relaxes their liquid assets in the future. Large banks show a steady increasing path of liquid assets, which is slightly lower than small banks because of their investment in illiquid assets.

Illiquid assets decision rules reveal that small banks raise illiquid assets steeply and faster at low levels of current illiquid assets, since at this level of current illiquid assets the price of illiquid assets is high, however, their level is smaller than large banks. By contrast, large banks raise future illiquid assets in accordance with their better initial conditions of holding liquid and

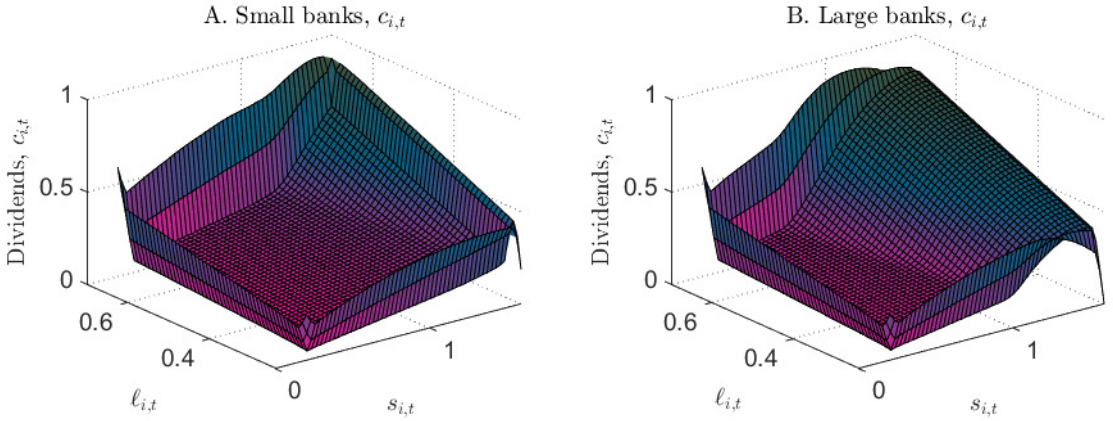


FIG. J.3.—Decision rules for dividends by bank size.

NOTE. The figure shows decision rules for dividends for small banks in panel A and large banks in panel B, respectively.

illiquid assets. Nevertheless, when they have reached a larger size, any further increases in liquid assets do not increase future illiquid assets, and the decision rule for this type of banks becomes flat, while being higher than small banks at all current liquid and illiquid assets values.

Figure J.3 reports decision rules of dividends for small banks in Panel A and large banks in Panel B. Large banks' dividends are higher and smoother than the dividends of small banks, and show a constant path over both current liquid and illiquid assets. They also show a more pronounced hump-shaped path at high values of current illiquid assets but low levels of liquid assets than small banks. Small banks build their buffer of liquidity with dividends to expand illiquid assets and show a hump-shaped path in the grid of illiquid assets. Dividends for small banks increase until the marginal value of holding additional liquid assets falls.

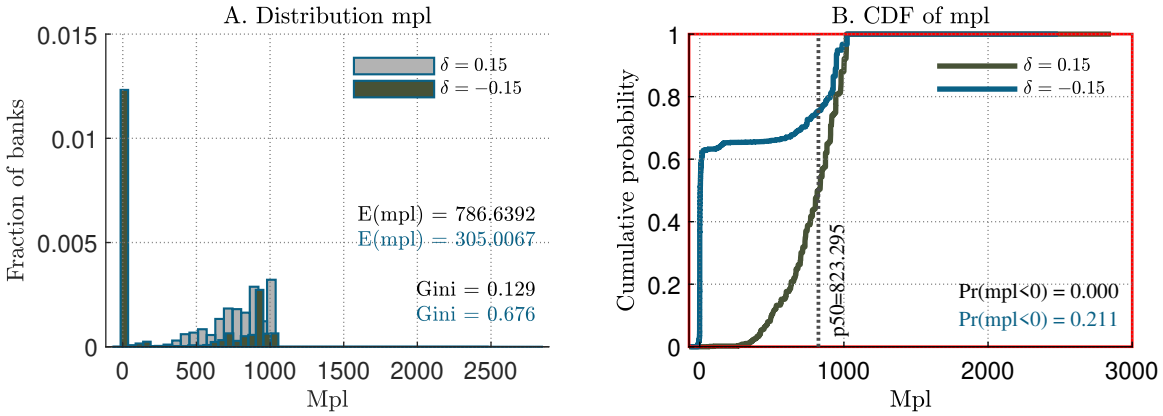


FIG. J.4.—Distribution banks' marginal propensity to lend at different submodularity.

NOTE. Distribution of the banks' marginal propensity to lend in panel A and its CDF in panel B for different values of the submodularity value. In panel A light gray distribution reports the result for $\delta = 0.15$ and the green distribution presents the result for the model with $\delta = -0.15$. In panel B the vertical line reports the 50th percentile of the marginal propensity to lend for the model with $\delta = 0.15$. The first line of reported quantities in black refers to the baseline model with $\delta = 0.15$ and the green lines to the model with $\delta = -0.15$.

Figure J.4 shows the distribution of the marginal propensity to lend for different values of the strength of submodularity between liquid assets and short term debt. Panel A shows in light gray the distribution with $\delta = 0.15$. Relative to the baseline value, the results reveal that as the strength of submodularity rises, the distribution of the marginal propensity to lend contains more mass toward the middle of the distribution becoming less concentrated. This is the essence of negative assortative matching, because the distribution of the marginal propensity to lend becomes more balanced toward the middle, this means that matching between large and small banks is the supported matching in the banking sector. The increase in δ makes holding of liquid assets costly for large banks, therefore, the viable way to increase their liquid assets and invest more in illiquid assets is by matching with small banks who, instead, benefit from the relaxation of the constraint.

Since inequality in the marginal propensity to lend significantly reduces to 12.9% when measured through the Gini coefficient the system becomes more stable with a fraction of 0.3% of banks centered with a marginal propensity to lend of $1,008.76$. The distribution has therefore not only shifted to the right, but also more mass is created and the additional mass comes from small banks matching with large banks, thereby redefining the market structure. Consequently, the probability of observing a negative marginal propensity to lend is zero almost surely.

By contrast, when the strength of submodularity takes on negative values ($\delta = -0.15$), small banks are massively liquidity constrained impairing their ability to expand illiquid assets, while only few large banks can sustain a high marginal propensity to lend. The distribution becomes dispersed toward the middle with a fraction of 1.23% of banks having a marginal propensity to lend of 10.801 and only 0.23% of banks holding more than three times the average marginal propensity to lend in the economy (928.867). The cumulative distribution function associated with it becomes very steep since now there is a 21.1% probability of observing a negative marginal propensity to lend.

J.2 Supplement of Assortative Matching

Figure J.5 shows how the market structure is endogenously formed across banks by their type. In particular, it points out the V-shaped path of the marginal propensity to lend along the deposits' distribution for large and medium banks, while it is mainly flat and slightly decreasing for small banks. In particular, the impact of the submodularity term is shown in the decision rule of liquidity assets for large banks which depicts a very clear picture. The optimal choice for large banks is to increase illiquid assets more than liquid assets since illiquid assets bring a higher return. Yet, the submodularity in the liquidity constraint tights liquid assets of large banks as deposits rise. Banks can run down their liquid assets. At high deposits states, large banks use dividends to satisfy the tighter liquidity constraint and they become decreasing as deposits rise. In contrast to small banks, their tighter constraint is brought by the strength of submodularity, whereby deposits and liquid assets become complement and large banks find

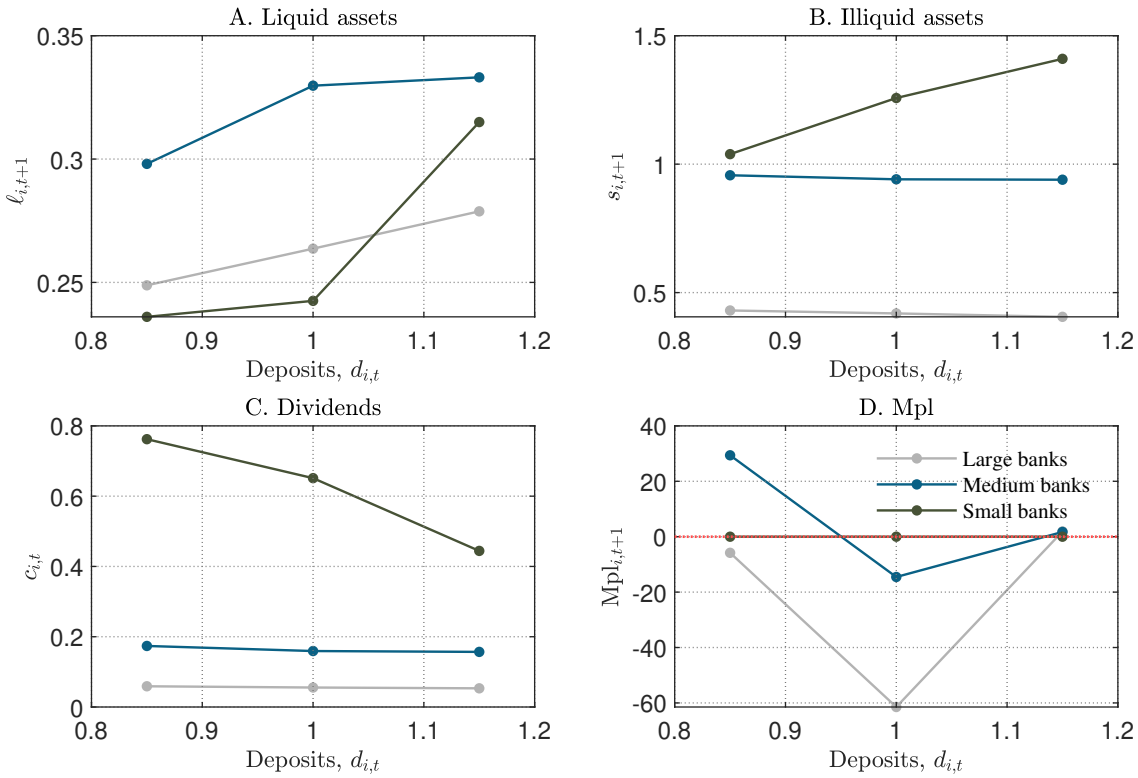


FIG. J.5.—Negative assortative matching by banks' size.

NOTE. The figure shows liquid assets, illiquid assets, dividends and the marginal propensity to lend by banks' size in the space of deposits.

optimal to match with small banks to relax the constraint.

For small banks, high deposits increase liquid and illiquid assets significantly, while dividends fall to satisfy the liquidity constraint. Since small banks have less stable funding inflow, any upward change in the interest rate on loans generates a lower marginal propensity to lend as deposits increase. By matching with large banks, small banks can stabilise their deposits base by raising illiquid assets with a positive marginal propensity to lend.

Figure J.6 shows the negative assortative matching in the banking sector by varying the strength of submodularity between deposits and liquid assets in the liquidity constraint. Relative to the baseline model, a higher δ, makes liquid assets more costly for large banks as deposits grow, as a consequence liquid assets fall while dividends rise to satisfy the liquidity constraint. While liquid assets rise by a small amount, the marginal propensity to lend is very sensitive to the strength of submodularity, particularly for medium and large banks. The matching generates the expansion in liquid assets driven by small banks, and a larger marginal propensity to lend driven by medium banks that are more sensitive to the price of illiquid assets.

By contrast, a lower value of δ, mainly relaxes the liquidity constraint for large banks that increase liquid assets more strongly. High deposits states are now associated with a much lower marginal propensity to lend, while lower dividends generate lower illiquid assets for medium and small banks.

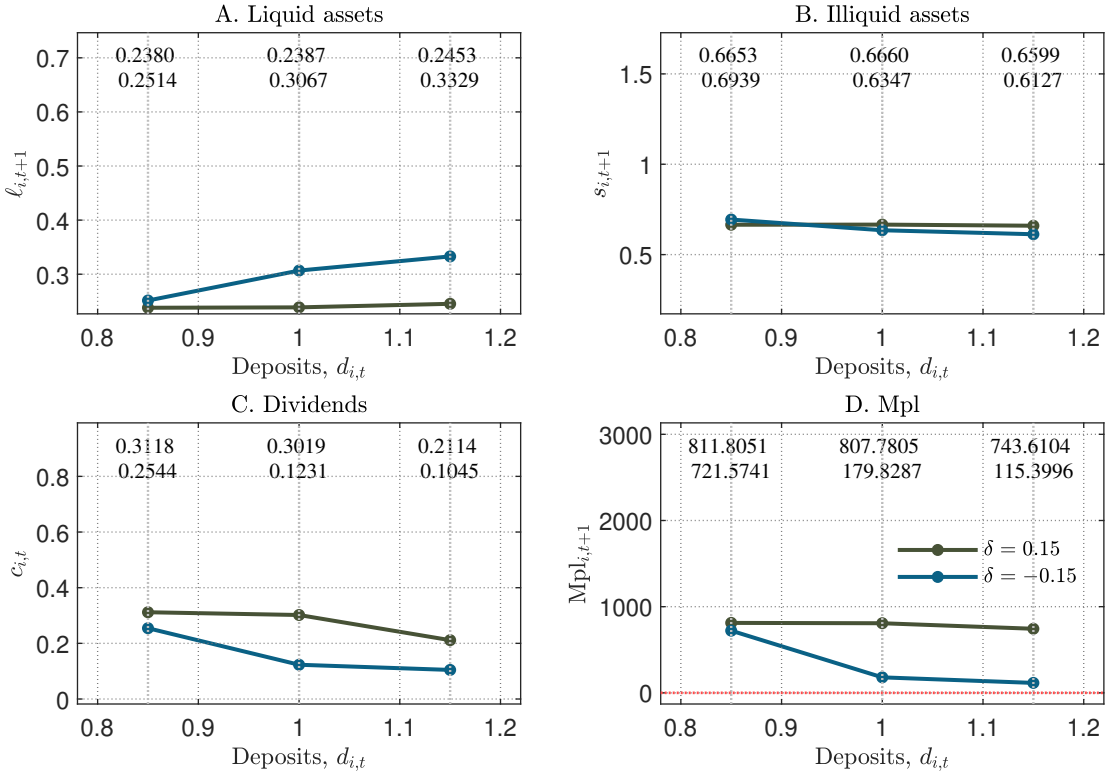


FIG. J.6.—Negative assortative matching by strength of submodularity.

NOTE. The figure shows ergodic means across states for liquid assets, illiquid assets, dividends and the marginal propensity to lend in the space of deposits, under different strength of submodularity, $\delta = 0.15$ and $\delta = -0.15$. Numbers on each state denotes mean values of decision rules across states. First line of numbers is for the model with $\delta = 0.15$, and second line is for the model with $\delta = -0.15$.

Figure J.7 shows how the market structure of banks changes by varying the leverage constraint, κ, first at the mean value of illiquid assets observed in the data, $\kappa = 0.823$ and then above double its value, $\kappa = 1.900$. By varying the leverage constraint, a high value of κ relaxes the constraint of banks since they can hold more illiquid assets per units of deposits, whereas when κ is low, banks are leverage constrained and any increase in deposits does not translate into more illiquid assets, relative to the baseline.

With a low leverage constraint parameter, banks need to build the buffer of liquidity and expand illiquid assets with dividends, because the leverage constraint is binding, consequently leading to a fall in the marginal propensity to lend. By contrast with a high value of κ, banks can expand illiquid assets through the additional deposits and dividends, keeping liquid assets unchanged relative to the baseline economy. While large at low deposit states, the marginal propensity to lend decreases at large deposits states. The market exhibits negative assortative matching between large banks who are less leverage constrained, and small banks for whom the leverage constraint is tight, and whose matching generates a larger and more stable surplus, since they can expand illiquid assets without significantly impairing dividends and liquid assets.

Figure J.8 reports the negative assortative matching pattern between banks for the model

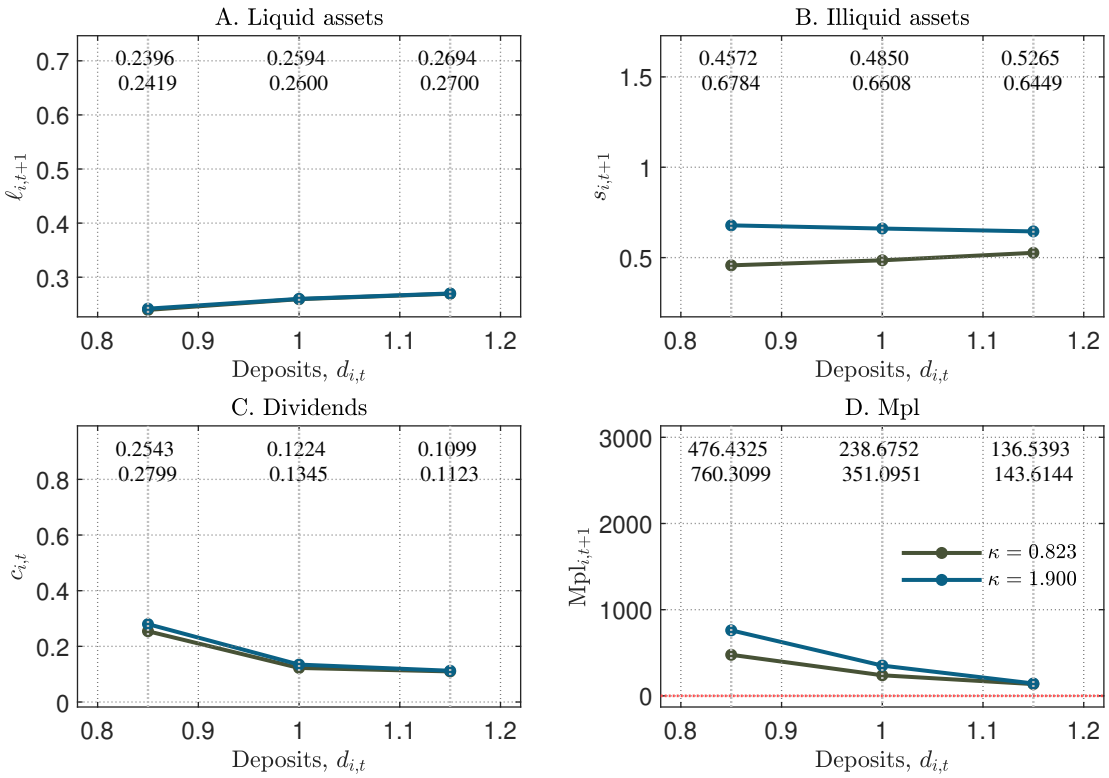


FIG. J.7.—Negative assortative matching by leverage constraint.

NOTE. The figure shows ergodic means across states for liquid assets, illiquid assets, dividends and the marginal propensity to lend in the space of deposits, under different values of the leverage constraint, $\kappa = 0.823$ and $\kappa = 1.90$. Numbers on each state denotes mean values of decision rules across states. First line of numbers is for the model with $\kappa = 0.823$, and second line is for the model with $\kappa = 1.90$.

with a higher fraction of exit, $\delta_e = 0.30$, than the baseline model. Increasing the fraction of banks' exit, raises current dividends and decreases the marginal propensity to lend. Essentially, banks become less patient and while leaving liquid and illiquid assets unchanged, the effective discount factor changes downward the incentive banks have in extending illiquid assets, thorough the marginal propensity to lend.

J.3 Supplement Distributional Adjustment Costs in Illiquid Assets

Figure J.9 shows the path of liquid assets (Panel A), illiquid assets (Panel B), the price of illiquid assets (Panel C), and the marginal propensity to lend (Panel D) as a function of adjustment costs, for different values of the fraction of exiting banks. Without exit, banks discount future dividends more strongly, they are more patient and long-lived; therefore, they care about future values of dividends. Liquid and illiquid assets, the price of illiquid assets and the marginal propensity to lend are higher than the baseline model in magnitude. As $\delta_e \to 0$ and adjustment costs rise, the price of illiquid assets increases, giving banks incentives to extend loans, thus, increasing the marginal propensity to lend. The economy is mainly dominated by large banks who can hold large liquid and illiquid assets.

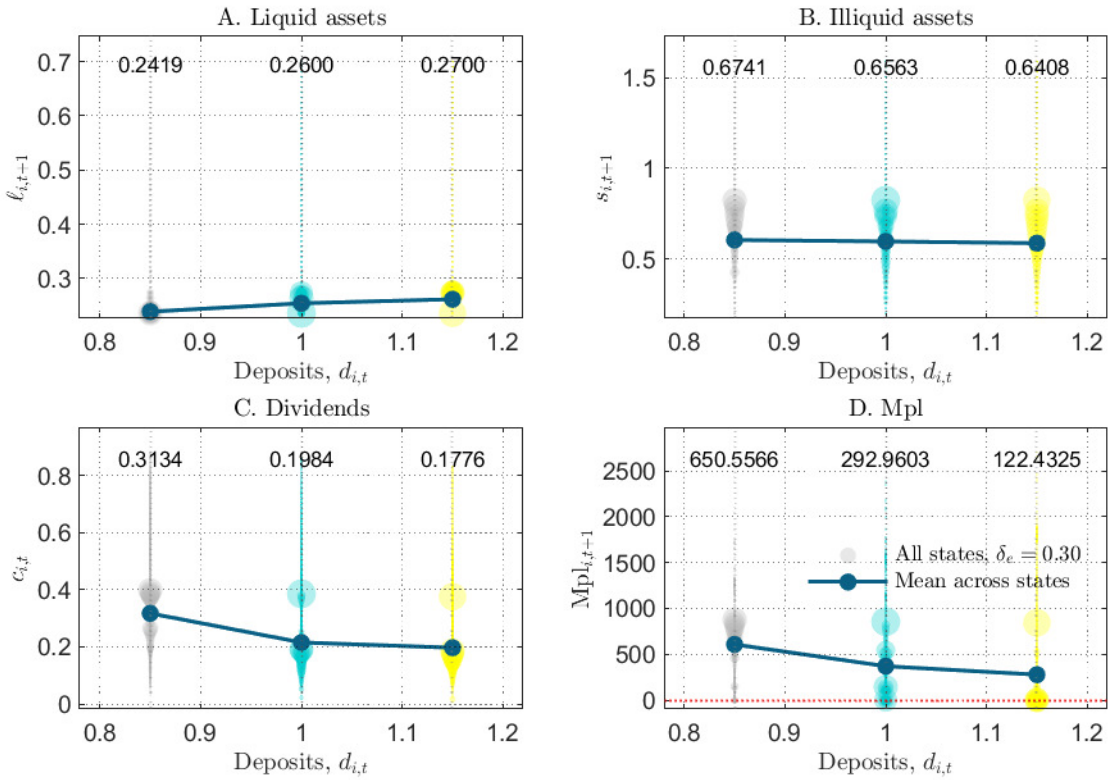


FIG. J.8.—Negative assortative matching by entry and exit.

NOTE. The figure shows ergodic means across states for liquid assets, illiquid assets, dividends and the marginal propensity to lend in the space of deposits, for the model with $\delta_e = 0.30$. Circles shows the states dimensions for each decision rule over the three states of deposits. Numbers on each state denotes mean values of decision rules across states.

As δ_e increases, banks become short-lived, the effective discount factor falls, therefore, they care less about future dividends, and optimise giving more weight to current liquid and illiquid assets. Because banks are short lived, their balance sheet is smaller since they have less time to expand liquid and illiquid assets through the future, therefore their holding of liquid and illiquid assets is substantially lower than the baseline values. As adjustment costs increase, the price of illiquid assets falls, however, when δ_e increases, the marginal propensity to lend increases toward an economy without exit. This occurs because the system becomes dominated by small banks who are sensitive to changes in the price of illiquid assets, the constraint is less binding at the margin, and the marginal return to a small rise in the interest rate on loans is higher. The negative assortative matching pattern is even stronger since, as δ_e increases, the marginal propensity to lend converges to the marginal propensity to lend of the model without exit dominated by large banks matching with small banks and reconciling the increasing arm of the V-shaped path of the marginal propensity to lend analysed in Section 6.3.4.

Figure J.10 reports the path of liquid assets (Panel A), illiquid assets (Panel B), the price of illiquid assets (Panel C), and the marginal propensity to lend (Panel D) as a function of

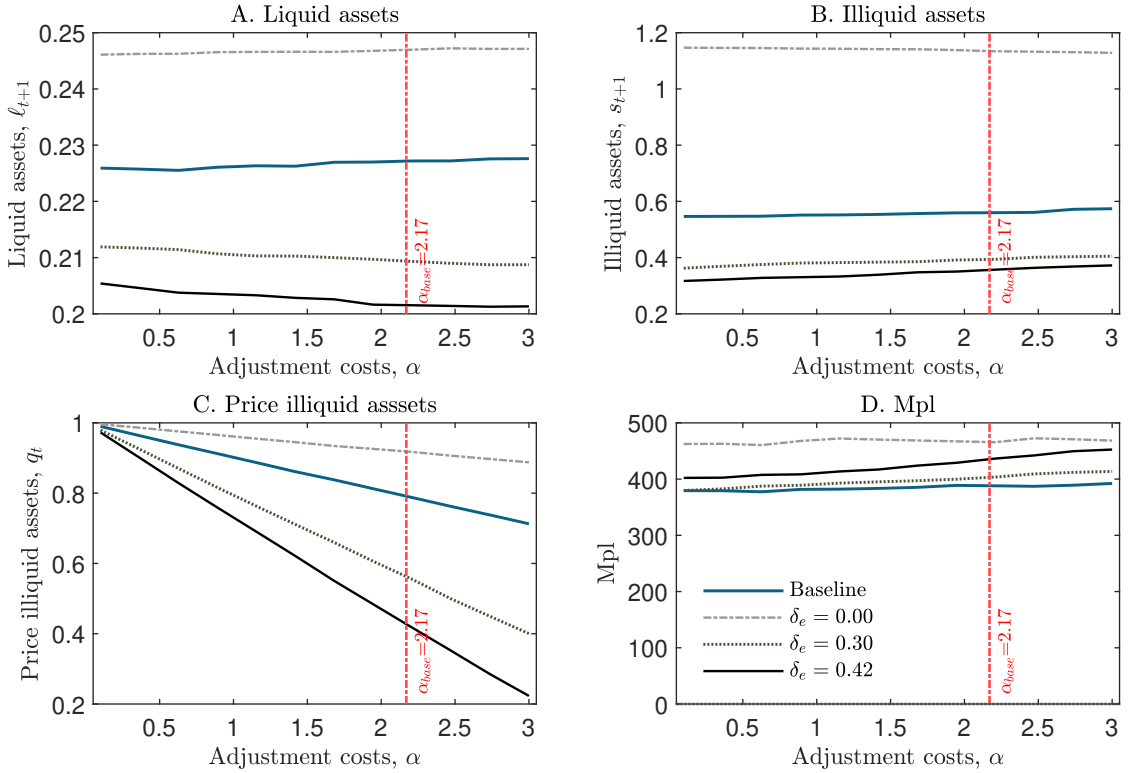


FIG. J.9.—Adjustment costs sensitivity to entry and exit.

NOTE. The figure shows the path of liquid assets, illiquid assets, the price of illiquid assets and the marginal propensity to lend along different values of adjustment costs, for the baseline economy (solid blue lines), for the economy without exit, $\delta_e = 0.00$ (dashed-dot black line), the model with $\delta_e = 0.30$ (dotted black line) and $\delta_e = 0.42$ solid black lines. Vertical red lines denote the adjustment costs parameter under the baseline calibration.

adjustment costs, at lower and higher values of the leverage constraint. While the economy with high leverage resembles the paths of the baseline economy along the adjustment costs dimension, a lower leverage, constrains the banks' ability to expand liquid assets for each unit of deposits. Consequently, not only the accumulation of liquid and illiquid assets is smaller and the price of illiquid assets falls, but also the gap in the marginal propensity to lend between the baseline and the economy with $\kappa = 0.823$ widens, since banks are less sensitive to movements in the interest rate on loans at each point of the adjustment costs.

J.4 MIT Shock Robustness Analysis

Figure J.11 shows the transition paths of the main variables according to the strength of liquidity regulations and variations in adjustment costs. The redistribution of assets between banks becomes more apparent, since changes in the strength of submodularity δ mainly changes the price of illiquid assets and the marginal propensity to lend relative to the baseline. In line with the predictions of the stationary equilibrium, a larger negative value of the submodularity parameter, $\delta = -0.15$ combined with the expansion in the liquidity constraint, leads to a

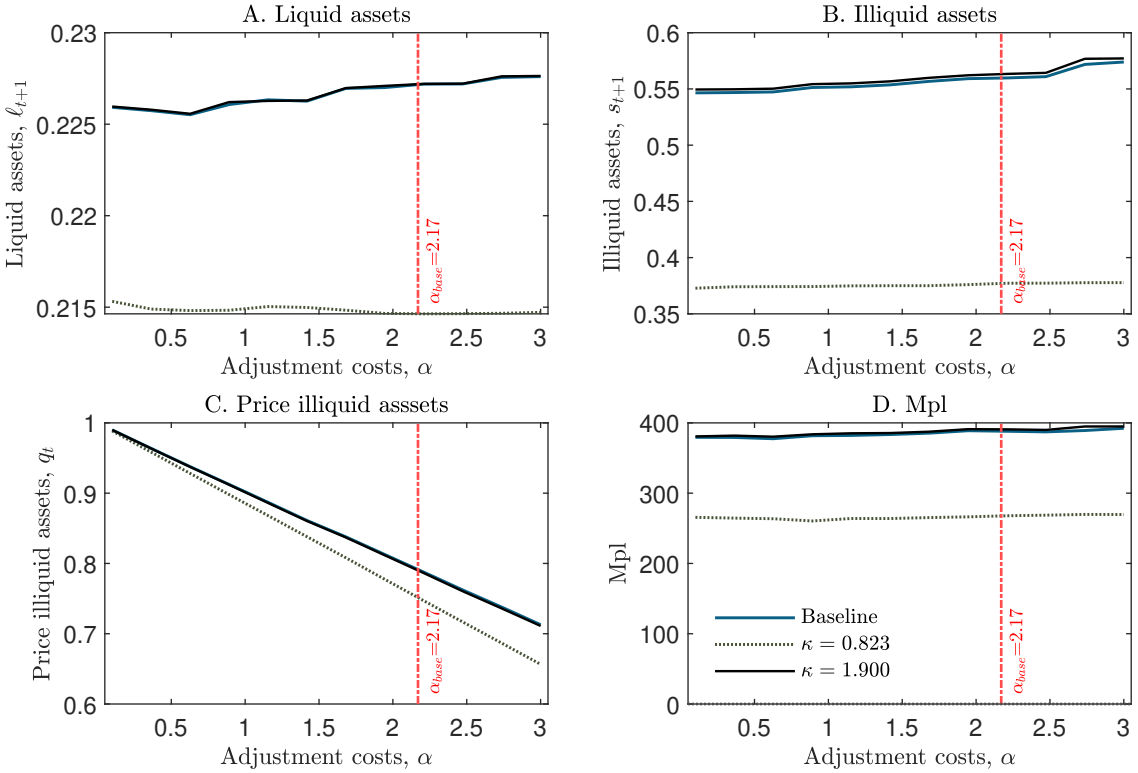


FIG. J.10.—Adjustment costs sensitivity to leverage constraint.

NOTE. The figure shows the path of liquid assets, illiquid assets, the price of illiquid assets and the marginal propensity to lend along different values of adjustment costs, for the baseline economy (solid blue lines), for the economy with lower leverage constraint, $\kappa = 0.823$ (dotted black line) and higher leverage, $\kappa = 0.823$ (solid black line). Vertical red lines denote the adjustment costs parameter under the baseline calibration.

downward path of liquid assets that reaches a 4.61% fall in the second period, before reverting back to its steady state value. Banks redistribute the additional liquidity to expand illiquid assets that rise to 2.15% and occurs with a one period delay relative to the baseline model.

On impact the price of illiquid assets rises to 21.86%, while banks expand lending by 3.36% only about 23 times less than the baseline economy. The reason is that with a less tighter liquidity constraint, the economy is dominated by large banks who have more stable funding inflows and expand lending less aggressively than small banks. In equilibrium the price of illiquid assets increases as the constraints are relaxed, increasing the marginal value of illiquid assets of large banks. The rise in illiquid assets is mainly due to the upward change in illiquid assets of large banks, effectively illiquid assets are redistributed from small banks to large banks. Moreover, as the strength of submodularity between liquid assets and deposits falls, responses along the transition path become smoother reflecting the fact that the liquidity constraint is binding less tightly than the baseline economy for large banks.

Liquidity regulations then determine the market structure by characterising decision rules of banks and the types of banks present in the economy. The lower (higher) the degree of submodularity is, the less (more) tighter the liquidity constraint is, the more the economy is

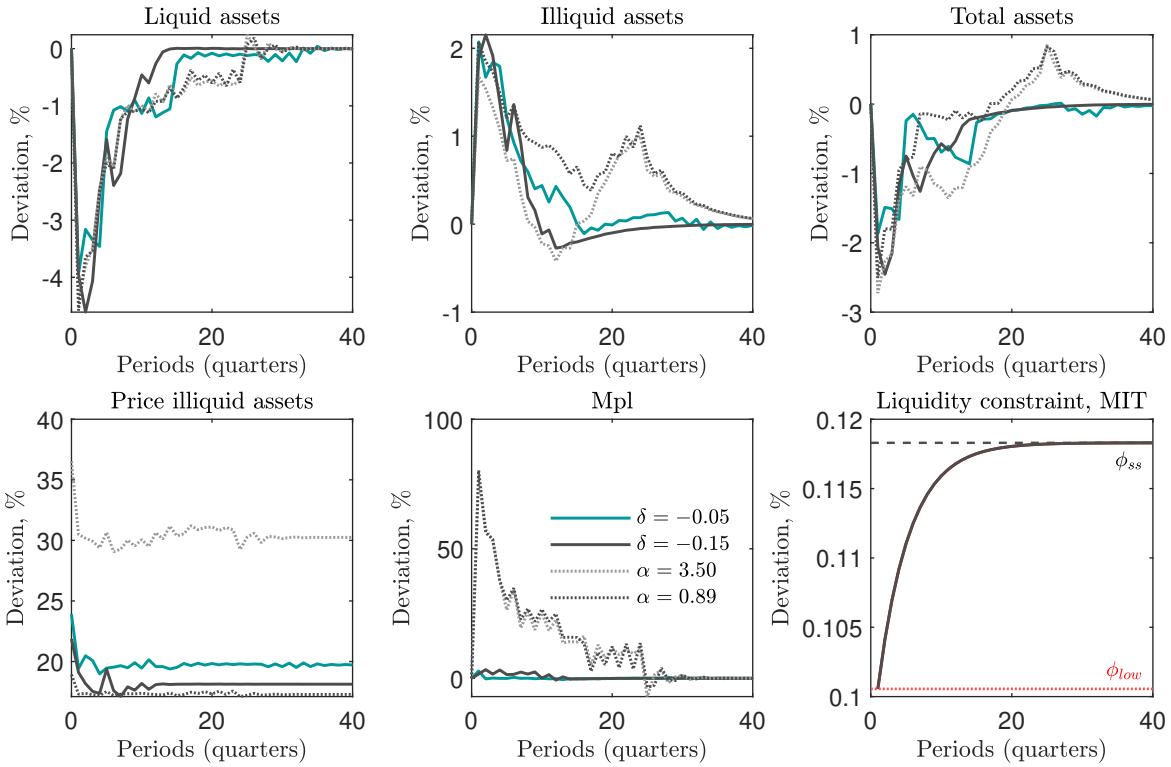


FIG. J.11.—MIT shock and liquidity regulations.

NOTE. The figure shows transition paths to a MIT shock to the liquidity constraint that falls 15% below its steady state value and it is solved along the transition path with geometrically speed of recovery set to $\rho_\phi = 0.80$ in a horizon of 40 periods. The y-axis shows responses in percentage deviation from their steady state values. Transition paths are shown for the model with $\delta = -0.05$ (light blue lines), $\delta = -0.15$ (solid dark gray lines), $\alpha = 3.50$ (dotted light gray lines), and $\alpha = 0.89$ (dotted black lines). Boundaries of the liquidity constraint are denoted in dashed black line, ϕ_{ss}, and dotted red line, ϕ_{low}.

populated by large banks, and the lower (higher) the adjustments of the marginal propensity to lend are, thereby making the economy more stable.

Figure J.11 reports also transition paths to changes in the adjustment costs parameter, $\alpha = 3.50$, and $\alpha = 0.899$, in light gray and black dotted lines, respectively. While the initial impact of the relaxation of the liquidity constraint is the reallocation of assets from liquid assets to illiquid assets, the larger upward changes occur for the price of illiquid assets and the marginal propensity to lend. By increasing illiquid assets, a higher α makes more costly for banks to adjust illiquid assets, thus the fundamental value of illiquid assets decreases and the price of illiquid assets strongly increases to 36.53% that is 10 percentage points higher than the baseline. Over time the transition of the price becomes more gradual and smooth converging to its new steady state value. The increase in the price because of the relaxation of the liquidity constraint redistributes illiquid assets from large to small banks and the marginal propensity to lend rises by 3 percentage points more than the baseline.

A lower adjustment costs parameter set to $\alpha = 0.899$, generates an almost identical response for the marginal propensity to lend to the baseline economy, but it attenuates the response of the

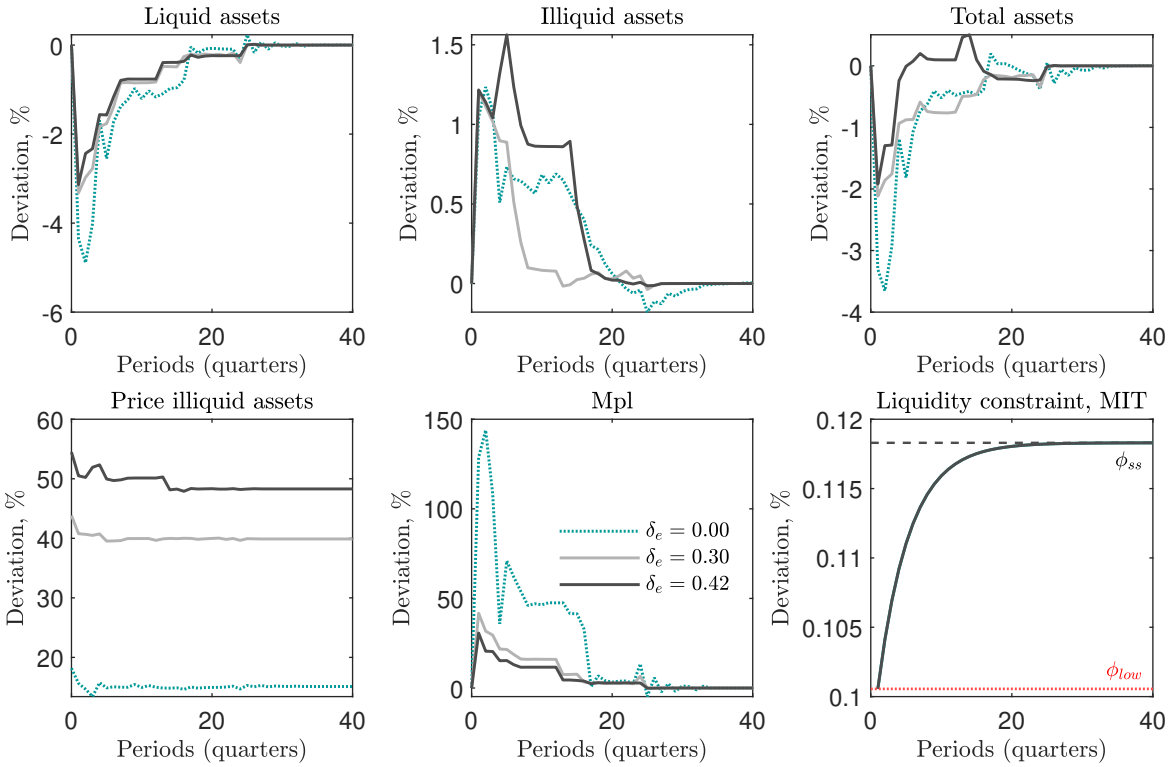


FIG. J.12.—MIT shock and the impact of entry and exit.

NOTE. The figure shows transition paths to a MIT shock to the liquidity constraint that falls 15% below its steady state value and it is solved along the transition path with geometrically speed of recovery set to $\rho_\phi = 0.80$ in a horizon of 40 periods. The y-axis shows responses in percentage deviation from their steady state values. Transition paths are shown for the model with entry value $\delta_e = -0.00$ (light blue lines), $\delta = 0.30$ (solid light gray lines), $\delta_e = 0.42$ (solid dark gray lines). Boundaries of the liquidity constraint are denoted in dashed black line, ϕ_{ss}, and dotted red line, ϕ_{low}.

price of illiquid assets along the transition path, making it less sluggish and converging quickly after one period to its steady state value. Changes in adjustment costs change the speed of adjustment since variables now need more time to converge back to their steady states, while the liquidity constraint is binding for small banks and is less tighter for large banks leading to the less smooth paths along the transition.

Therefore, the joint interaction of liquidity regulations and their submodularity with adjustment costs determine the redistribution of assets between banks and endogenously defines the market structure of banks determining which type of banks dominate the economy.

Figure J.12 shows the transition paths of the main variables of the model to a shock to the liquidity constraint which falls 15% below its steady state value for different values of the fraction of exit. Without exit, banks strongly discount the future value of dividends, as a result the shock generates a gradual decay of liquid assets until the second period which fall by 4.89%, while illiquid rise by 1.23%, less than the baseline economy. The price of illiquid assets increases to 18% on impact, before returning to its steady state value. While the liquidity constraint continues to be tighter for large banks, small banks adjust their lending margin more strongly

than large banks. Consequently, the marginal propensity to lend gradually jumps to 143.93 at $t = 2$, and slowly returns to its steady state.

The larger the share of banks' exit up to $\delta_e = 0.42$, the lower the fall in liquid assets since banks are now less patient and care less about the future. More assets can be redistributed to illiquid assets, while the price of illiquid is 3 times higher than the price of the model without exit. Likewise, the relaxation of the constraint and the more myopic banks lead to a redistribution of illiquid assets toward large banks whose response on the marginal propensity to lend dominates the one of small banks, thereby leading a rise at $t = 1$ that is less than a half the rise of the marginal propensity to lend in the baseline economy. More precisely, the marginal propensity to lend rises by 30.67%. The higher exit rate also generates a simultaneous much smoother response of the marginal propensity to lend and of the price of illiquid assets.

K The Bank Term Funding Program

The Bank Term Funding Program is an emergency loan program implemented by the Fed launched the 12 March 2023 with the aim to provide additional loans to banks following the bank runs in 2023. The program provided loans to eligible banks' institutions in exchange for collateral in the form of quality assets such as treasury securities for up to one year. The program was officially closed March 11, 2024.

The program was designed to put additional liquidity into the financial system in response to Signature Bank and SVB failure as well as subsequent US banks' failures. The failure of these banks followed the erosion of value in Treasury securities as the interest rates rose with inflation. As the liquidity needs of banks increased, the available Treasury securities were sold at a loss. The Fed introduced the Bank Term Funding Program to accomplish the needs of these depositor institutions to rescue the financial system.

The financial pressure in the banking system led banking institutions to borrow $136 billion in one week by December 2023. As the outstanding loans in the program in January 17, 2024 were $161.5 billion, the interest rate on this type of loans that are part of the Bank Term Funding Programme also increased. The loans were borrowed by banks at face value instead of market value of the pledged collateral and the treasury securities accepted as collateral were purchased by the Fed in open market operations. By providing additional source of liquidity, the program eliminated the need of the institution to sell assets quickly in time of financial stress.[18]

The scope of my study is to examine the Bank Term Funding Programme had it been implemented in 2010 following the 2008-09 financial crisis. To accomplish it, I investigate the quantitative performance of the model driven by the designed economic features developed in the main text, with an unanticipated MIT shock to the liquidity capacity of banks where banks re-optimise taking into account the future paths of the main variables of the model.

[18]More information on the Bank Term Funding Program are available on the Fed's website: https://www.federalreserve.gov/financial-stability/bank-term-funding-program.htm.

L Data Details

I collected banks' microdata for a large panel of banks from 1998 to 2016. The data source is the Consolidated Report of Condition and Income submitted by commercial banks to the Federal Reserve, and provided by Wharton Research Data Services (2024). Data are collected for banks located in the fifty states and the District of Columbia following Corbae and D'Erasmo (2021) and I consider individual bank level data. The balance sheet variables are:

i) Liquid assets

1. RCFD0010 Cash and balances due from depository institutions.
2. RCFD0211 Amortised cost of held to maturity treasury securities.
3. RCFD1287 Fair vale available for sale treasury securities.
4. RCFD1289 Amortised cost of held to maturity US government agencies issued by US government agencies.
5. RCFD1293 Fair vale of available for sale US government agency and corporation obligations issued by US government agencies.
6. RCFD1294 Amortised cost of held to maturity US government agency and corporation obligations issued by US government sponsored agencies.
7. RCFD1298 Fair value of available for sale US government agency and corporation obligations issued by US government agencies.
8. RCFD1350 Federal funds sold and securities purchased under agreements to resell in domestic offices of the bank and of its edge and agreement subsidiaries.
9. RCFD1698 Amortised cost of held to maturity mortgage passthrough securities guaranteed by GNMA.
10. RCFD1702 Fair value of for sale mortgage passthrough securities guaranteed by GNMA.
11. RCFD1703 Amortised cost of held to maturity mortgage passthrough securities issued by FNMA and FHLMC.
12. RCFD1707 Fair value of available for sale mortgage passthrough securities issued by FNMA and FHLMC.
13. RCFD1714 Amortised cost of other held to maturity mortgage backed securities issued or guaranteed by FNMA, FHLMC or GNMA.
14. RCFD1717 Fair value of other available for sale mortgage backed passthrough securities issued by FNMA and FHLMC.
15. RCFD1718 Amortised cost of other held to maturity mortgage backed securities collateralised by MBS or guaranteed by FNMA, FHLMC or GNMA.
16. RCFD1732 Fair value of other available for sale mortgage backed securities collateralised by MBS issued or guaranteed by FNMA, FHLMC or GNMA.
17. RCFD1754 Held to maturity securities total.
18. RCFD2170 Total assets.

19. RCFD3545 Trading assets total.
20. RCFD8274 Tier 1 capital allowable under risk based capital guidelines.
21. RCFDB987 Federal funds sold in domestic offices.
22. RCFDB989 Securities purchased under agreements to resell.
23. RCFDG300 Mortgage backed securities, residential passthrough securities guaranteed by GNMA, held to maturity amortised cost.
24. RCFDG303 Mortgage backed securities, residential passthrough securities guaranteed by GNMA, available for sale fair value.
25. RCFDG307 Mortgage backed securities, residential passthrough securities issued by FNMA and FHLMC, available for sale fair value.
26. RCFDG312 Mortgage backed securities, other residential MBS issued or guaranteed by GNMA, FNMA and FHLMC, held to maturity amortised cost.
27. RCFDG315 Mortgage backed securities, other residential MBS issued or guaranteed by GNMA, FNMA and FHLMC, available for sale fair value.
28. RCFDG316 Mortgage backed securities, other residential MBS, collaterilised by MBS issued or guaranteed by GNMA, FNMA and FHLMC, held to maturity amortised cost.
29. RCFDG319 Mortgage backed securities, other residential MBS, collateralised by MBS issued or guaranteed by GNMA, FNMA and FHLMC, available for sale fair value.
30. RCFDG324 Mortgage backed securities, commercial mortgage backed securities, commercial passthrough securities, held to maturity amortised cost.
31. RCFDG327 Mortgage backed securities, commercial mortgage backed securities, commercial passthrough securities, available for sale fair value.

ii) Illiquid assets

1. RCFD1400 Total loans and leases. This item from 2000 have been replaced by summing RCON5369, RCONB528-3123, RCONB529.
2. RCFD1410 Loans secured by real estate.
3. RCFD1766 Commercial and industrial loans.
4. RCFD2170 Total assets.
5. RCFD3190 Other borrowed money.
6. RCFD3816 Commercial real estate construction and land development, commitments to fund loans secured by real estate.
7. RCFDB538 Loans to individuals for household, family and other personal expenditures, credit cards. Introduced in 2009.
8. RCFDB539 Loans to individuals for household, family and other personal expenditures, other revolving credit plans. Introduced in 2009.

iii) Liabilities

1. RCFD2200 Total deposits.

TABLE L.1.—

Mathematical definition of variables

Variable	Equation
Liquidity 1	(Cash+ Federal Funds Sold + Securities held to maturity + Trading assets)
Liquidity 2	Sum of variables in *i)* of banks' balance sheet
Illiquid Assets	Sum of variables in *ii)* of banks' balance sheet
Price of Loans	(Interest and fee income loans + Interest and fee income real estate + Interest and fee income commercial and industrial loans + Income from lease financing)/ Illiquid Assets
Loans Adjust. Costs	(Loan recoveries - Loan charge-offs)
Return on Loans	(Interest and fee income from loans - Loans adjustment costs)
Mpl	(Unit change in illiquid assets/ Unit change in return on loans)

NOTE. Mathematical derivation of variables used in the paper. 'Loans Adjust. Costs' stands for loans adjustment costs.

2. RCFD2800 Federal Funds Purchased and Securities Repurchased. This item has been replaced since 2002 by RCONB993 and RCONB995 summing these two items.

The income statements variables are:

i) Income and cost

1. RIAD4010 Interest and fee income from loans.
2. RIAD4011 Interest and fee income on loans secured by real estate.
3. RIAD4012 Interest and fee income on commercial and industrial loans.
4. RIAD4065 Income from lease financing receivables.
5. RIAD4170 Interest on deposits.
6. RIAD4605 Total loan recoveries.
7. RIAD4635 Total loan charge-offs.

The aggregate variables for excess reserves and interest rate follow the definitions and the construction outlined in Gigante (2026). I use two measures for individual bank liquidity microdata. The first measure contains a subset of collected variables in *i)* of the balance sheet, which are RCFD0010, RCFD1350, RCFD1754 and RCFD3545. The second measure contains all variables in *i)* of the balance sheet variables. I compute illiquid assets as the sum of all variables in *ii)* of the balance sheet. The price of loans is constructed as ratio of the sum of RIAD4010, RIAD4011, RIAD4012 and RIAD4065 to total illiquid assets. Loans adjustment costs are computed as the difference between RIAD4605 and RIAD4635. The return on loans is the difference between interest income on loans minus adjustment costs. In computing the price of loans, loan adjustment costs and the return on loans, and their respective variables in units of deposits I control for outliers and hence, I eliminate observations at the bottom of the deposit distribution. The marginal propensity to lend is computed as the ratio between the change in illiquid assets in two consecutive years and the change in the return on loans in two consecutive years.